Upgrade Dell PowerFlex to v3.6.x

June 2025
Rev. 06

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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1

Introduction

Upgrade PowerFlex software and firmware components to the latest compatible versions, using automated and manual processes.

1

Introduction

This guide provides the upgrade procedures for PowerFlex components.

The following table lists the upgrade support matrix:

Table 1. Upgrade support matrix

ScaleIO / VxFlex operating system / PowerFlex base version	Upgrade path	Supporting documentation
v2.0.1.4, 2.6.x	Two-step upgrade flow: 1. Upgrade to latest version of v3.0.1.x 2. Upgrade to v3.6.0.x	*Upgrade VxFlex OS to v3.0.x Guide* *Upgrade PowerFlex to v3.6 Guide*
v3.0, v3.0.0.x, v3.0.1.x, v3.5.x, v3.6	Single-step upgrade from: • VxFlex operating system v3.0/v3.0.0.x/ v3.0.1.x • PowerFlex v3.5.x • PowerFlex v3.6 to PowerFlex v3.6.0.x	*Upgrade PowerFlex v3.6 Guide*

NOTE: On ESXi-based PowerFlex v3.0, or v3.0.0.x systems that contain CentOS 7.5-based SVMs, the CentOS operating system must be upgraded after upgrade of system components.

NOTE: Upgrade of AMS managed PowerFlex systems to v3.6:

AMS is not supported on PowerFlex v3.5 and later versions. For guidance on AMS removal and the option to move to PowerFlex Manager, reach out to PowerFlex Product Management.

Automated and manual upgrade of software components

You can upgrade PowerFlex software components with automated tools, in a non-disruptive manner, or manually upgrade components.

Instead of upgrading manually, Dell recommends that upgrades be performed using the PowerFlex Installer or the vSphere PowerFlex plug-in. The PowerFlex Installer resides on the PowerFlex Gateway.

NOTE: When performing a manual upgrade (as opposed to using the PowerFlex Installer), it is crucial to follow all the described upgrade procedures step by step. Failure to do so might result in data loss and system instability.

Before upgrading the system, check that the following issues are addressed:

- Ensure that all servers meet the system requirements for the new version, as described in the *Getting to Know Dell PowerFlex Guide*. For ESXi-based systems, SVM memory requirements must also be taken into account.
- Ensure that sufficient spare capacity is available to support the upgrade. The spare capacity will be necessary while nodes are temporarily offline and being upgraded. For more information, see the *Getting to Know Dell PowerFlex Guide*.
- Ensure that there are no failed disks (failed capacity).
- Verify that the cluster is not in Degraded state.
- Ensure that all data NICS are connected.
- If distributed vSwitches are used, ensure that there are enough ports in each of the distributed vSwitches used for PowerFlex (Host, SVM and new SVM). The number of ports needed is equal to three times the number of nodes in the system.

Dell recommends that upgrades be performed using the PowerFlex Installer (which or the vSphere PowerFlex plug-in.

When using the PowerFlex Installer to upgrade, you can configure the following settings as part of the upgrade process:

- Set high-performance profile—if the existing system components used the compact profile (default in versions earlier than v3.0) you can choose to set the high-performance profile for MDM, SDS, or SDC components. Before upgrading multiple SDS nodes and changing their performance profile to high-performance, ensure that there is adequate available RAM on each node. If there is not enough RAM, you can change the performance profile after upgrading, one node at a time.

  When upgrading a system, the previous performance profiles are retained, and are not changed to the new default setting (high-performance). They only override the existing configuration when you select the option to set the performance profile during upgrade.

  NOTE: If you select to apply the profile again, the profile settings will be applied and will override existing settings. If no profile is selected, existing settings will be retained.

- Restart Windows-based servers—enable the automatic restart of Windows servers on which SDC is being installed. This option should not be used if there are components of another PowerFlex cluster on the same server.
- Enable parallel SDS upgrades—enable the upgrading of all SDS nodes in an entire Fault Set at the same time
- Do not upgrade any SDCs (ESXi nodes will not be upgraded, in any case. SDCs on ESXi are upgraded using the PowerFlex plug-in or manually.)
- Resume an upgrade that was interrupted when using PowerFlex Installer on a different PowerFlex Gateway (allow upgrade even when already in Upgrade state)

After upgrading the system, you can configure the following settings:

- Install RFcache
- Extend the MDM cluster from 3-node to 5-node mode
- Add virtual IP addresses

These options, and others, are described in the *Configure and Customize Dell PowerFlex*.

The upgrade procedure for your environment is determined by your current configuration. Proceed to the section that matches your configuration:

Table 2. Automated upgrades

Configuration	See...
Physical Linux—The base system resides solely on bare metal Linux-based servers.	"Automated upgrade of Linux-based nodes"
ESXi—The base system resides solely on bare metal ESXi-based servers.	"Automated upgrade of ESXi-based nodes"
Two-layer (ESX and Linux)—The MDM and SDS reside on bare metal Linux-based servers, and the SDCs reside on ESXi-based servers.	"Automated upgrade of nodes in a two-layer configuration"
Two-layer (Windows and Linux)—The MDM and SDS reside on bare metal Linux-based servers, and the SDCs reside on Windows-based servers	"Automated upgrade of nodes in a two-layer configuration"

Table 3. Manual upgrades

Configuration	See...
Physical Linux—The base system resides solely on bare metal Linux-based servers.	Manual upgrade of Linux-based nodes
ESXi and two-layer (ESX and Linux, Windows and Linux)	Contact Customer Support

Upgrade limitations

Check that you are aware of upgrade limitations before commencing the upgrade process.

Upgrades of SDCs or LIA LDAP configuration in large scale systems

The PowerFlex Gateway has an upgrade limit of 512 SDCs per non-disruptive upgrade. The same limit applies when configuring LIA LDAP in large scale systems. For these scenarios, follow the recommendations provided in the "Reference" section of this guide.

Multiple SDSs on the same server

The installation of multiple SDS instances on the same server is no longer supported. To guarantee a better customer experience and improved stability, PowerFlex currently only supports a single SDS per server.

As previous releases of PowerFlex supported the installation of multiple SDS instances per server (up to four), existing servers using the multiple-SDS feature should be reconfigured in order to be eligible for an upgrade. Please consult with your Dell Professional Services contact to assist with this configuration change, before you upgrade your system.

NVDIMM capacity requirements for future upgrades

Increased NVDIMM capacity per SDS is mandatory in PowerFlex version 3.5 or higher that contain Fine Granularity Storage Pools. If your system has insufficient NVDIMM capacity for PowerFlex v3.5 or higher, you will not be able to upgrade your system, and a warning alert will be generated.

An upgrade readiness tool is provided with this release of PowerFlex to help you check your system. Instructions for doing so are described in the "Post-Upgrade" section of this guide, and in the *Configure and Customize Dell PowerFlex*. If the tool recommends more NVDIMM capacity, contact your account manager for assistance before you upgrade your system.

NVDIMM 4k alignment

On nodes configured with NVDIMMs, ndctl and daxio packages must installed before performing the upgrade:

- ndctl package—for example: `ndctl-64.1-2.el7.x86_64.rpm`
- daxio package—for example: `daxio-1.5.1-2.1.el7.x86_64.rpm`

Replication feature

Upgrades of systems where replication is used using the PowerFlex Installer are supported, but manual upgrades of replicated systems are not allowed. Contact your account manager for assistance before you manually upgrade your system.

Simultaneous NDU of source and destination sites is not allowed. Each PowerFlex system should be upgraded separately at different times.

Upgrade to new major ESXi version

Upgrade of the ESXi operating system from version 6.7 to 7.0 where PowerFlex version 3.5 or higher is deployed requires SDC removal and re-installation. For more information, see the *Reference* section of this guide.

RHEL/CentOS upgrade to version 8.3

In cases where you are upgrading the operating system from RHEL/CentOS 8.0 to RHEL/CentOS 8.3 on a node containing preconfigure NVDIMMs (devdax devices), you are required to use the upgrade command for no dependencies.

About this task

NOTE: If you are upgrading multiple nodes, perform the following steps one node at a time.

Steps

1. From the PowerFlex GUI, place the node into Protected Maintenance Mode and wait for the operating to complete.
2. From the system console, enter:

```
dnf upgrade --nobest
```

3. Answer **Yes** to all questions that occur during the upgrade process.
4. Wait for the upgrade process to complete, including the reboot, if required.
5. From the PowerFlex GUI, remove the node from Protected Maintenance Mode.
   Wait for the rebuild to complete.
6. Check the upgrade log on the node to determine whether any packages require upgrading manually.
7. Proceed to the next node, if relevant.

Configure the SDS for optimal memory usage

For certain system configurations, the SDS needs to be configured to be able to use more than one Non-Uniform Memory Access (NUMA) domain.

About this task

In many cases, memory is split into a few NUMA domains. For example, in a two-CPU-socket server configuration, each of the CPUs is in a different NUMA domain. The SDS is configured by default to have affinity to socket 0 in a server or VM, and therefore, by default, the SDS is connected only to NUMA 0, and only has access to the memory in NUMA 0.

In ESXi in HCI configurations, the NUMA definition requires special attention. When configuring the SVM, there are cases where the SVM includes two vNUMAs. For example, if the number of vCPUs required for the SVM is larger than the number of physical cores in the CPU, ESXi will configure the SVM with two vNUMAs. Other cases are ESXi HCI with replication, where the SVM includes both the SDR and the SDS. The SDS will be configured on vNUMA 0 and the SDR on vNUMA 1. The total memory will be split equally between the two vNUMAs, and therefore, the SDS may not have enough memory on NUMA 0.

The memory size required for the SDS depends on the configuration, and on other factors. In order to determine the required size for the SDS, and whether NUMA configuration is required, follow the size recommendations provided in the *Getting to Know Dell PowerFlex Guide*.

If the memory supported by NUMA 0 (usually half the total memory) is less than the amount of memory required by the SDS, you need to allow the SDS access to the memory in the other NUMA. For example, if a node has 96 GiB of physical memory, divided into 48 GiB on each CPU, and the SDS requires 58 GiB of memory, this NUMA modification is required, because 48 GiB is not sufficient for the SDS.

This operation will have a small effect on performance (less than 10%), and for this reason, is not the default setting.

NOTE: This setting can also be modified on operational SDSs that are expected to use more memory in the future, for example, due to adding SDS devices. Note that this procedure requires a graceful SDS restart in order to take effect.

Steps

1. Use one of the following options to edit the NUMA configuration:

Option	Description
Use the echo command	On the node where the SDS is deployed, issue the command: `echo "numa_memory_affinity=0" >> /opt/emc/scaleio/sds/cfg/conf.txt`
Use a text editor	a. On the node where the SDS is deployed, navigate to the file: `/opt/emc/scaleio/sds/cfg/conf.txt` b. Open the file with a text editor, and add the following line to the file: *numa_memory_affinity=0* c. Save the file.

2. Gracefully restart the SDS:
   a. Use one of the PowerFlex management interfaces to place the SDS in instant maintenance mode.
   b. Restart the SDS process:

```
/opt/emc/scaleio/sds/bin/delete_service.sh
/opt/emc/scaleio/sds/bin/create_service.sh
```

   c. Exit instant maintenance mode.
   d. Wait for the rebuild to finish before performing other tasks.

Results

Configuration of memory usage for the SDS is complete.

Next steps

Repeat this procedure on all SDSs that require this modification.

Special requirements for upgrades from v2.x to v3.x

Follow these special guidelines for a non-disruptive upgrade from PowerFlex v2.X to v3.X (in addition to the supported upgrade matrix):

Prerequisites

- IP addresses of SDCs to be excluded during the non-disruptive upgrade
- Admin access credentials for the PowerFlex Gateway

About this task

To upgrade SDCs along with the other system components during a non-disruptive upgrade, use the regular recommended procedures. However, if SDCs need to be upgraded separately, at a different time, perform the following steps before commencing any upgrade procedures. These steps will exclude the SDCs from the non-disruptive upgrade process.

Steps

1. Use a text editor to open the `gatewayUser.properties` file, located in the following directory on the PowerFlex Installer/PowerFlex Gateway:

Operating system	File location
Windows	`C:\Program Files\EMC\ScaleIO\Gateway\webapps\ROOT\WEB-INF\classes\`
Linux	`/opt/emc/scaleio/gateway/webapps/ROOT/WEB-INF/classes`

2. Add the IP addresses of the SDCs to be excluded to the property *im.ip.ignore.list=*
   For example: im.ip.ignore.list=10.0.0.1,10.0.0.2,10.0.0.3
3. Save and close the file.
4. Restart the scaleio-gateway service:

Operating system	Action
Windows	Restart the EMC ScaleIO Gateway service
Linux	Type the following command: `service scaleio-gateway restart`

5. Proceed with the regular upgrade steps described in this guide.

Results

The SDCs listed in the ignore list will be excluded from non-disruptive upgrades.

II

Automated Upgrade

Upgrade PowerFlex components using PowerFlex automated management tools.

2

Automated Upgrade of Linux-based Nodes

Use automated management tools to upgrade PowerFlex components on Linux-based nodes.

Automated upgrade of physical Linux-based configurations

Upgrade PowerFlex in an automated process, when all components reside on physical Linux-based nodes.

Physical machine upgrades use the PowerFlex Installer (part of the PowerFlex Gateway), together with the LIA, to orchestrate the upgrade.

NOTE: To verify if LIA is installed, or to add LIA to a system where it is not installed, see "Add LIA to a system to enable automated maintenance and upgrades" in the *Configure and Customize Dell PowerFlex*.

Upgrades are non-disruptive; upgrades can be performed while I/Os are running and volumes are mapped.

The PowerFlex Installer can be used to upgrade MDM clusters from a 3-node cluster to a 3-node cluster, or from a 5-node to a 5-node cluster. Extension of a 3-node MDM cluster to a 5-node MDM cluster can be performed after the system upgrade. For more information, see "Extend an existing PowerFlex system" in the *Configure and Customize Dell PowerFlex*.

NOTE: Automated upgrades are not supported on systems managed by a single-node MDM cluster.

Linux package names

PowerFlex Linux installation packages have several name formats.

Package formats

Throughout this section, Linux packages are displayed in the following format:

`EMC-ScaleIO-<component>-3.6-X.<build>.<flavor>.x86_64.rpm`

where:

- *<component>* is the PowerFlex component: MDM, LIA, SDS, SDR, and so on.
- *<flavor>* is the operating system for your environment, according to the following table:

Table 4. Linux package formats

Linux flavor	Package format	Example
RHEL/OL/CentOS	el<version>	`EMC-ScaleIO-mdm-3.6-X.<build>.el<version>.x86_64.rpm` Example: `EMC-ScaleIO-mdm-3.6-<build>.X.el6.x86_64.rpm`
SUSE	sles<version>	`EMC-ScaleIO-mdm-3.6-X.<build>.sles<version>.x86_64.rpm` Example: `EMC-ScaleIO-mdm-3.6-<build>.X.sles12.2.x86_64.rpm`
Ubuntu	Ubuntu.<version>	`EMC-ScaleIO-mdm-3.6-X.<build>.Ubuntu.<version>.x86_64.tar` Example: `EMC-ScaleIO-mdm-3.6-X.<build>.Ubuntu.16.04.x86_64.tar` Ubuntu packages may need to be extracted before use. This is described where relevant.

Table 4. Linux package formats (continued)

Linux flavor	Package format	Example
Citrix Hypervisor (XenServer)	xs<version>	`EMC-ScaleIO-mdm-3.6-X.<build>.xs<version>.x86_64.rpm` Example: `EMC-ScaleIO-mdm-3.6-X.<build>.xs7.0.0.x86_64.rpm`

NOTE: RedHat CoreOS and Fedora CoreOS are supported on CSI containers. Refer to CSI documentation for more information.

Use the packages and the installation commands that match your Linux operating system environment.

On Citrix Hypervisor (XenServer)-based nodes, the syntax for PowerFlex CLI commands is `siocli`, instead of `scli`.

NOTE: Before deploying or upgrading Oracle Linux (OL) or Ubuntu-based systems, ensure that the PowerFlex environment is prepared, as described in the preparation tasks for Oracle Linux, SLES and Ubuntu in the *Deploy Dell PowerFlex* .

Preparing to upgrade physical Linux-based configurations

Steps to perform before upgrading PowerFlex on physical Linux-based nodes.

About this task

Before upgrading the PowerFlex components, perform the following steps:

Steps

1. Install one of the following on the PowerFlex Gateway and PowerFlex presentation server nodes, as described in detail in the system requirements in the *Getting to Know Dell PowerFlex Guide*:

Option	Example

Oracle JRE 1.8 build 301	• Linux:
	`rpm -i jre-8uXX-linux-x64.rpm`
	• Windows (for the PowerFlex Gateway only): Double-click the file:
	`jre-8uXX-windows-x64.exe`

OpenJDK JDK 8 build 302.b08 or OpenJDK 11.0.12.0.7 **NOTE:** JAVA 8 is not supported on PowerFlex v3.6.3 and later versions. For more information, see *Dell PowerFlex 3.6.2 Release Notes*.	Refer to the following information: • Linux: Oracle Java SE Archive • Windows (for the PowerFlex Gateway only): Downloads for Windows x86_64

2. Download and extract the PowerFlex packages.
   You need the packages for your base version and for the latest version (the version to which you are upgrading).
3. If you plan to use a PowerFlex Gateway that did not initially deploy your PowerFlex system, add the IP addresses of the system's primary and secondary MDMs to the `gatewayUser.properties` file, `mdm.ip.addresses` property, separated by semicolons. Both management and data IP addresses must be listed.
   Use a text editor to open the `gatewayUser.properties` file, located in the following directory on the PowerFlex Gateway node:

PowerFlex Gateway operating system	Location of gatewayUser.properties file
Windows	C:\Program Files\EMC\ScaleIO\Gateway\webapps\ROOT\WEB-INF\classes\
Linux	/opt/emc/scaleio/gateway/webapps/ROOT/WEB-INF/classes

4. For data networks using IPv6, if you plan to implement a floating virtual IP address for the MDM, disable the IPv6 DAD setting on the node's relevant interfaces, using the command:

```
sysctl net.ipv6.conf.<interface_name>.dad_transmits=0
```

5. If your SDSs contain NVDIMMs, ensure that the *ndctl* and *daxio* utilities are installed on each node before you commence the upgrade.
   - ndctl package—for example: `ndctl-64.1-2.el7.x86_64.rpm`
   - daxio package—for example: `daxio-1.5.1-2.1.el7.x86_64.rpm`

   CAUTION: Failure to meet this requirement could cause data loss.

6. If your SDSs contain NVDIMMs, spare capacity equivalent to the capacity of the NVDIMMs is required during the upgrade process. Ensure that sufficient capacity is available in the `/tmp` directory of each SDS. If you would like to use a different directory, you can configure this in the `gatewayUser.properties` file.
   a. Using a text editor, open the `gatewayUser.properties` file, located in the following directory on the PowerFlex Gateway node:

PowerFlex Gateway operating system	Location of gatewayUser.properties file
Windows	C:\Program Files\EMC\ScaleIO\Gateway\webapps\ROOT\WEB-INF\classes\
Linux	/opt/emc/scaleio/gateway/webapps/ROOT/WEB-INF/classes

   a. Add the following line to the file:
   `nvdimm.granularity.format.backup.path=<PATH>`

   where *<PATH>* is the path to the directory that will be used for NVDIMM backup.

   b. Save and close the file.
   c. Restart the PowerFlex Gateway service:
      - Windows: From the Windows **Services** window, restart the **EMC ScaleIO Gateway**.
      - Linux: Type the following command:

```
service scaleio-gateway restart
```

7. If you intend to use the CHAP component authentication feature after the upgrade, make the necessary preparations to ensure that you upgrade all system components now. Systems containing any components running PowerFlex versions lower than v3.5 do not support CHAP authentication.

   NOTE: Upgrade to v3.5 sets the CHAP authentication feature to DISABLED, and passwords will be all-zero. To enable the feature, after the upgrade is complete on all system components, follow the post-installation procedures described in the *Configure and Customize Dell PowerFlex*.

8. In SDSs containing NVDIMMs, verify that there are no bad blocks on the NVDIMMs. Bad blocks might prevent upgrade of the corresponding SDS. Bad blocks might be present when an NVDIMM's status is Normal. If there are bad blocks, replace the NVDIMMs with bad blocks before upgrading the node.
   a. Using shell, run the following command on each Linux-based node that has an NVDIMM:

```
ndctl list --media-errors
```

   b. If the output for any of the hosts displays a result for `"badblock_count":`, one or more NVDIMMs (devdax device) have bad blocks. For example:

```
{
 "dev":"namespace1.0",
 "mode":"devdax",
 "map":"dev",
 "size":62914560,
 "uuid":"47072fee-1447-4e04-b4ca-f28e2c03af80",
 "chardev":"dax1.0",
 "align":4096,
 "badblock_count":13,
```

```
    "badblocks":
}
```

c. Refer to the Linux NVDIMM replacement procedure for the specific server model.

Non-default certificate use before and after a PowerFlex Gateway upgrade

When using a non-default security certificate, you must perform certain actions before and after upgrading the PowerFlex Gateway.

About this task

If a non-default security certificate is used on the PowerFlex Gateway in Windows and Linux environments (for example, if the certificate is signed by the user organization CA), you must save a copy of the certificate and the catalina file before the upgrade, and restore them to their original location after the upgrade.

Steps

1. Before commencing the PowerFlex Gateway upgrade, locate the `*.keystore` file and the `catalina.properties` file, and save a copy of them in another location.

   The default file locations, per operating system, are:

   - Linux:
     - `/opt/emc/scaleio/gateway/conf/catalina.properties`
     - `/opt/emc/scaleio/gateway/conf/certificates/.keystore`
   - Windows (64 bit):
     - `C:\Program Files\EMC\ScaleIO\Gateway\conf\catalina.properties`
     - `C:\Program Files\EMC\ScaleIO\Gateway\conf\certificates\.keystore`

2. After the upgrade is complete, copy these files back to their original location.

Upgrading PowerFlex—Linux-based nodes

Use the PowerFlex Installer to upgrade PowerFlex components to the latest version, on Linux-based nodes.

Prerequisites

Software packages are required for both current and target versions. The current version is required to facilitate rollback, should it be necessary.

Ensure that you have authentication credentials for the primary MDM (IP address or host name, and password) and for LIA (native password, or for LDAP, the server URI, Group, BaseDN, user name and password).

About this task

- This procedure can be used for upgrading a 3 or 5-node cluster on all Linux flavor nodes.
- When SNMP is enabled in the 2.x base version, SNMP MDM credentials remain in the lockbox after the upgrade.

NOTE: During this procedure, when you mark the upgrade operation as completed, an MDM service restart occurs.

Steps

1. Upgrade the Linux-based PowerFlex Gateway:

   ```
   rpm -U EMC-ScaleIO-gateway-3.6-X.<build>.x86_64.rpm
   ```

   The following variables can be added to the command, if required:

   - *SIO_GW_KEYTOOL=<keytool_path>* allows override of the location for the Java keytool utility. For example: `SIO_GW_KEYTOOL=/usr/java/jre1.8.0_XX/bin/`
   - *SIO_JAVA=<Java version>* specifies a different Java version

Example:

```
rpm -U /tmp/EMC-ScaleIO-gateway-3.6-X.<build>.x86_64.rpm
```

A success message is displayed, and the PowerFlex Installer has been updated to the new version. Log in to the updated PowerFlex Installer will be required.

2. When upgrading Citrix Hypervisor (XenServer)-based nodes, perform the following to upgrade the LIA manually. For other Linux flavors, skip to the next step.
   a. Upgrade (or install) LIA on one of the nodes in the system:

      ```
      rpm -U EMC-ScaleIO-lia-3.6-X.<build>.xsX.X.X.x86_64.rpm --nosignature
      ```

   b. Repeat for every server in the system.
      After all LIAs are installed/upgraded, continue to the next step.
3. Log in to the PowerFlex Installer (https://*<INSTALLER_IP_ADDRESS>*).
4. Click **Packages**.

   The **Manage Installation Packages** window appears.

   NOTE: When you use this PowerFlex Installer to install packages for Windows and Linux, you can upload all the files at once.

5. Browse to where the PowerFlex packages are located for the operating system of the nodes that you are deploying.
6. Upload the installation packages from the current and the target versions—select all the relevant files for the operating systems you want to deploy on, then click **Open**.

   File formats can be RPM, TAR, or MSI. Minimally, you must select these packages:

   a. MDM
   b. SDS
   c. SDC
   d. LIA

   Packages from the current version: lia, mdm, sdc, sds, xcache (optional), sdr (optional), VMware VVol (VASA provider; optional). For example:

   - `EMC-ScaleIO-lia-3.0.x-XXX.<build>.<flavor>.x86_64.rpm`
   - `EMC-ScaleIO-mdm-3.0.x-XXX.<build>.<flavor>.x86_64.rpm`

   Packages from the latest target version: lia, mdm, sdc, sds, xcache (optional), sdr (optional), VMware VVol (VASA provider; optional). For example:

   - `EMC-ScaleIO-lia-3.6-X.<build><flavor>.x86_64.rpm`
   - `EMC-ScaleIO-mdm-3.6-X.<build>.<flavor>.x86_64.rpm`

   The **Browse** button changes its appearance to reflect the selected files.
7. Click **Upload**.

   The uploaded installation packages (RHEL, in this example) are listed in the file table.

8. Upgrade the PowerFlex components:
   a. From the PowerFlex Installer, click the **Maintain** tab.
   b. From the **Maintenance operation** screen, perform the following:
      i. Type the primary MDM's credentials.
      ii. Type the LIA password (upgrades from versions prior to v3.0 only allow native authentication for LIA).
      iii. Click **Retrieve system topology**. A success message is displayed.
      iv. Click **Upgrade**.

      The **Device Media Type for PowerFlex system** window is displayed.

   c. Select the device media type (mandatory) and external acceleration type (optional) for each Storage Pool in the system, and then click **Proceed**.

      For external acceleration, options include:
      - none—No devices are accelerated by a non-PowerFlex read or write cache
      - read—All devices are accelerated by a non-PowerFlex read cache
      - write—All devices are accelerated by a non-PowerFlex write cache

- read_and_write—All devices are accelerated by both non-PowerFlex read and write cache

This input is required in order to prevent the generation of false alerts for media type mismatches. For example, if an HDD device is added which the SDS perceives as being too fast to fit the HDD criteria, alerts might be generated. External acceleration/caching is explained in the *Getting to Know Dell PowerFlex Guide*.

d. Set upgrade options:
   - To prevent upgrade of SDCs, select **Do not upgrade any SDCs**. You might want to use this option if the SDCs are running critical applications that are currently in use. You can upgrade the SDCs later.
   - To continue an upgrade that was interrupted on a different gateway, select **Allow upgrade even when already in Upgrade state**.

     Use this option when you started an upgrade using a different gateway, but that gateway failed or is offline now.
   - Set performance profiles.

     High performance profiles are mandatory when using Fine Granularity Storage Pools. For other configurations, they are optional. You can modify these profiles later.
   - Perform automatic restart of Windows-based servers (required for Windows-based nodes on which SDCs are deployed).
   - To upgrade SDSs in the same Fault Set together, select **Allow parallel SDS upgrades**.

     Parallel upgrade of SDSs in the same Fault Set saves time; the upgrade will be performed faster.

e. Confirm the upgrade operation by typing the MDM admin password, and click **Upgrade**.

   NOTE: If a certificate notice is displayed, review and approve it.

f. Click the **Monitor** tab, and approve each step of the upgrade process.
g. When the upgrade is complete, click **Mark operation completed**.

Results

The upgrade is complete.

Upgrade the PowerFlex GUI from versions earlier than v3.5

Use these steps to upgrade the PowerFlex GUI from versions earlier than v3.5.

Prerequisites

NOTE: From PowerFlex version 3.5, the PowerFlex GUI is a web-based application installed on a server. Therefore, the new PowerFlex GUI is a fresh installation known as the PowerFlex presentation server, and the GUI client from VxFlex OS versions of the product is neither upgraded nor supported going forward.

- Ensure that the workstation complies with the PowerFlex system requirements.
- Ensure that you have the PowerFlex presentation server installation file. It is included in the PowerFlex complete software package that you downloaded from Dell Technologies support site.

NOTE: It is not recommended to install the PowerFlex presentation server on the same server as the PowerFlex Gateway. If you must install them on the same server, the PowerFlex Gateway must be installed first. The PowerFlex presentation server listens on port 8443, and the PowerFlex Gateway installer checks that port 8443 is available, although it does not use this port.

Steps

1. Install the PowerFlex presentation server on a server in your system:
   - Linux:

```
rpm -i EMC-ScaleIO-mgmt-server-3.6-X.<build>.noarch.rpm
```

- Debian (run with administrator privileges):

```
sudo dpkg -i emc-scaleio-mgmt-server_3.6-X.<build>_amd64.deb
```

2. Prior to using the PowerFlex GUI, link the PowerFlex presentation server to the PowerFlex system. For more information, about linking the system, login and use of the PowerFlex GUI, refer to the *Configure and Customize Dell PowerFlex*.

Upgrade the PowerFlex presentation server (PowerFlex GUI) from v3.5 or later

Upgrade the PowerFlex presentation server from v3.5 or later.

Prerequisites

Ensure that you have the package for the PowerFlex presentation server, and access to the node hosting the PowerFlex presentation server.

Steps

1. Save a copy of the PowerFlex presentation server package on the node where PowerFlex presentation server is installed.
2. On the node running the PowerFlex presentation server, run the following command:
   - Linux:

```
rpm -Uvh EMC-ScaleIO-mgmt_server-3.6-X.<build>.noarch.rpm
```

   - Debian:

```
dpkg -i emc-scaleio-mgmt_server-3.6-X.<build>.amd64.deb
```

3. On a Centos 8-based nodes only, when the installation script has finished, type Q.

Results

Upgrade of the PowerFlex presentation server is complete. Link the PowerFlex presentation server to the MDM cluster, as described in the *Configure and Customize Dell PowerFlex*. Existing credentials can be used, provided they belong to either admin or configure + monitor (LDAP or native) users.

Prepare server operating systems for support of large-scale systems (up to 2048 SDCs)

For large-scale systems with up to 2048 SDCs, prepare the operating systems of servers that will host MDM, SDS, SDR, and PowerFlex Gateway components.

Prerequisites

Superuser permissions are required for this procedure.

About this task

Configuration of servers hosting MDM, SDS, SDR, and PowerFlex Gateway components is required, in order to support large-scale systems with up to 2048 SDCs.

- A system setting must be validated and configured on servers hosting MDM, SDS, SDR, and the PowerFlex Gateway.
- For all servers hosting MDM, SDS, and SDR components, certain threshold parameters must be configured. This change enables a larger number of records in the operating system's ARP Cache table.

Steps

1. On servers hosting MDM, SDS, SDR, or PowerFlex Gateway components, do the following:

a. Run this command:

```
cat /proc/sys/fs/file-max
```

The output for the command is the number of files that can be opened by the operating system.

b. If the number is 24576 or higher, go to step 2. If the number is less than 24576, go to the next sub-step.
c. Open the file `/etc/sysctl.conf` using vi.
d. If the following line exists, edit it with vi. If it does not exist, add the full line with the following setting:

```
fs.file-max = 24576
```

2. On servers hosting MDM, SDS, or SDR components, if the following lines exist in the file `/etc/sysctl.conf`, edit them with vi. If they do not exist, add the full lines with the following settings:

```
net.ipv4.neigh.default.gc_thresh1 = 8192
net.ipv4.neigh.default.gc_thresh2 = 16384
net.ipv4.neigh.default.gc_thresh3 = 32768
```

3. Save the file.
4. Apply the changes using the command:

```
sysctl -p
```

5. Repeat this procedure on the next server, until all servers are configured.

Post-upgrade activities

Tasks to perform after the automated upgrade on Linux-based servers is complete.

After the upgrade, you can perform many other tasks, including:

- Install RFcache, if necessary.

  NOTE: From v3.0 and later, the commands and workflow for adding RFcache are different from earlier versions. For more information, see the *Configure and Customize Dell PowerFlex* and the *Dell PowerFlex CLI Reference Guide*.

- Extend the MDM cluster from 3-node to 5-node mode
- If not already configured, add virtual IP addresses to the data network
- Take advantage of the new features added to this release

These options, and others, are described in the *Configure and Customize Dell PowerFlex*.

3

Automated Upgrade of ESXi-based Nodes

Use the PowerFlex automated management tools to upgrade PowerFlex components on ESXi-based nodes.

Upgrading ESXi-based configurations

This section describes how to upgrade when all PowerFlex components reside on ESXi-based nodes.

The upgrade uses the PowerFlex Installer (part of the PowerFlex Gateway), together with the LIA (on an SVM), to orchestrate the upgrade.

The MDM cluster can be extended from 3-node mode to 5-node mode after the system upgrade. For more information, see "Extend an existing PowerFlex system" in the *Configure and Customize Dell PowerFlex*.

Download installation packages

Download and extract the complete software package for this version.

About this task

Steps

1. From the Online Support site (Dell Support for ScaleIO Drivers), download the complete software for this version.
2. Extract the packages from the downloaded file.
3. From the `PowerFlex <version> Complete VMware SW` directory, save the following software artifacts. These artifacts must be saved on the server that will be used to register the vSphere PowerFlex plug-in and to upload the OVA template.

   NOTE: PowerCLI from VMware must also be installed on this server.

   a. PowerFlex <version> ESX
   b. PowerFlex <version> vSphere plug-in

      The PowerFlex plug-in will be used to deploy PowerFlex on the servers in the system.

      NOTE: The SDC package for ESXi 7 cannot be installed in an automated process by the PowerFlex plug-in.

4. From the `PowerFlex <version> Windows` directory, save the software artifacts for the PowerFlex GUI.

   The GUI will be installed and used after deployment to configure some of the system features.

Preparing to upgrade ESXi configurations

Steps to perform before upgrading PowerFlex on ESXi-based nodes.

Steps

1. Download the PowerFlex Gateway and PowerFlex GUI packages.
2. When upgrading a node with ESXi 6.0U2:
   a. Ensure that the following line does not appear in the `etc/ssh/sshd_config` file of the ESXi-based node:

      `kexalgorithms diffie-hellman-group1-sha1,diffie-hellman-group-exchange-sha1`

      If the referred line appears, delete it. The line deletion does not relate to any configuration and you do not need to restart either the ESXi node, nor the sshd service.

b. Repeat the procedure for the rest of the nodes with ESXi 6.0U2.

3. The upgrade process requires entering the ESXi-based nodes into maintenance mode. To do so, ensure that Admission Control is not enabled on the VMware cluster:

Client	Procedure
vSphere Web Client	a. From the vSphere Web Client, log in to the vCenter. b. Select the cluster that contains the PowerFlex nodes. c. Select the **Configure** tab. d. Select **Services** > **vSphere Availability**. e. At the bottom of the window, expand **Admission Control**, and ensure that it is **Disabled**.
vSphere Client	a. From the vSphere client, log in to the vCenter. b. Right-click a cluster that contains PowerFlex nodes and select **Edit Settings**. c. Click the **vSphere HA** menu, and ensure that **Admission Control** is set to **Disable**.

4. For data networks using IPv6, if you plan to implement a floating virtual IP address for the MDM, disable the IPv6 DAD setting on the node's relevant interfaces, using the command:

```
sysctl net.ipv6.conf.<interface_name>.dad_transmits=0
```

5. If you intend to use the CHAP component authentication feature after the upgrade, make the necessary preparations to ensure that you upgrade all system components now. Systems containing any components running PowerFlex versions lower than v3.5 do not support CHAP authentication.

   NOTE: Upgrade to v3.5 sets the CHAP authentication feature to DISABLED, and passwords will be all-zero. To enable the feature, after the upgrade is complete on all system components, follow the post-installation procedures described in the *Configure and Customize Dell PowerFlex*.

6. Increase SVM system disk size of all SVMs in the system to at least 16 GB. For details, see the "Reference" section of this guide.

Next steps

If there are Stateless ESXi nodes (SDCs that are not managed by the PowerFlex vSphere plug-in), you must add their IP addresses to the PowerFlex Gateway IP ignore list before upgrading. For more information, see "Upgrade the PowerFlex Gateway with Stateless ESXi IP addresses."

Continue the upgrade according to your operating system environment.

Non-default certificate use before and after a PowerFlex Gateway upgrade

When using a non-default security certificate, you must perform certain actions before and after upgrading the PowerFlex Gateway.

About this task

If a non-default security certificate is used on the PowerFlex Gateway in Windows and Linux environments (for example, if the certificate is signed by the user organization CA), you must save a copy of the certificate and the catalina file before the upgrade, and restore them to their original location after the upgrade.

Steps

1. Before commencing the PowerFlex Gateway upgrade, locate the `*.keystore` file and the `catalina.properties` file, and save a copy of them in another location.

   The default file locations, per operating system, are:

   - Linux:
     - `/opt/emc/scaleio/gateway/conf/catalina.properties`
     - `/opt/emc/scaleio/gateway/conf/certificates/.keystore`
   - Windows (64 bit):
     - `C:\Program Files\EMC\ScaleIO\Gateway\conf\catalina.properties`

- C:\Program Files\EMC\ScaleIO\Gateway\conf\certificates\.keystore

2. After the upgrade is complete, copy these files back to their original location.

Upgrade the PowerFlex Gateway with Stateless ESXi IP addresses

When upgrading a PowerFlex system with Stateless ESXI nodes, you must add the node IP addresses to the PowerFlex Gateway IP ignore list before upgrading.

About this task

Failure to perform this step will prevent the upgrade from starting.

Steps

1. Edit `gatewayUser.properties`, located in `/opt/emc/scaleio/gateway/webapps/ROOT/WEB-INF/classes/`
2. In the `im.ip.ignore.list=` section, add all the IP addresses of the Stateless nodes, comma-separated.
3. Save the file.
4. Restart the PowerFlex Gateway service:

```
service scaleio-gateway restart
```

Results

The IP addresses are added to the PowerFlex Gateway IP ignore list. The system can now be upgraded.

Upgrading PowerFlex—ESXi-based nodes

This section describes how to upgrade PowerFlex to the latest version. This procedure can be used to upgrade from MDM 3-node or 5-node cluster mode.

Unregister the current PowerFlex plug-in

Prerequisites

PowerCLI is required for unregistering the plug-in. Administrator privileges are required.

About this task

In this procedure, the currently installed vSphere PowerFlex plug-in will be removed in preparation for installing the latest version.

Steps

1. From the folder where you extracted the `EMC-ScaleIO-vSphere-plugin-installer-3.6-X.<build>.zip` file, use PowerCLI, as administrator, to run the `VxFlexOSPluginSetup.ps1` script.
2. Enter the vCenter credentials and confirm the script actions. If you are asked to trust an untrusted publisher, type **A** to continue.
3. When prompted to choose a mode, select option **2**, `Unregister ScaleIO plug-in`.
4. Follow the progress on the screen, and when prompted to do so, log out, then log back in to the vSphere web client.
5. Verify that the plug-in is no longer registered (VxFlex OS icon should no longer be visible).

   NOTE: If the icon has not disappeared, even though PowerCLI reports that it was successfully unregistered, reboot the vCenter.

6. When the process is complete, enter **4** to exit the plug-in script.

Register the new PowerFlex plug-in

Register the new vSphere PowerFlex plug-in.

Prerequisites

Before you begin, ensure that there is communication between the vSphere web client server (usually installed on the vCenter) and the web server storing the PowerFlex plug-in.

About this task

Register the new PowerFlex plug-in.

To use your own web server, see Register the PowerFlex plug-in manually.

Steps

1. Copy the following files to a designated host (preferably a vCenter):
   - `ScaleIOVM_3nics_3.6.XXX.XXX.ova`
   - `EMC-ScaleIO-vSphere-plugin-installer-3.6-X.<build>.zip`
2. Extract the contents of the zip file.
3. Using PowerCLI for VMware, set to Run as administrator, run the following script: `VxFlexOSPluginSetup.ps1`.
   a. Enter the vCenter name or IP address, user name, and password.
   b. For `Choose Mode`, select option **1, Register PowerFlex plugin**.
   c. Read the upgrade notice, and enter **`y`** to continue.
   d. For `Select Registration Mode`, select **`Standard`** (simplified, using embedded Tomcat).
      This step may take a few minutes.
   e. If necessary, accept the thumbprint.
4. Log in to the vSphere web client.
   a. If you are already logged in, log out, then log in again.
   b. Verify that the PowerFlex icon appears in the web client.
   c. Open the PowerFlex plug-in in the client, and ensure that it is the correct version.
      If the version did not update, clear your browser cookies, log out, then log in to the PowerFlex plug-in again.
5. In the PowerCLI window, press **Enter** to finish the PowerFlex plug-in download and return to the menu.

Upload the OVA template

Use PowerCLI for VMware to upload the OVA template to a vCenter.

Prerequisites

- Ensure that there is communication between the vSphere web client server (usually installed on the vCenter) and the host storing the vSphere PowerFlex plug-in file.
- Ensure that the server with PowerCLI for VMware can run the PowerFlex plug-in setup script.
- Obtain the vCenter credentials.
- Ensure that you have the required software packages that were downloaded from the Customer Support site.

Steps

1. Copy the following file to your designated host (preferably your vCenter) which has PowerCLI for VMware installed:
   `ScaleIOVM_2nics_3.6.X.ova`
2. Using PowerCLI, set to `Run as administrator`, run the following script:

```
PowerFlexPluginSetup-3.6-<build>.X.ps1
```

   a. Enter the vCenter name or IP address, user name, and password.
   b. From the PowerCLI script, select **3** to `Create SVM template`.

c. Enter the parameters described in the following table. (If you are already logged in, some of these parameters may not be necessary.)

Table 5. vCenter configuration parameters

Parameter	Description
vcenter	vCenter name or IP address
user name	vCenter user name
password	vCenter password
datacenter	The name of the data center where the datastore that will store the template resides
ova_path	The path of the SVM's OVA
datastores	A list of datastores, up to eight, on which the templates will be created. Press ENTER to stop specifying datastores. **NOTE:** Each template requires approximately 8 GB of space in the datastore.

NOTE: For best results, enter a local (not shared) datastore for each ESXi server.

For faster, parallel, deployment in large-scale environments, you can use the OVA to create SVM templates on as many as eight datastores. To do so, type the datastore names, and when you are done, leave the next line blank. The following example shows how to enter two datastores:

```
datastores[0]: datastore1
datastores[1]: datastore1 (1)
datastores[2]:
```

The upload procedure can take several minutes, during which time a temporary SVM is created, the templates are created, and then the temporary SVM is deleted.

When each template is created, a message similar to the following appears:

```
The template VxFlexOS SVM Template (v3.6.X) was successfully created
```

d. When the process is complete, type **4** to exit the PowerFlex plug-in script.

Add or replace an OVA file

If you have already used the OVA file to create a template, you cannot create another template with the same name in the same datacenter.

Either remove the original template first, or use the `VxFlexOSPluginSetup-3.6-<build>.X.ps1` script, option #3, to assign a different name to the new template.

You can also upload the OVA manually using VMware OVA upload tools, however this method is not recommended. Configure the networks manually after deployment, or during the wizard workflow. For more information, see VMware documentation.

Upgrade the PowerFlex Gateway

Upgrade the PowerFlex Gateway. This process must be performed before upgrading the other PowerFlex system components.

Prerequisites

Ensure the following:

- You have the installation package for the new version of the PowerFlex Gateway to which you are upgrading.
- You have the installation packages for RHEL7 for the current PowerFlex version installed in your system (for example, for PowerFlex v3.0.1, get PowerFlex v3.0.1 packages for RHEL7).

- If the installation packages for the currently installed version of PowerFlex are not loaded in your PowerFlex Installer, get them (for example, packages for PowerFlex v3.0.1 on SLES 15). You will need to upload them to the PowerFlex Installer during this procedure.
- You have credentials for an admin user for the PowerFlex Installer.
- You have credentials for the cluster's vCenter: IP address or domain name, username, and password.
- You have the LDAP connection details and credentials for the PowerFlex Gateway.
- You have the MDM password, which is required for configuring the lockbox and LIA.
- You have the root user and password for all SVMs.
- There must be at least one PowerFlex node that does not have MDM or tiebreaker installed.

About this task

Limitations:

- DAS Cache writeback caching of HDDs is no longer supported in PowerFlex systems.

  Before beginning, gracefully stop the DAS Cache and restart the SDS nodes, one-at-a-time. This will stop the use of `/dev/fio-X` device names and revert to the underlying device names. For details, refer to *VxFlex Ready Node DAS Cache Maintenance Operations Technical Notes*.

Steps

1. Upload the new PowerFlex Gateway RPM installation package to the PowerFlex Gateway node.
2. On the PowerFlex Gateway, in command line, run the command:

   ```
   rpm -U EMC-ScaleIO-gateway-3.6-X.<build>.x86_64.rpm
   ```

   The gateway has been upgraded.

Next steps

- Proceed to "Upgrade PowerFlex components".

Upgrade PowerFlex components

Upgrade PowerFlex using the PowerFlex Gateway.

Prerequisites

Software packages are required for both current and target versions. The current version is required to facilitate rollback, should it be necessary.

About this task

Steps

1. From the plug-in **VxFlex OS Systems** screen, launch the PowerFlex Installer by clicking **Open PowerFlex Gateway**.
2. In the PowerFlex Installer, enter the default credentials.
3. Click **Packages**.

   You may need to re-authenticate with your login credentials. The **Manage Installation Packages** window appears.

   NOTE: To use this PowerFlex Installer to install packages for Windows and Linux, you can upload all the files at once.

4. Browse to the location where the PowerFlex packages for Linux operating systems are saved.
5. Select all the relevant files for the operating systems on which you want to upgrade. Both your current base version of PowerFlex, and the version to which you are upgrading are required.
   For example, for the current SVMs which are now based on CentOS 7.6, select the Linux RHEL 7.x files.

   File formats can be RPM, TAR, or MSI. Minimally, you must select these packages:

   a. MDM
   b. SDS

c. SDC
d. LIA

The **Browse** button changes its appearance to reflect the selected files.

Packages from the base version, for all components that are deployed: LIA, MDM, SDS, XCACHE (optional), SDR (optional), VASA Provider (optional, for VMware VASA provider for VVols), for example:

- `EMC-SCALEIO-LIA-3.0.x.<build>.<flavor>.x86_64.RPM`
- `EMC-SCALEIO-MDM-3.0.x.<build>.<flavor>.x86_64.RPM`

Packages from the new version: LIA, MDM, SDS, XCACHE (optional, for Medium Granularity storage acceleration), SDR (optional, for replication), VASA Provider (optional, for VMware VASA provider for VVols), for example:

- `EMC-SCALEIO-LIA-3.6-X.<build>.<flavor>.x86_64.RPM`
- `EMC-SCALEIO-MDM-3.6-X.<build>.<flavor>.x86_64.RPM`

6. When all files are selected, click **Open**.
7. Click **Upload**.

   The uploaded installation packages (RHEL, in this example) are listed in the file table.

8. Upgrade the PowerFlex components:
   a. From the PowerFlex Installer, click the **Maintain** tab.
   b. From the **Maintenance operation** screen, perform the following:
      i. Type the primary MDM's credentials.
      ii. Select the LIA authentication type (**Native** or **LDAP**), and if **Native** is selected, type the LIA password. If **LDAP** is selected, type the LDAP user name and password.
      iii. Click **Retrieve system topology**. A success message is displayed.
      iv. Click **Upgrade**.

      The **Device Media Type for VxFlex OS system** window is displayed.

   c. Select the device media type (mandatory) and external acceleration type (optional) for each Storage Pool in the system, and then click **Proceed**.

      For external acceleration, options include:
      - none—No devices are accelerated by a non-PowerFlex read or write cache
      - read—All devices are accelerated by a non-PowerFlex read cache
      - write—All devices are accelerated by a non-PowerFlex write cache
      - read_and_write—All devices are accelerated by both non-PowerFlex read and write cache

      This input is required in order to prevent the generation of false alerts for media type mismatches. For example, if an HDD device is added which the SDS perceives as being too fast to fit the HDD criteria, alerts might be generated. External acceleration/caching is explained in the *Getting to Know Dell PowerFlex Guide*.

   d. Set upgrade options:
      - To prevent upgrade of SDCs, select **Do not upgrade any SDCs**.

        If you choose this option, you can upgrade the SDCs later, using the vSphere PowerFlex plug-in. You might want to use this option if the SDCs are running critical applications that are currently in use.

      - To continue an upgrade that was interrupted on a different gateway, select **Allow upgrade even when already in Upgrade state**.

        Use this option when you started an upgrade using a different PowerFlex Gateway, but that PowerFlex Gateway failed or is offline now.

      - Set performance profiles.

        High performance profiles are mandatory when using Fine Granularity Storage Pools. For other configurations, they are optional. You can modify these profiles later.

      - Perform automatic restart of Windows-based nodes (required for Windows-based nodes on which SDCs are deployed).
      - To upgrade SDSs in the same Fault Set together, select **Enable parallel SDS upgrades**.

        Parallel upgrade of SDSs in the same Fault Set saves time; the upgrade will be performed faster.

   e. Confirm the upgrade operation and click **Upgrade**.

NOTE: If a certificate notice is displayed, review and approve it.

f. Click the **Monitor** tab, and approve each step of the upgrade process.
g. When complete, click **Mark operation completed**.

Next steps

If you are upgrading from PowerFlex v3.0.0.x to v3.0.1 or later, you must upgrade the SVMs' operating system next. Go to "Upgrade the CentOS operating system on SVMs."

Upgrade the CentOS operating system on SVMs

Use the PowerFlex Installer to upgrade the SVM's CentOS operating system to the latest security level of operating system security patching. This procedure only upgrades the operating system on the SVM. It does not upgrade PowerFlex.

NOTE: SVM operating system patching is applicable even if the SVM replacement procedure was performed. The latest patch bundle has the latest security patches that were released after the latest version of PowerFlex was officially launched (the refresh of the package is done on a monthly basis based on continuous security scans and input from customers).

Use the PowerFlex Installer to upgrade the CentOS operating system on multiple SVMs. This procedure is required when upgrading an ESXi-based PowerFlex system where SVMs are currently running on CentOS 7.5. (PowerFlex v3.0, or v3.0.0.x)

Ensure that you have the latest SVM operating system upgrade ZIP file. It is available from the support site. You can opt to download the latest file from Dell Support for ScaleIO Drivers.

The `patch_script` file that you require is part of the VMware complete software package (which is in the Complete SW download ZIP) and also as a separate artifact that is refreshed from time to time (usually once a month). The ZIP file name is: `SVM_OS_Patching_package_MMDDYYYY.zip`

Running scripts on hosts—overview

PowerFlex can be used to run user-provided scripts on servers hosting MDM or SDS components. This feature is supported on Linux-based (bare-metal or virtual) nodes only.

The PowerFlex Installer can be used to run a user-provided script on a node where PowerFlex is deployed. This feature can be used for any purpose external to the PowerFlex system, such as running a set of CLI commands, patching an operating system, and more. The feature allows the running of scripts in a safe manner, both from a security and a data integrity perspective.

PowerFlex Installer orchestrates the running of the script, ensuring that SDSs are placed in Maintenance Mode, to protect data during the process. In addition, parallel execution of scripts is only permitted on SDSs located in different Protection Domains. After the scripts have been run on an SDS, it exits Maintenance Mode.

Optionally, servers can be set to reboot after execution of the script. The process can also run a verification script either after the reboot, or after execution of the script, when no reboot is required.

Workflow overview (for reference purposes only)

The execution phase of this feature can be summarized as follows:

1. The system validates the following:
   - The patching script exists on the node after it was uploaded from the PowerFlex Gateway
   - No failed capacity exists
   - Sufficient spare capacity exists
   - The MDM cluster is in a valid state
2. Run the script on one host, using the following priorities:
   a. SDS only hosts, each time on a single SDS, unless the option `In parallel on different Protection Domains` is enabled
   b. Tiebreaker
   c. MDMs

   NOTE: The script will not run on a primary MDM. A switch-over MDM command will be run prior to running the script on a primary MDM. The script will not be run in parallel on multiple MDMs or tiebreakers.
3. SDS enters Maintenance Mode.
4. The script runs on the host.

5. The host reboots (if configured to do so).
6. The validation script runs on the node (if configured to do so).
7. SDS exits Maintenance Mode.

Run a script on one or more hosts—procedures

Run a script on one or multiple nodes where PowerFlex is deployed, using the PowerFlex Installer.

Prerequisites

- Prepare the script file to be run on one or more nodes hosting SDSs or MDMs, and optionally, a verification script to be run automatically after execution of the script. Specific file names are required for use of this feature. PowerFlex Installer can only upload scripts up to 500 MB in size. Name the files as follows:
  - Script: `patch_script`
  - Verification script: `verification_script`
- This feature is only supported on Linux-based (bare-metal or virtual) nodes.
- Ensure that you have the IP address and login credentials for the PowerFlex Gateway and the primary MDM.
- Ensure that you have the login credentials for the LIA.
- Ensure that you have enough disk space to run the script.

About this task

- This procedure is a long operation that cannot be rolled back automatically.
- Certificate approval messages may be displayed during the procedure. Approve all the certificates, and click **Retry failed** to complete the operation.

Steps

1. Configure the PowerFlex Installer to upload files from the PowerFlex Gateway node to all selected nodes.

   NOTE: Alternatively you can copy the script file to each node in the system under the file path `/opt/emc/scaleio/lia/bin/<folder>`. Follow the steps to make sure the script file runs on each of the nodes.

   a. Verify that all components in the PowerFlex system are upgraded to the latest version.
   b. On the PowerFlex Gateway node, go to `/opt/emc/scaleio/gateway/webapps/ROOT/WEB-INF/classes/gatewayUser.properties` file.
   c. Uncomment the following lines (verification script is optional):

   ```
   os.patching.is.upload.needed=true (Default is false)
   os.patching.patch.script.source.path=<enter the full path to the file>
   os.patching.verification.script.source.path=<enter the full path to the file>
   ```

   Example of PowerFlex Gateway on Linux:

   ```
   os.patching.is.upload.needed=true
   os.patching.patch.script.source.path=/opt/patch_script
   os.patching.verification.script.source.path=/opt/verification_script
   ```

   Example of PowerFlex Gateway on Windows:

   ```
   os.patching.is.upload.needed=true
   os.patching.patch.script.source.path=C:\\temp\\patch_script
   os.patching.verification.script.source.path=C:\\temp\\verification_patch
   ```

   d. Restart the PowerFlex Gateway service.

   For example, in ESXi:

   `systemctl stop scaleio-gateway.service` and `systemctl start scaleio-gateway.service`

   e. Copy the `patch_script` and `verification_script` files to the relevant folder on the PowerFlex Gateway node.
   f. In the Security settings of the files, allow read, write and execute permissions.
2. In your browser, navigate to the IP address of the PowerFlex Gateway, and log in.
3. Click the **Maintain** tab.

4. Enter the IP address and login credentials for the primary MDM, and for LIA.
5. At the bottom right of the screen, click **Retrieve system topology**.
   The system topology is displayed.
6. Click **System Logs & Analysis**, and select the **Run Script on Hosts** option.
   The **Run script on hosts** dialog box is displayed.
7. Enter the MDM password again.
8. For **Running script on** options, select one from the following:

Option	Description
Entire System	Run the script on all MDM and SDS nodes in the system. If you choose this option, you can also choose whether to run the script at the same time on SDSs that belong to different Protection Domains. To do so, select the check box for **In parallel on different Protection Domains**.
Protection Domain	Run the script on MDM and SDS nodes in a single Protection Domain. Select the required Protection Domain from the drop-down list.
Fault Set	Run the script on MDM and SDS nodes in a single Fault Set. Select the required Fault Set from the drop-down list.
SDS	Run the script on a single SDS. Select the required SDS from the drop-down list.

9. For **Running configuration**, select the **Stop process on script failure** option, if desired.
   If problems occur, see the troubleshooting notes at the end of this procedure.
10. In the **Script time-out** box, enter the number of minutes that should elapse before the PowerFlex Installer stops waiting for a response about the running script, and prints a timeout message.
11. In the **Verification script** box, select one of the following:

Option	Action
Run a verification script after the script. If a reboot is performed, the verification script is executed after the reboot.	Select `Run`
Do not run a verification script after the script	Select `Do not run`

12. In the **Post script action** box, select one of the following:

Option	Action
Reboot the node after execution of the script	Select `Reboot`
Do not reboot the node after execution of the script	Select `Do not reboot`

13. Click **Run Script on Hosts**.
14. Click the **Monitor** tab, - validate state starts.
    During this phase, the system verifies certificates and builds an execution phase for the nodes list.
15. Click **Start** execute phase - the execute phase starts.
16. When the execute phase is complete, click **Mark operation completed**.

    Troubleshooting notes:

    If a failure in executing the patch_script occurs, it may be due to one of the following reasons. Rectify the problem and then try again.

    - Script timeout
    - Script file permissions
    - Syntax error in the script
    - Degraded capacity, or MDM cluster in degraded state
    - An SDS is already in Maintenance Mode

Results

Upgrade of the CentOS operating system on all SVMs except for the PowerFlex Gateway is now complete. Upgrade the PowerFlex Gateway using the steps described in "Deploy and replace the PowerFlex Gateway SVM operating system using the PowerFlex plug-in" in the *Upgrade Dell PowerFlex*.

Troubleshoot upload phase when running scripts on hosts

Overcome the PowerFlex Gateway not being able to upload files using the "Run script on host" feature.

About this task

If the "Upload" phase fails with a timeout error, check if the following appears in the scaleio.log file on the PowerFlex Gateway, and the LIA log file of the relevant node. If the error appears, follow the remediation steps listed below. This issue is caused by a lack of resources in the PowerFlex Gateway's operating system, and in the JAVA process sustaining the upload.

Typical scaleio.log file:

```
2020-07-30 11:35:28,333 [executor-23] INFO  c.e.s.s.d.i.c.UploadFileCommand - activating
serial upload force algorithm due to upload of package UploadFileCommand[9.20.25.23,
9.20.125.23] that has failed due to memory problem java.lang.OutOfMemoryError: Java heap
space
2020-07-30 11:35:28,333 [executor-23] ERROR c.e.s.s.d.i.c.UploadFileCommand - UNEXPECTED
Error java.lang.OutOfMemoryError: Java heap space executing command .UploadFileCommand
(abort) :
```

Typical LIA log file:

```
30/07 11:51:19.035533 7f9f98115db0:liaCmd_CallCmdWithTimer:00253: Executing...
                 Command: chmod 744 /opt/emc/scaleio/lia/bin/patch_script
                 Timeout: [0]Sec (if zero - timeout is default)
30/07 11:51:19.040428 7f9f98115db0:liaNet_RecvRequest:00427: UPLOAD PACKAGE END
30/07 12:19:55.027051 7f9f98198db0:netSec_Recv_CK:00847: ERROR: Failed to read data from
handle 0x7f9f981a3ce0 (ERROR)
30/07 12:19:55.027066 7f9f98198db0:netPath_RcvLoop_CK:01611: WARNING:  :: Connected Live
SERVER path 0x7f9f981dd950 of portal 0x7f9f981ff170 net 0xa3a3c0 socket 11 inflights 0
HS:0 socket receive failed, Status ERROR
30/07 12:19:55.027076 7f9f98198db0:netCon_AbortInflightRpcsUptoDummyNode:05869: Aborted
0 inflight RPC messages due to disconnect.
```

Steps

1. Check if the PowerFlex Gateway's operating system has sufficient memory: minimum 8 GiB, recommended 12 GiB.
   - In case of SVM this requires shutdown of the SVM to modify the memory
   - This can be done at any time as long as no gateway orchestration is running
   - If "Run Script on Host" is running abort the operation prior to shutdown of the OS
2. After adding memory we need to increase the Java settings:
   a. Using file editor (such as vi) open the file: /opt/emc/scaleio/gateway/bin/setenv.sh and modify the following line:
      From:

      ```
      CATALINA_OPTS="-Xms128m -Xmx2048m"
      ```

      To:

      ```
      CATALINA_OPTS="-Xms128m -Xmx4096m"
      ```

   b. Restart the gateway service:
      Linux:

      ```
      systemctl scaleio-gateway restart
      ```

      Windows: Restart the service from the **services.msc** window.
3. Log in to the PowerFlex Gateway, and run the "Run Script on Host" feature with upload capability again.

Upgrade the PowerFlex Gateway operating system

Upgrade the PowerFlex Gateway CentOS operating system with the latest security patches.

About this task

Upgrade the CentOS operating system on the SVM hosing the PowerFlex Gateway.

Steps

1. Copy the `patch_script` file to the SVM hosting the PowerFlex Gateway.
2. In the Security settings of the file, allow read, write, and execute permissions.
3. Run the file, using the command:

   ```
   ./patch_script
   ```

   The script will run on the SVM, and will display a message when the operation is complete.
4. Restart the SVM.

Upgrade and restart the SDCs

About this task

This process involves restarting the ESXi-based node where SDC is deployed. The restart may cause I/O failures. It is recommended to migrate VMs before starting this procedure.

Steps

1. From the VMware plug-in main screen, under **PowerFlex Systems**, click **SDCs**.
2. On the **SDCs** screen, right-click an SDC and select **Upgrade SDC**.
3. In the **Upgrade SDC** dialog box, perform the following for each ESXi:
   a. Enter the ESXi password and click **OK**.
   b. When the upgrade is complete, click **Finish**, then **Close**.

   NOTE: If the node has an SVM, place the SDS in maintenance mode.

   In addition, if you have not migrated VMs manually, it is recommended to do so manually or to use ESXi maintenance mode. If you want to use ESXi maintenance mode, power off the Storage VM (SVM) prior to entering ESXi maintenance mode.
4. From the vCenter, reboot each ESXi node, one at a time.

   NOTE: If the ESXi node was placed in maintenance mode, exit ESXi maintenance mode and power on the SVM.

   After each ESXi reboot, take the SDS out of maintenance mode and wait for the rebuild and rebalance operation to finish prior to upgrading the next SDC node.

   NOTE: It is possible to follow SDCs' upgrade process through the main PowerFlex plug-in window. Under **Advanced tasks**, click **Show Upgrade Process**. Check the status of the SDC upgrade, and if successful, for each node, click **Finish**, and then **Close**.

Upgrade the PowerFlex presentation server (PowerFlex GUI) from v3.5 or later

Upgrade the PowerFlex presentation server from v3.5 or later.

Prerequisites

Ensure that you have the package for the PowerFlex presentation server, and access to the node hosting the PowerFlex presentation server.

Steps

1. Save a copy of the PowerFlex presentation server package on the node where PowerFlex presentation server is installed.
2. On the node running the PowerFlex presentation server, run the following command:
   - Linux:

     ```
     rpm -Uvh EMC-ScaleIO-mgmt_server-3.6-X.<build>.noarch.rpm
     ```

   - Debian:

     ```
     dpkg -i emc-scaleio-mgmt_server-3.6-X.<build>.amd64.deb
     ```

3. On a Centos 8-based nodes only, when the installation script has finished, type Q.

Results

Upgrade of the PowerFlex presentation server is complete. Link the PowerFlex presentation server to the MDM cluster, as described in the *Configure and Customize Dell PowerFlex*. Existing credentials can be used, provided they belong to either admin or configure + monitor (LDAP or native) users.

Upgrade the PowerFlex GUI from versions earlier than v3.5

Use these steps to upgrade the PowerFlex GUI from versions earlier than v3.5.

Prerequisites

NOTE: From PowerFlex version 3.5, the PowerFlex GUI is a web-based application installed on a server. Therefore, the new PowerFlex GUI is a fresh installation known as the PowerFlex presentation server, and the GUI client from VxFlex OS versions of the product is neither upgraded nor supported going forward.

- Ensure that the workstation complies with the PowerFlex system requirements.
- Ensure that you have the PowerFlex presentation server installation file. It is included in the PowerFlex complete software package that you downloaded from Dell Technologies support site.

NOTE: It is not recommended to install the PowerFlex presentation server on the same server as the PowerFlex Gateway. If you must install them on the same server, the PowerFlex Gateway must be installed first. The PowerFlex presentation server listens on port 8443, and the PowerFlex Gateway installer checks that port 8443 is available, although it does not use this port.

Steps

1. Install the PowerFlex presentation server on a server in your system:
   - Linux:

     ```
     rpm -i EMC-ScaleIO-mgmt-server-3.6-X.<build>.noarch.rpm
     ```

- Debian (run with administrator privileges):

```
sudo dpkg -i emc-scaleio-mgmt-server_3.6-X.<build>_amd64.deb
```

2. Prior to using the PowerFlex GUI, link the PowerFlex presentation server to the PowerFlex system. For more information, about linking the system, login and use of the PowerFlex GUI, refer to the *Configure and Customize Dell PowerFlex*.

After upgrading an ESXi-based system

Tasks that can be performed after upgrading the system.

After the upgrade, you can perform many other tasks, including:

- Add virtual IP addresses, which are mandatory for PowerFlex plug-in setups from PowerFlex v2.0 and later. For more information, see *Configure and Customize Dell PowerFlex*.
- Migrate an RDM-based PowerFlex system to DirectPath-based. This option is described in the "Manual migration of an RDM-based PowerFlex system to DirectPath-based" section of this guide.
- Install RFcache, if necessary.

  NOTE: From v3.0 and later, the commands and workflow for adding RFcache are different from earlier versions. For more information, see the *Configure and Customize PowerFlex Guide* and the *PowerFlex CLI Reference Guide*.

- Take advantage of the new features added to this release.

  These options, and others, are described in the *Configure and Customize PowerFlex Guide*.

4

Automated Upgrade of Nodes in a Two-Layer Configuration

Use the PowerFlex automated management tools to upgrade PowerFlex components in a two-layer configuration.

Preparing to upgrade two-layer configuration—Linux and ESX-based

Steps to perform before upgrading PowerFlex components in a two-layer configuration, where SDCs are deployed on ESXi-based servers, and the other components are deployed on Linux-based bare metal servers.

About this task

Before upgrading the PowerFlex components, perform the following steps:

Steps

1. Install one of the following on the PowerFlex Gateway and PowerFlex presentation server nodes, as described in detail in the system requirements in the *Getting to Know Dell PowerFlex Guide*:

Option	Example

Oracle JRE 1.8 build 301	• Linux:
	`rpm -i jre-8uXX-linux-x64.rpm`
	• Windows (for the PowerFlex Gateway only): Double-click the file:
	`jre-8uXX-windows-x64.exe`

OpenJDK JDK 8 build 302.b08 or OpenJDK 11.0.12.0.7 **NOTE:** JAVA 8 is not supported on PowerFlex v3.6.3 and later versions. For more information, see *Dell PowerFlex 3.6.2 Release Notes*.	Refer to the following information: • Linux: Oracle Java SE Archive • Windows (for the PowerFlex Gateway only): Downloads for Windows x86_64

2. Download and extract the PowerFlex packages.

   You need the packages for your base version and for the latest version (the version to which you are upgrading).

3. For data networks using IPv6, if you plan to implement a floating virtual IP address for the MDM, disable the IPv6 DAD setting on the node's relevant interfaces, using the command:

   ```
   sysctl net.ipv6.conf.<interface_name>.dad_transmits=0
   ```

4. If you intend to use the CHAP component authentication feature after the upgrade, make the necessary preparations to ensure that you upgrade all system components now. Systems containing any components running PowerFlex versions lower than v3.5 do not support CHAP authentication.

   NOTE: Upgrade to v3.5 sets the CHAP authentication feature to DISABLED, and passwords will be all-zero. To enable the feature, after the upgrade is complete on all system components, follow the post-installation procedures described in the *Configure and Customize Dell PowerFlex*.

Non-default certificate use before and after a PowerFlex Gateway upgrade

When using a non-default security certificate, you must perform certain actions before and after upgrading the PowerFlex Gateway.

About this task

If a non-default security certificate is used on the PowerFlex Gateway in Windows and Linux environments (for example, if the certificate is signed by the user organization CA), you must save a copy of the certificate and the catalina file before the upgrade, and restore them to their original location after the upgrade.

Steps

1. Before commencing the PowerFlex Gateway upgrade, locate the `*.keystore` file and the `catalina.properties` file, and save a copy of them in another location.

   The default file locations, per operating system, are:

   - Linux:
     - `/opt/emc/scaleio/gateway/conf/catalina.properties`
     - `/opt/emc/scaleio/gateway/conf/certificates/.keystore`
   - Windows (64 bit):
     - `C:\Program Files\EMC\ScaleIO\Gateway\conf\catalina.properties`
     - `C:\Program Files\EMC\ScaleIO\Gateway\conf\certificates\.keystore`
2. After the upgrade is complete, copy these files back to their original location.

Upgrading two-layer configurations (Linux-based and ESX-based)

Upgrade a system when the MDM, SDS, and SDR (optional) components are deployed on bare metal servers, and the SDCs reside on ESXi-based nodes.

This type of upgrade uses the PowerFlex Installer (part of the PowerFlex Gateway), together with the LIA of the new version, to orchestrate the upgrade.

The ESXi-based upgrade process uses the vSphere PowerFlex plug-in.

NOTE: To verify if LIA is installed, or to add LIA to a system where it is not installed, see "Add LIA to a system to enable automated upgrade" in the *Configure and Customize Dell PowerFlex*.

Upgrading is non-disruptive; you can upgrade while I/Os are running, and volumes are mapped.

The MDM cluster can be extended from 3-node mode to 5-node mode after upgrading. For more information, see the *Configure and Customize Dell PowerFlex*.

Upgrading a PowerFlex two-layer system

Upgrade a two-layer PowerFlex system to the latest version.

About this task

This procedure can be used for upgrading a 3-node or 5-node MDM cluster that are deployed on all Linux flavor-based nodes.

Steps

1. Edit the following file:
   `/opt/emc/scaleio/gateway/webapps/ROOT/WEB-INF/classes/gatewayUser.properties`
2. Add the complete list of SDC IP addresses to `im.ip.ignore.list`.

3. Restart the PowerFlex Gateway:

```
/etc/init.d/scaleio-gateway restart
```

4. Upgrade the Linux-based PowerFlex Gateway:

```
rpm -U EMC-ScaleIO-gateway-3.6-X.<build>.x86_64.rpm
```

The following variables can be added to the command, if required:

- *SIO_GW_KEYTOOL=<keytool_path>* allows override of the location for the Java keytool utility. For example: `SIO_GW_KEYTOOL=/usr/java/jre1.8.0_XX/bin/`
- *SIO_JAVA=<Java version>* specifies a different Java version

Example:

```
rpm -U /tmp/EMC-ScaleIO-gateway-3.6-X.<build>.x86_64.rpm
```

A success message is displayed, and the PowerFlex Installer has been updated to the new version. Log in to the updated PowerFlex Installer will be required.

5. Log in to the PowerFlex Installer (In a web browser, go to https://<PowerFlex Gateway_IP_ADDRESS>).
6. Click **Packages**.

You may need to re-authenticate with your login credentials. The **Manage Installation Packages** window appears.

NOTE: When using the PowerFlex Installer to install packages for Windows and Linux, you can upload all the files at once.

7. Browse to the package locations for the operating systems of the nodes that you want to upgrade.
8. Select all the relevant files for the operating systems of the nodes which you want to upgrade. Both your current version and the target version to which you are upgrading are required.

File formats can be RPM, TAR, or MSI. Minimally, you must select these packages:

a. MDM
b. SDS
c. SDC
d. LIA

The **Browse** button changes its appearance to reflect the selected files.

Packages from the current version: lia, mdm, sds, sdc, xcache (optional), sdr (optional). For example:

- `EMC-ScaleIO-lia-3.0.x-XXX.<build>.<flavor>.x86_64.rpm`
- `EMC-ScaleIO-mdm-3.0.x-XXX.<build>.<flavor>.x86_64.rpm`

Packages from target version: lia, mdm, sds, sdc, xcache (optional), sdr (optional). For example:

- `EMC-ScaleIO-lia-3.6-X.<build>.<flavor>.x86_64.rpm`
- `EMC-ScaleIO-mdm-3.6-X.<build>.<flavor>.x86_64.rpm`

9. When all files are selected, click **Open**.
10. Click **Upload**.

The uploaded installation packages (RHEL, in this example) are listed in the file table.

11. Upgrade the PowerFlex components:
    a. From the PowerFlex Installer, click the **Maintain** tab.
    b. From the **Maintenance operation** screen, perform the following:
        i. Type the primary MDM's credentials.
        ii. Type the LIA password (upgrades from versions prior to v3.0 only allow native authentication for LIA).
        iii. Click **Retrieve system topology**. A success message is displayed.
        iv. Click **Upgrade**.

        The **Device Media Type for PowerFlex system** window is displayed.

    c. Select the device media type (mandatory) and external acceleration type (optional) for each Storage Pool in the system, and then click **Proceed**.

For external acceleration, options include:

- none—No devices are accelerated by a non-PowerFlex read or write cache
- read—All devices are accelerated by a non-PowerFlex read cache
- write—All devices are accelerated by a non-PowerFlex write cache
- read_and_write—All devices are accelerated by both non-PowerFlex read and write cache

This input is required in order to prevent the generation of false alerts for media type mismatches. For example, if an HDD device is added which the SDS perceives as being too fast to fit the HDD criteria, alerts might be generated. External acceleration/caching is explained in the *Getting to Know Dell PowerFlex Guide*.

d. Set upgrade options:
   - To prevent upgrade of SDCs, select **Do not upgrade any SDCs**. You might want to use this option if the SDCs are running critical applications that are currently in use. You can upgrade the SDCs later.
   - To continue an upgrade that was interrupted on a different gateway, select **Allow upgrade even when already in Upgrade state**.

     Use this option when you started an upgrade using a different gateway, but that gateway failed or is offline now.
   - Set performance profiles.

     High performance profiles are mandatory when using Fine Granularity Storage Pools. For other configurations, they are optional. You can modify these profiles later.
   - Perform automatic restart of Windows-based servers (required for Windows-based nodes on which SDCs are deployed).
   - To upgrade SDSs in the same Fault Set together, select **Allow parallel SDS upgrades**.

     Parallel upgrade of SDSs in the same Fault Set saves time; the upgrade will be performed faster.

e. Confirm the upgrade operation by typing the MDM admin password, and click **Upgrade**.

   NOTE: If a certificate notice is displayed, review and approve it.

f. Click the **Monitor** tab, and approve each step of the upgrade process.
g. When the upgrade is complete, click **Mark operation completed**.

12. The following steps involve restarting the SDC node, which might cause I/O failures. It is recommended to migrate VMs before starting.
13. Remove the old plug-in:
    a. From the folder where you extracted the `EMC-ScaleIO-vSphere-plugin-installer-3.6-X.<build>.X.zip` file, use PowerCLI to run the `VxFlexOSPluginSetup.ps1 script`.
    b. Select option **2, Unregister ScaleIO plugin**.
    c. Enter the vCenter credentials and confirm the script actions.
    d. Log out, then log back in to the vSphere web client.

    The plug-in is no longer registered.
14. Register the new plug-in, as described in the "Register the new plug-in" topic.
15. Upgrade SDCs on the ESXi-based nodes, as described in the "Upgrade and restart the SDCs" topic.

    This step requires restarting the SDC node, which may cause I/O failures. It is recommended to migrate VMs before starting.

Register the new PowerFlex plug-in

Register the new vSphere PowerFlex plug-in.

Prerequisites

Before you begin, ensure that there is communication between the vSphere web client server (usually installed on the vCenter) and the web server storing the PowerFlex plug-in.

About this task

Register the new PowerFlex plug-in.

To use your own web server, see Register the PowerFlex plug-in manually.

Steps

1. Copy the following files to a designated host (preferably a vCenter):
   - `ScaleIOVM_3nics_3.6.XXX.XXX.ova`
   - `EMC-ScaleIO-vSphere-plugin-installer-3.6-X.<build>.zip`
2. Extract the contents of the zip file.
3. Using PowerCLI for VMware, set to Run as administrator, run the following script: `VxFlexOSPluginSetup.ps1`.
   a. Enter the vCenter name or IP address, user name, and password.
   b. For `Choose Mode`, select option **1, Register PowerFlex plugin**.
   c. Read the upgrade notice, and enter **`y`** to continue.
   d. For `Select Registration Mode`, select **`Standard`** (simplified, using embedded Tomcat).
      This step may take a few minutes.
   e. If necessary, accept the thumbprint.
4. Log in to the vSphere web client.
   a. If you are already logged in, log out, then log in again.
   b. Verify that the PowerFlex icon appears in the web client.
   c. Open the PowerFlex plug-in in the client, and ensure that it is the correct version.
      If the version did not update, clear your browser cookies, log out, then log in to the PowerFlex plug-in again.
5. In the PowerCLI window, press **Enter** to finish the PowerFlex plug-in download and return to the menu.

Upgrade and restart the SDCs

About this task

This process involves restarting the ESXi-based node where SDC is deployed. The restart may cause I/O failures. It is recommended to migrate VMs before starting this procedure.

Steps

1. From the VMware plug-in main screen, under **PowerFlex Systems**, click **SDCs**.
2. On the **SDCs** screen, right-click an SDC and select **Upgrade SDC**.
3. In the **Upgrade SDC** dialog box, perform the following for each ESXi:
   a. Enter the ESXi password and click **OK**.
   b. When the upgrade is complete, click **Finish**, then **Close**.

   NOTE: If the node has an SVM, place the SDS in maintenance mode.

   In addition, if you have not migrated VMs manually, it is recommended to do so manually or to use ESXi maintenance mode. If you want to use ESXi maintenance mode, power off the Storage VM (SVM) prior to entering ESXi maintenance mode.

4. From the vCenter, reboot each ESXi node, one at a time.

   NOTE: If the ESXi node was placed in maintenance mode, exit ESXi maintenance mode and power on the SVM.

   After each ESXi reboot, take the SDS out of maintenance mode and wait for the rebuild and rebalance operation to finish prior to upgrading the next SDC node.

   NOTE: It is possible to follow SDCs' upgrade process through the main PowerFlex plug-in window. Under **Advanced tasks**, click **Show Upgrade Process**. Check the status of the SDC upgrade, and if successful, for each node, click **Finish**, and then **Close**.

Upgrade the PowerFlex GUI from versions earlier than v3.5

Use these steps to upgrade the PowerFlex GUI from versions earlier than v3.5.

Prerequisites

NOTE: From PowerFlex version 3.5, the PowerFlex GUI is a web-based application installed on a server. Therefore, the new PowerFlex GUI is a fresh installation known as the PowerFlex presentation server, and the GUI client from VxFlex OS versions of the product is neither upgraded nor supported going forward.

- Ensure that the workstation complies with the PowerFlex system requirements.
- Ensure that you have the PowerFlex presentation server installation file. It is included in the PowerFlex complete software package that you downloaded from Dell Technologies support site.

NOTE: It is not recommended to install the PowerFlex presentation server on the same server as the PowerFlex Gateway. If you must install them on the same server, the PowerFlex Gateway must be installed first. The PowerFlex presentation server listens on port 8443, and the PowerFlex Gateway installer checks that port 8443 is available, although it does not use this port.

Steps

1. Install the PowerFlex presentation server on a server in your system:
   - Linux:

     ```
     rpm -i EMC-ScaleIO-mgmt-server-3.6-X.<build>.noarch.rpm
     ```

   - Debian (run with administrator privileges):

     ```
     sudo dpkg -i emc-scaleio-mgmt-server_3.6-X.<build>_amd64.deb
     ```

2. Prior to using the PowerFlex GUI, link the PowerFlex presentation server to the PowerFlex system. For more information, about linking the system, login and use of the PowerFlex GUI, refer to the *Configure and Customize Dell PowerFlex*.

Upgrade the PowerFlex presentation server (PowerFlex GUI) from v3.5 or later

Upgrade the PowerFlex presentation server from v3.5 or later.

Prerequisites

Ensure that you have the package for the PowerFlex presentation server, and access to the node hosting the PowerFlex presentation server.

Steps

1. Save a copy of the PowerFlex presentation server package on the node where PowerFlex presentation server is installed.
2. On the node running the PowerFlex presentation server, run the following command:
   - Linux:

     ```
     rpm -Uvh EMC-ScaleIO-mgmt_server-3.6-X.<build>.noarch.rpm
     ```

   - Debian:

     ```
     dpkg -i emc-scaleio-mgmt_server-3.6-X.<build>.amd64.deb
     ```

3. On a Centos 8-based nodes only, when the installation script has finished, type Q.

Results

Upgrade of the PowerFlex presentation server is complete. Link the PowerFlex presentation server to the MDM cluster, as described in the *Configure and Customize Dell PowerFlex*. Existing credentials can be used, provided they belong to either admin or configure + monitor (LDAP or native) users.

Upgrade the PowerFlex GUI from versions earlier than v3.5

Use these steps to upgrade the PowerFlex GUI from versions earlier than v3.5.

Prerequisites

NOTE: From PowerFlex version 3.5, the PowerFlex GUI is a web-based application installed on a server. Therefore, the new PowerFlex GUI is a fresh installation known as the PowerFlex presentation server, and the GUI client from VxFlex OS versions of the product is neither upgraded nor supported going forward.

- Ensure that the workstation complies with the PowerFlex system requirements.
- Ensure that you have the PowerFlex presentation server installation file. It is included in the PowerFlex complete software package that you downloaded from Dell Technologies support site.

NOTE: It is not recommended to install the PowerFlex presentation server on the same server as the PowerFlex Gateway. If you must install them on the same server, the PowerFlex Gateway must be installed first. The PowerFlex presentation server listens on port 8443, and the PowerFlex Gateway installer checks that port 8443 is available, although it does not use this port.

Steps

1. Install the PowerFlex presentation server on a server in your system:
   - Linux:

     ```
     rpm -i EMC-ScaleIO-mgmt-server-3.6-X.<build>.noarch.rpm
     ```

   - Debian (run with administrator privileges):

     ```
     sudo dpkg -i emc-scaleio-mgmt-server_3.6-X.<build>_amd64.deb
     ```

2. Prior to using the PowerFlex GUI, link the PowerFlex presentation server to the PowerFlex system. For more information, about linking the system, login and use of the PowerFlex GUI, refer to the *Configure and Customize Dell PowerFlex*.

Prepare server operating systems for support of large-scale systems (up to 2048 SDCs)

For large-scale systems with up to 2048 SDCs, prepare the operating systems of servers that will host MDM, SDS, SDR, and PowerFlex Gateway components.

Prerequisites

Superuser permissions are required for this procedure.

About this task

Configuration of servers hosting MDM, SDS, SDR, and PowerFlex Gateway components is required, in order to support large-scale systems with up to 2048 SDCs.

- A system setting must be validated and configured on servers hosting MDM, SDS, SDR, and the PowerFlex Gateway.
- For all servers hosting MDM, SDS, and SDR components, certain threshold parameters must be configured. This change enables a larger number of records in the operating system's ARP Cache table.

Steps

1. On servers hosting MDM, SDS, SDR, or PowerFlex Gateway components, do the following:
   a. Run this command:

      ```
      cat /proc/sys/fs/file-max
      ```

      The output for the command is the number of files that can be opened by the operating system.
   b. If the number is 24576 or higher, go to step 2. If the number is less than 24576, go to the next sub-step.
   c. Open the file `/etc/sysctl.conf` using vi.
   d. If the following line exists, edit it with vi. If it does not exist, add the full line with the following setting:

      ```
      fs.file-max = 24576
      ```

2. On servers hosting MDM, SDS, or SDR components, if the following lines exist in the file `/etc/sysctl.conf`, edit them with vi. If they do not exist, add the full lines with the following settings:

   ```
   net.ipv4.neigh.default.gc_thresh1 = 8192
   net.ipv4.neigh.default.gc_thresh2 = 16384
   net.ipv4.neigh.default.gc_thresh3 = 32768
   ```

3. Save the file.
4. Apply the changes using the command:

   ```
   sysctl -p
   ```

5. Repeat this procedure on the next server, until all servers are configured.

After upgrading a two-layer system

Tasks that can be performed after upgrading.

After the upgrade, you can perform many other tasks, including:

- Install RFcache, if necessary.

  NOTE: From v3.0 and later, the commands and workflow for adding RFcache are different from earlier versions. For more information, see the *Configure and Customize PowerFlex Guide* and the *PowerFlex CLI Reference Guide*.

- Take advantage of the new features added to this release.

  These options, and others, are described in the *Configure and Customize PowerFlex Guide*.

III

Manual Upgrade

Some aspects of the upgrade process can be performed manually, though best practice is to use automated processes whenever possible.

Manual Upgrade of Hosts

The hosts upgrade process can be performed manually, though best practice is to use automated processes whenever possible.

NOTE: While upgrading the system, the SDC component is not loaded as part of the automatic upgrade. Reboot the SDC node.

Upgrade SDC on Windows-based server

Use the below procedures to perform the upgrade SDC on a Windows-based server.

Prerequisites

The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the customer support site).

Steps

1. Double-click the MSI file and click **Next** to finish.
2. Perform a graceful reboot of the server (accounting for migration or stopping of Virtual Machines and applications as needed).

Upgrade SDC on Linux and AIX

Use the below procedures to upgrade SDC on Linux and AIX.

Prerequisites

The required installation software package for your SDC's operating system (available from the zipped software packages that can be downloaded from the customer support site)

NOTE: Do not reboot cluster members that are serving as part of a co-res cluster after installing a new SDC. For more information, see the *Dell PowerFlex 3.6.x Administration Guide*.

Steps

Install the appropriate package for your Linux distribution.

For example:

```
rpm -Uvh EMC-ScaleIO-sdc-3.6-x.x.rpm
```

NOTE: If the SDC service fails after the PowerFlex gateway upgrade, perform a graceful reboot of the server (accounting for migration or stopping of Virtual Machines and applications as needed).

Upgrade SDC on VMware ESXi

Use the below procedures to upgrade SDC on VMware ESXi.

Prerequisites

The required installation software package for your operating system of SDC (available from the zipped software packages that can be downloaded from the Customer Support site)

NOTE: This procedure is only applicable for unmanaged SDC that may require manual updates.

Steps

1. Copy the .VIB file to the VMware ESXi.
2. Run the following command:

   ```
   esxcli software vib update -d <path_to_zip>
   ```

   Where <path_to_zip> is the path to the SDC .zip file.
3. Perform a graceful reboot of the server (accounting for migration or stopping of Virtual Machines and applications as needed).

5

Manual Upgrade of Linux-based Nodes

Use command line to manually upgrade PowerFlex components on a Linux-based node.

Manual upgrade of physical Linux-based configurations

This section describes how to perform a manual upgrade when all PowerFlex components reside on physical Linux-based nodes.

It is highly recommended to perform upgrades using the PowerFlex Installer.

NOTE: When performing a manual upgrade, it is crucial to follow the described upgrade procedures step by step. Failure to do so might result in data loss and system instability.

Upgrading is non-disruptive; you can upgrade while I/Os are running, and volumes are mapped.

Proceed to the section that matches your system environment.

Linux package names

PowerFlex Linux installation packages have several name formats.

Package formats

Throughout this section, Linux packages are displayed in the following format:

`EMC-ScaleIO-<component>-3.6-X.<build>.<flavor>.x86_64.rpm`

where:

- *<component>* is the PowerFlex component: MDM, LIA, SDS, SDR, and so on.
- *<flavor>* is the operating system for your environment, according to the following table:

Table 6. Linux package formats

Linux flavor	Package format	Example
RHEL/OL/CentOS	el<version>	`EMC-ScaleIO-mdm-3.6-X.<build>.el<version>.x86_64.rpm` Example: `EMC-ScaleIO-mdm-3.6-<build>.X.el6.x86_64.rpm`
SUSE	sles<version>	`EMC-ScaleIO-mdm-3.6-X.<build>.sles<version>.x86_64.rpm` Example: `EMC-ScaleIO-mdm-3.6-<build>.X.sles12.2.x86_64.rpm`
Ubuntu	Ubuntu.<version>	`EMC-ScaleIO-mdm-3.6-X.<build>.Ubuntu.<version>.x86_64.tar` Example: `EMC-ScaleIO-mdm-3.6-X.<build>.Ubuntu.16.04.x86_64.tar` Ubuntu packages may need to be extracted before use. This is described where relevant.
Citrix Hypervisor (XenServer)	xs<version>	`EMC-ScaleIO-mdm-3.6-X.<build>.xs<version>.x86_64.rpm` Example: `EMC-ScaleIO-mdm-3.6-X.<build>.xs7.0.0.x86_64.rpm`

NOTE: RedHat CoreOS and Fedora CoreOS are supported on CSI containers. Refer to CSI documentation for more information.

Use the packages and the installation commands that match your Linux operating system environment.

On Citrix Hypervisor (XenServer)-based nodes, the syntax for PowerFlex CLI commands is `siocli`, instead of `scli`.

NOTE: Before deploying or upgrading Oracle Linux (OL) or Ubuntu-based systems, ensure that the PowerFlex environment is prepared, as described in the preparation tasks for Oracle Linux, SLES and Ubuntu in the *Deploy Dell PowerFlex* .

Non-default certificate use before and after a PowerFlex Gateway upgrade

When using a non-default security certificate, you must perform certain actions before and after upgrading the PowerFlex Gateway.

About this task

If a non-default security certificate is used on the PowerFlex Gateway in Windows and Linux environments (for example, if the certificate is signed by the user organization CA), you must save a copy of the certificate and the catalina file before the upgrade, and restore them to their original location after the upgrade.

Steps

1. Before commencing the PowerFlex Gateway upgrade, locate the `*.keystore` file and the `catalina.properties` file, and save a copy of them in another location.

   The default file locations, per operating system, are:

   - Linux:
     - `/opt/emc/scaleio/gateway/conf/catalina.properties`
     - `/opt/emc/scaleio/gateway/conf/certificates/.keystore`
   - Windows (64 bit):
     - `C:\Program Files\EMC\ScaleIO\Gateway\conf\catalina.properties`
     - `C:\Program Files\EMC\ScaleIO\Gateway\conf\certificates\.keystore`
2. After the upgrade is complete, copy these files back to their original location.

Upgrade the PowerFlex GUI from versions earlier than v3.5

Use these steps to upgrade the PowerFlex GUI from versions earlier than v3.5.

About this task

NOTE: From PowerFlex version 3.5, the PowerFlex GUI is a web-based application installed on a server. Therefore, the new PowerFlex GUI is a fresh installation known as the PowerFlex presentation server, and the GUI client from VxFlex OS versions of the product is neither upgraded nor supported going forward.

Steps

1. Install the PowerFlex presentation server on a server in your system:
   - Linux:

     ```
     rpm -i EMC-ScaleIO-mgmt-server-3.6-X.<build>.noarch.rpm
     ```

- Debian (run with administrator privileges):

```
sudo dpkg -i emc-scaleio-mgmt-server_3.6-X.<build>_amd64.deb
```

2. Prior to using the PowerFlex GUI, link the PowerFlex presentation server to the PowerFlex system. For more information, about linking the system, login and use of the PowerFlex GUI, refer to the *Configure and Customize Dell PowerFlex*.

Upgrade the PowerFlex presentation server (PowerFlex GUI) from v3.5 or later

Upgrade the PowerFlex presentation server from v3.5 or later.

Prerequisites

Ensure that you have the package for the PowerFlex presentation server, and access to the node hosting the PowerFlex presentation server.

Steps

1. Save a copy of the PowerFlex presentation server package on the node where PowerFlex presentation server is installed.
2. On the node running the PowerFlex presentation server, run the following command:
   - Linux:

```
rpm -Uvh EMC-ScaleIO-mgmt_server-3.6-X.<build>.noarch.rpm
```

   - Debian:

```
dpkg -i emc-scaleio-mgmt_server-3.6-X.<build>.amd64.deb
```

3. On a Centos 8-based nodes only, when the installation script has finished, type Q.

Results

Upgrade of the PowerFlex presentation server is complete. Link the PowerFlex presentation server to the MDM cluster, as described in the *Configure and Customize Dell PowerFlex*. Existing credentials can be used, provided they belong to either admin or configure + monitor (LDAP or native) users.

Manual upgrade of Linux-based nodes—3-node MDM cluster

Manual upgrade of PowerFlex components, on a Linux-based nodes (all flavors) when the MDM cluster is in 3-node mode.

Prerequisites

- Ensure that the nodes that will be upgraded meet the requirements described in the *Getting to Know Dell PowerFlex Guide*. In addition, LIA on each node requires an additional 350 MB RAM. Note that SDS components could require up to 19 GB if the maximum capacity of 128 TB is used.
- If needed, install the `GPG-RPM-KEY` file (with the `rpm --import <path_to_new_public_key>` command). This is used to authenticate the RPM packages.
- For data networks using IPv6, if you plan to implement a floating virtual IP address for the MDM, disable the IPv6 DAD setting on the node's relevant interfaces, using the command:

```
sysctl net.ipv6.conf.<interface_name>.dad_transmits=0
```

- If you intend to use the CHAP component authentication feature after the upgrade, make the necessary preparations to ensure that you upgrade all system components now. Systems containing any components running PowerFlex versions lower than v3.5 do not support CHAP authentication.

NOTE: Upgrade to v3.5 sets the CHAP authentication feature to DISABLED, and passwords will be all-zero. To enable the feature, after the upgrade is complete on all system components, follow the post-installation procedures described in the *Configure and Customize Dell PowerFlex*.

About this task

When SNMP is enabled in the 2.x base version, SNMP MDM credentials remain in the lockbox after the upgrade.

When performing this procedure on XenServer nodes, use `siocli` in place of `scli`.

For the purposes of this procedure, the following naming conventions are used:

MDM 1—primary MDM

MDM 2—secondary MDM

TB 1—tiebreaker 1

NOTE: During this procedure, the `finalize_upgrade` command initiates an MDM service restart. This should have no other effect on the system.

Steps

1. Download and extract the latest PowerFlex packages to `/tmp/`.
2. Upgrade (or install) LIA on all machines in the system:

   ```
   rpm -U EMC-ScaleIO-lia-3.6-X.<build>.elX.x86_64.rpm --nosignature
   ```

3. Switch to MDM single mode, by running the following commands on MDM 1:

   ```
   scli --login --username <USER> --password <PASSWORD>
   ```

   ```
   scli --start_upgrade
   ```

   ```
   scli --allow_commands_during_upgrade
   ```

   ```
   scli --switch_cluster_mode --cluster_mode 1_node --remove_slave_mdm_ip <MDM2 IP> --
   remove_tb_ip <TB1 IP>
   ```

   In the `--switch_cluster_mode` command, type the data (not management) IP addresses.

4. Upgrade MDM 2 and TB 1, by running the following commands on both:
   a. Stop service:

      ```
      /opt/emc/scaleio/mdm/bin/delete_service.sh
      ```

   b. Upgrade RPM:

      ```
      I_AM_SURE=1 UPGRADE_PRECHECK_SCRIPT_DONE=1 rpm -U EMC-ScaleIO-mdm-3.6-
      X.<build>.<flavor>.x86_64.rpm
      ```

   c. Start service:

      ```
      /opt/emc/scaleio/mdm/bin/create_service.sh
      ```

5. Add the upgraded MDM 2 and TB 1, by running the following commands on MDM 1:
   a. Add the components to the cluster:

      ```
      scli --switch_cluster_mode --cluster_mode 3_node --add_slave_mdm_ip <MDM2 IP> --
      add_tb_ip <TB1 IP>
      ```

b. Verify that the cluster is in normal state:

```
scli --query_cluster
```

6. Switch MDM ownership, by running the following command on MDM 1:

```
scli --switch_mdm_ownership --new_master_mdm_ip <MDM2 IP>
```

7. Run the following commands on MDM 2:
   a. Log in:

```
scli --login --username <USER> --password <PASSWORD>
```

   b. Enable commands:

```
scli --allow_commands_during_upgrade
```

   c. View cluster information:

```
scli --query_cluster
```

   d. Switch to single mode:

```
scli --switch_cluster_mode --cluster_mode 1_node --remove_slave_mdm_ip <MDM1 IP> --
remove_tb_ip <TB1 IP>
```

8. Upgrade MDM 1, by running the following commands on MDM 1:
   a. Stop service:

```
/opt/emc/scaleio/mdm/bin/delete_service.sh
```

   b. Upgrade RPM:

```
I_AM_SURE=1 UPGRADE_PRECHECK_SCRIPT_DONE=1 rpm -U EMC-ScaleIO-mdm-3.6-
X.<build>.<flavor>.x86_64.rpm
```

   c. Start service:

```
/opt/emc/scaleio/mdm/bin/create_service.sh
```

9. Add the MDM as a standby MDM, and switch to 3-node cluster mode, by running these commands from the primary MDM (MDM 2):
   a. Log in (if not logged in):

```
scli --login --username <USER> --password <PASSWORD>
```

   b. Enable commands (if not already done):

```
scli --allow_commands_during_upgrade
```

   c. Switch mode:

```
scli --switch_cluster_mode --cluster_mode 3_node --add_slave_mdm_ip <MDM1 IP> --
add_tb_ip <TB1 IP>
```

   d. Wait for the cluster to finish synchronizing the repository:

```
scli --query_cluster
```

   Wait until cluster state is Normal.

e. Switch cluster ownership:

```
scli --switch_mdm_ownership --new_master_mdm_ip <MDM1 IP>
```

10. Display list of all SDSs, by running the following commands from primary MDM 1:

```
scli --login --username <USER> --password <PASSWORD>
```

```
scli --query_all_sds
```

11. Upgrade SDSs. Repeat this step on each SDS:
    a. Enter an SDS into maintenance mode, by running the following command from MDM 1:

```
scli --enter_maintenance_mode --sds_id <SDS ID>
```

    Validate that the operation succeeded.
    b. Disable RFcache (if installed), by running the following command from MDM 1:

```
scli --disable_rfcache --sds_id <SDS ID>
```

    c. Upgrade the SDS, by running the following command on the machine where the SDS resides:

```
rpm -U EMC-ScaleIO-sds-3.6-X.<build>.<flavor>.x86_64.rpm
```

    d. Upgrade RFcache (if installed), by running the following command on the machine where the SDS resides:

```
rpm -U EMC-ScaleIO-xcache-3.6-X.<build>.<flavor>.x86_64.rpm
```

    e. Enable RFcache (if installed), by running the following command from MDM 1:

```
scli --enable_rfcache --sds_id <SDS ID>
```

    f. Exit the SDS from maintenance mode, by running the following command from MDM 1:

```
scli --exit_maintenance_mode --sds_id <SDS ID>
```

    g. Before continuing, validate that the SDS is in the proper mode, by running the following command from MDM 1:

```
scli --query_properties --object_type SDS --object_id <SDS ID> --properties
MAINTENANCE_MODE_STATE
```

    Ensure that the state is NO_MAINTENANCE

    The SDS is upgraded. Repeat these steps for every SDS.
12. Upgrade all SDCs:

```
rpm -U EMC-ScaleIO-sdc-3.6-X.<build>.<flavor>.x86_64.rpm
```

    If the SDC kernel module is processing I/O, its upgrade will occur only after the node is restarted.

    NOTE: An SDC may appear to be running I/O even though no I/O is actually occurring. In such cases, perform a reboot and return to operation of each SDC one by one, preferably after all other system components have been upgraded.

    The SDC is upgraded. Repeat this step for every SDC.
13. Validate the upgrade's current status, by running the following command on MDM 1:

```
scli --query_upgrade
```

    Ensure that the output includes the following:

```
0 SDSs need to be upgraded in total
```

14. Define the expected device media type for each Storage Pool, by running the following command for each one:

```
scli --set_storage_pool_media_type --protection_domain_name <PROTECTION_DOMAIN_NAME>
--storage_pool_name <STORAGE_POOL_NAME> --media_type <MEDIA_TYPE>
```

Device media type for a Storage Pool has been set. Repeat this step for every Storage Pool.

15. Finalize the upgrade, by running the following command on MDM 1:

```
scli --finalize_upgrade
```

Ensure that the output includes the following:

```
Successfully finalized the upgrade process
```

16. Upgrade the currently installed PowerFlex Gateway by running the following command (optional):

```
rpm -U EMC-ScaleIO-gateway-3.6-X.<build>.x86_64.rpm
```

The following variables can be added to the command, if required:

- *SIO_GW_KEYTOOL=<keytool_path>* allows override of the location for the Java keytool utility. For example: `SIO_GW_KEYTOOL=/usr/java/jre1.8.0_XX/bin/`
- *SIO_JAVA=<Java version>* specifies a different Java version

Example:

```
rpm -U /tmp/EMC-ScaleIO-gateway-3.6-X.<build>.x86_64.rpm
```

A success message is displayed, and the PowerFlex Installer interface now reflects the updated PowerFlex Gateway version.

NOTE: If your PowerFlex Gateway uses a non-default certificate, you must copy the certificate and other files before upgrading the PowerFlex Gateway. For more information, see the "Upgrading the PowerFlex Gateway when a non-default certificate is used" topic.

Next steps

If you are upgrading from v2.6.x, use the CLI or the upgraded GUI to add RFcache devices back to the system. First, create an RFcache SSD Acceleration Pool for the RFcache devices, and then add the devices to the Acceleration Pool. For more information, see the *Configure and Customize PowerFlex* guide.

NOTE: From v3.0 and later, the commands and workflow for adding RFcache are different from earlier versions.

Post-upgrade activities

Tasks to perform after the automated upgrade on Linux-based nodes is complete.

After the upgrade, you can perform many other tasks, including:

- Install RFcache, if necessary.

  NOTE: From v3.0 and later, the commands and workflow for adding RFcache are different from earlier versions. For more information, see the *Configure and Customize Dell PowerFlex* and the *Dell PowerFlex CLI Reference Guide*.

- Extend the MDM cluster from 3-node to 5-node mode
- If not already configured, add virtual IP addresses to the data network
- Take advantage of the new features added to this release

These options, and others, are described in the *Configure and Customize Dell PowerFlex*.

Manual upgrade of Linux-based nodes—5-node MDM cluster

Manual upgrade of PowerFlex components, on a Linux-based node (all flavors) when the MDM cluster is in 5-node mode.

Prerequisites

- Ensure that the nodes that will be upgraded meet the requirements described in the *Getting to Know Dell PowerFlex Guide*. In addition, LIA on each node requires an additional 350 MB RAM. Note that SDS components could require up to 19 GB if the maximum capacity of 128 TB is used.
- If needed, install the `GPG-RPM-KEY` file (with the `rpm --import <path_to_new_public_key>` command). This is used to authenticate the RPM packages.
- For data networks using IPv6, if you plan to implement a floating virtual IP address for the MDM, disable the IPv6 DAD setting on the node's relevant interfaces, using the command:

```
sysctl net.ipv6.conf.<interface_name>.dad_transmits=0
```

- If you intend to use the CHAP component authentication feature after the upgrade, make the necessary preparations to ensure that you upgrade all system components now. Systems containing any components running PowerFlex versions lower than v3.5 do not support CHAP authentication.

  NOTE: Upgrade to v3.5 sets the CHAP authentication feature to DISABLED, and passwords will be all-zero. To enable the feature, after the upgrade is complete on all system components, follow the post-installation procedures described in the *Configure and Customize Dell PowerFlex*.

About this task

When SNMP is enabled in the 2.x base version, SNMP MDM credentials remain in the lockbox after the upgrade.

When performing this procedure on Citrix Hypervisor (XenServer)-based nodes, use `siocli` in place of `scli`.

For the purposes of this procedure, the following naming conventions are used:

MDM 1—primary MDM

MDM 2—secondary 1 MDM

MDM 3—secondary 2 MDM

TB 1—tiebreaker 1

TB 2—tiebreaker 2

NOTE: During this procedure, the `finalize_upgrade` command initiates an MDM service restart.

Steps

1. Download and extract the latest PowerFlex packages to `/tmp/`.
2. Upgrade (or install) LIA on all nodes in the system:

```
rpm -U EMC-ScaleIO-lia-3.6-X.<build>.<flavor>.x86_64.rpm --nosignature
```

3. Start the upgrade, by running the following commands on MDM 1:

```
scli --login --username <USER> --password <PASSWORD>
```

```
scli --start_upgrade
```

4. Upgrade the MDM on the nodes (except for the primary MDM).

   NOTE: Perform the upgrade steps on each node in the MDM cluster, one-at-a-time in the following order, on MDM 2, MDM 3, TB 1 and then TB 2. Start with MDM 2.

a. On MDM 2, stop the service:

```
/opt/emc/scaleio/mdm/bin/delete_service.sh
```

b. Upgrade RPM:

```
I_AM_SURE=1 UPGRADE_PRECHECK_SCRIPT_DONE=1 rpm -U EMC-ScaleIO-mdm-3.6-
X.<build>.<flavor>.x86_64.rpm
```

c. Start service:

```
/opt/emc/scaleio/mdm/bin/create_service.sh
```

d. Verify that the cluster is in normal state:

```
scli --query_cluster
```

e. After validating MDM 2, repeat these sub-steps for MDM 3, TB 1 and then TB 2.

The MDM packages are upgraded, per node.

5. Switch MDM ownership, by running the following commands on MDM1:

```
scli --allow_commands_during_upgrade
```

```
scli --switch_mdm_ownership --new_master_mdm_ip <MDM2 IP>
```

6. Upgrade MDM 1, by running the following commands on it:
   a. Stop service:

```
/opt/emc/scaleio/mdm/bin/delete_service.sh
```

   b. Upgrade RPM:

```
I_AM_SURE=1 UPGRADE_PRECHECK_SCRIPT_DONE=1 rpm -U EMC-ScaleIO-mdm-3.6-
X.<build>.<flavor>.x86_64.rpm
```

   c. Start service:

```
/opt/emc/scaleio/mdm/bin/create_service.sh
```

7. Run the following command on MDM 1 to view cluster status, and wait for the cluster to complete the repository synchronization:

```
scli --query_cluster
```

   Wait until the cluster state is normal.

8. Switch MDM ownership, by running the following commands on MDM 2:

```
scli --login --username <USER> --password <PASSWORD>
```

```
scli --switch_mdm_ownership --new_master_mdm_ip <MDM1 IP>
```

9. Display list of all SDS, by running the following commands from primary MDM 1:

```
scli --login --username <USER> --password <PASSWORD>
```

```
scli --query_all_sds
```

10. Upgrade SDSs. Repeat this step on each SDS:

a. Enter an SDS into maintenance mode, by running the following command from MDM 1:

```
scli --enter_maintenance_mode --sds_id <SDS ID>
```

Validate that the operation succeeded.

b. Disable RFcache (if installed), by running the following command from MDM 1:

```
scli --disable_rfcache --sds_id <SDS ID>
```

c. Upgrade the SDS, by running the following command on the machine where the SDS resides:

```
rpm -U EMC-ScaleIO-sds-3.6-X.<build>.<flavor>.x86_64.rpm
```

d. Upgrade RFcache (if installed), by running the following command on the machine where the SDS resides:

```
rpm -U EMC-ScaleIO-xcache-3.6-X.<build>.<flavor>.x86_64.rpm
```

e. Enable RFcache (if installed), by running the following command from MDM 1:

```
scli --enable_rfcache --sds_id <SDS ID>
```

f. Exit the SDS from maintenance mode, by running the following command from MDM 1:

```
scli --exit_maintenance_mode --sds_id <SDS ID>
```

g. Before continuing, validate that the SDS is in the proper mode, by running the following command from MDM 1:

```
scli --query_properties --object_type SDS --object_id <SDS ID> --properties
MAINTENANCE_MODE_STATE
```

Ensure that the state is NO_MAINTENANCE

The SDS is upgraded. Repeat these steps for every SDS.

11. Upgrade all SDCs:

```
rpm -U EMC-ScaleIO-sdc-3.6-X.<build>.<flavor>.x86_64.rpm
```

If the SDC kernel module is processing I/O, its upgrade will occur only after the node is restarted.

NOTE: An SDC may appear to be running I/O even though no I/O is actually occurring. In such cases, perform a reboot and return to operation of each SDC one by one, preferably after all other system components have been upgraded.

The SDC is upgraded. Repeat this step for every SDC.

12. Validate the upgrade's current status, by running the following command on MDM 1:

```
scli --query_upgrade
```

Ensure that the output includes the following:

```
0 SDSs need to be upgraded in total
```

13. Define the expected device media type for each Storage Pool, by running the following command for each one:

```
scli --set_storage_pool_media_type --protection_domain_name <PROTECTION_DOMAIN_NAME>
--storage_pool_name <STORAGE_POOL_NAME> --media_type <MEDIA_TYPE>
```

Device media type for a Storage Pool has been set. Repeat this step for every Storage Pool.

14. Finalize the upgrade, by running the following command on MDM 1:

```
scli --finalize_upgrade
```

Ensure that the output includes the following:

```
Successfully finalized the upgrade process
```

15. Upgrade the currently installed PowerFlex Gateway by running the following command (optional):

```
rpm -U EMC-ScaleIO-gateway-3.6-X.<build>.x86_64.rpm
```

The following variables can be added to the command, if required:

- *SIO_GW_KEYTOOL=<keytool_path>* allows override of the location for the Java keytool utility. For example: `SIO_GW_KEYTOOL=/usr/java/jre1.8.0_XX/bin/`
- *SIO_JAVA=<Java version>* specifies a different Java version

Example:

```
rpm -U /tmp/EMC-ScaleIO-gateway-3.6-X.<build>.x86_64.rpm
```

A success message is displayed, and the PowerFlex Installer interface now reflects the updated PowerFlex Gateway version.

NOTE: If your PowerFlex Gateway uses a non-default certificate, you must copy the certificate and other files before upgrading the PowerFlex Gateway. For more information, see the "Upgrading the PowerFlex Gateway when a non-default certificate is used" topic.

Next steps

If you are upgrading from v2.6.x, use the CLI or the upgraded GUI to add RFcache devices back to the system. First, create an RFcache SSD Acceleration Pool for the RFcache devices, and then add the devices to the Acceleration Pool. For more information, see the *Configure and Customize Dell PowerFlex*.

NOTE: From v3.0 and later, the commands and workflow for adding RFcache are different from earlier versions.

Post-upgrade activities

Tasks to perform after the automated upgrade on Linux-based nodes is complete.

After the upgrade, you can perform many other tasks, including:

- Install RFcache, if necessary.

  NOTE: From v3.0 and later, the commands and workflow for adding RFcache are different from earlier versions. For more information, see the *Configure and Customize Dell PowerFlex* and the *Dell PowerFlex CLI Reference Guide*.

- Extend the MDM cluster from 3-node to 5-node mode
- If not already configured, add virtual IP addresses to the data network
- Take advantage of the new features added to this release

These options, and others, are described in the *Configure and Customize Dell PowerFlex*.

Prepare server operating systems for support of large-scale systems (up to 2048 SDCs)

For large scale-systems with up to 2048 SDCs, prepare the operating systems of servers that will host MDM, SDS, SDR, and PowerFlex Gateway components.

Prerequisites

Superuser permissions are required for this procedure.

About this task

Configuration of servers hosting MDM, SDS, SDR, and PowerFlex Gateway components is required, in order to support large-scale systems with up to 2048 SDCs.

- A system setting must be validated and configured on servers hosting MDM, SDS, SDR, and the PowerFlex Gateway.

- For all servers hosting MDM, SDS, and SDR components, certain threshold parameters must be configured. This change enables a larger number of records in the operating system's ARP Cache table.

Steps

1. On servers hosting MDM, SDS, SDR, or PowerFlex Gateway components, do the following:
   a. Run this command:

      ```
      cat /proc/sys/fs/file-max
      ```

      The output for the command is the number of files that can be opened by the operating system.
   b. If the number is 24576 or higher, go to step 2. If the number is less than 24576, go to the next sub-step.
   c. Open the file `/etc/sysctl.conf` using vi.
   d. If the following line exists, edit it with vi. If it does not exist, add the full line with the following setting:

      ```
      fs.file-max = 24576
      ```

2. On servers hosting MDM, SDS, or SDR components, if the following lines exist in the file `/etc/sysctl.conf`, edit them with vi. If they do not exist, add the full lines with the following settings:

   ```
   net.ipv4.neigh.default.gc_thresh1 = 8192
   net.ipv4.neigh.default.gc_thresh2 = 16384
   net.ipv4.neigh.default.gc_thresh3 = 32768
   ```

3. Save the file.
4. Apply the changes using the command:

   ```
   sysctl -p
   ```

5. Repeat this procedure on the next server, until all servers are configured.

6

Manual Upgrades in the ESXi Environment

Some aspects of upgrading components in the ESXi environment can be performed manually.

Manually upgrade an ESXi-based PowerFlex system

For use cases where automated upgrades are not possible, follow the manual upgrade procedures described in this section.

Use the following workflow to upgrade the PowerFlex system:

1. Collect information about system components.
2. Ensure that all prerequisites are met.
3. Upgrade Java.
4. Upgrade the PowerFlex GUI.
5. Upgrade the PowerFlex Gateway.
6. Upgrade the PowerFlex system components:
   - MDM
   - LIA
   - SDS
   - RFcache (xcache)—optional
   - SDC
7. Patch the operating system of the PowerFlex Gateway SVM.

Collect and record PowerFlex information

Collect and record information about your system's components before commencing upgrade procedures.

About this task

Use the procedures that follow to collect and record the information in this table:

Table 7. Resource and credential information

Resource	IP address	Credentials
VMware vSphere production cluster	-	-
Jump server	-	-
PowerFlex root password for SVMs	-	-
PowerFlex admin password	-	-
PowerFlex Gateway	-	-
Production datacenter name	-	-
Production vCenter name	-	-
MDM1 (primary)	-	-
MDM2 (secondary 1)	-	-
MDM3 (secondary 2)	-	-
Tiebreaker1	-	-
Tiebreaker2	-	-

Table 7. Resource and credential information (continued)

Resource	IP address	Credentials
PowerFlex virtual IP subnets	-	-
SDS IP addresses (SVM and RHEL)	-	-

Steps

1. Gather the datacenter and vCenter names:
   a. Log on to the customer vCenter server.
   b. Click the **Home** icon at the top of the window and select **Hosts and Clusters**.
   c. Record the **datacenter** name.
   d. Record the **vCenter** name.
   The vCenter name is displayed above the datacenter name.
2. Gather and record additional PowerFlex cluster and component information:
   a. Use a console application to log in to the primary MDM SVM.
   b. Log in to the PowerFlex CLI:

```
scli --login --username admin --password <PASSWORD>
```

   c. Perform a query to obtain cluster information:

```
scli --query_cluster
```

   d. Gather SDS IP addresses:

```
scli --query_all_sds
```

   e. Gather SDC IP addresses:

```
scli --query_all_sdc
```

3. Verify the state of the PowerFlex system:
   a. Log in to the PowerFlex GUI using the primary MDM IP address and admin user credentials.
   b. On the **Dashboard**, verify the following:
      - There is no rebuild or rebalance running in the background (display should show 0.0 KB/s)
      - Adequate capacity exists for the upgrade
      - No SDSs or SDCs are disconnected
      - SDS devices are error-free
      - The MDM cluster, including tiebreakers, is not in Degraded state
      - There are no active alerts shown in the **Alerts** view
4. Verify available disk space:
   a. Use a console application to log in to each MDM and each tiebreaker.
   b. On each one, run the following command to display available disk space:

```
df -h
```

   Each MDM and tiebreaker must have at least 1.2 GB of available disk space.

Prerequisites for manual upgrades in ESXi environments

In addition to general prerequisites for all PowerFlex systems, ensure that the specific requirements for upgrading PowerFlex manually in ESXI environments are met.

ESXi-based nodes and VMware cluster prerequisites

Ensure that the ESXi-based nodes and the VMware cluster meet the prerequisites before you perform the upgrade.

- When upgrading a node based on ESXi 6.5U2 and higher
  1. Ensure that the following line does not appear in the `/etc/ssh/sshd_config` file of the ESXi host:`kexalgorithms diffie-hellman-group1-sha1, diffie-hellmangroup-exchange-sha1`

     If this line does appear, delete it. The line deletion does not relate to any configuration, and you do not need to restart either the ESXi-based node, nor the sshd service.

  2. Repeat the procedure for the rest of the nodes based on ESXi 6.5U2 and higher.
- The upgrade process requires entering the ESXi-based nodes into maintenance mode. To do so, ensure that Admission Control is not enabled on the VMware cluster:

Table 8. Resource and credential information

Resource	IP address	Credentials
VMware vSphere production cluster	-	-
Jump server	-	-
PowerFlex root password for SVMs	-	-
PowerFlex admin password	-	-
PowerFlex Gateway	-	-
Production datacenter name	-	-
Production vCenter name	-	-
MDM1 (primary)	-	-
MDM2 (secondary 1)	-	-
MDM3 (secondary 2)	-	-
Tiebreaker1	-	-
Tiebreaker2	-	-
PowerFlex virtual IP subnets	-	-
SDS IP addresses (SVM and RHEL)	-	-

Java upgrade prerequisites

Before changing the Java version of a node that is running PowerFlex Gateway, you must prepare lockbox-related files.

Prerequisites

Ensure that you have Oracle JRE 1.8 build 301 or OpenJDK JDK 8 build 302.b08 installed before upgrading your system.

About this task

NOTE: If you want to upgrade OpenJDK to version 11.0.12.0.7, upgrade it after upgrade of the PowerFlex Gateway.

NOTE: DO NOT perform the following procedure when upgrading to OpenJDK to version 11.0.12.0.7.

The lockbox in PowerFlex uses files in the Java folder of the PowerFlex Gateway. These files need to be saved before any Java version update, then pasted back into the Java folder.

Steps

1. On the PowerFlex Gateway, in the `jre\lib\ext` (or `jre/lib/ext` for Windows) Java folder, copy these files to a different folder:
   - `commons-lang3-3.6.jar`
   - `cryptoj-6.2.3.jar`
2. Update the Java version.
3. Paste the files back to the folder from where you copied them.

PowerFlex GUI upgrade prerequisites

Before upgrading PowerFlex components from all previous versions, ensure that Oracle JRE 1.8 build 301, or OpenJDK JDK 8 build 302.b08 is installed on the server that will host the PowerFlex presentation server. The PowerFlex presentation server is an application that is used from PowerFlex v3.5 and higher. The VxFlex OS GUI client that was used in versions earlier than v3.5 will not be upgraded and is not supported going forward. If the PowerFlex presentation server and the PowerFlex Gateway will be installed on the same server, the PowerFlex Gateway must be installed before the PowerFlex presentation server. The PowerFlex presentation server listens on port 8443, and the PowerFlex Gateway checks that port 8443 is available, although it does not use this port.

NOTE: If you want to upgrade OpenJDK to version 11, do so after upgrade of the PowerFlex presentation server.

NOTE: If Java is already installed, check the existing version of Java by running the following command: `java -version`

For example, install the Oracle JRE file relevant for the server's operating system:

```
rpm -i /var/tmp/jre-8uXX-linux-x64.rpm
```

For information about downloading and installing OpenJDK, refer to: How to download and install prebuilt OpenJDK packages

PowerFlex Gateway upgrade prerequisites

Before upgrading the PowerFlex Gateway, ensure that Oracle JRE 1.8 build 301, or OpenJDK JDK 8 build 302.b08, is installed on the node.

NOTE: If you want to upgrade OpenJDK to version 11 (Linux only), do so after upgrade of the PowerFlex Gateway.

NOTE: If Java is already installed, check the existing version of Java by running the following command: `java -version`

For example, install the Oracle JRE file relevant for the node's operating system:

- Windows: `jre-8uXX-windows-x64.exe`
- Linux: `rpm -i /var/tmp/jre-8uXX-linux-x64.rpm`

For information about downloading and installing OpenJDK, refer to:

- Linux:

  How to download and install prebuilt OpenJDK packages
- Windows:

  Download OJDK for Windows x86_64

Non-default certificate use before and after a PowerFlex Gateway upgrade

When using a non-default security certificate, you must perform certain actions before and after upgrading the PowerFlex Gateway.

About this task

If a non-default security certificate is used on the PowerFlex Gateway in Windows and Linux environments (for example, if the certificate is signed by the user organization CA), you must save a copy of the certificate and the catalina file before the upgrade, and restore them to their original location after the upgrade.

Steps

1. Before commencing the PowerFlex Gateway upgrade, locate the `*.keystore` file and the `catalina.properties` file, and save a copy of them in another location.

   The default file locations, per operating system, are:

   - Linux:
     - `/opt/emc/scaleio/gateway/conf/catalina.properties`
     - `/opt/emc/scaleio/gateway/conf/certificates/.keystore`
   - Windows (64 bit):
     - `C:\Program Files\EMC\ScaleIO\Gateway\conf\catalina.properties`
     - `C:\Program Files\EMC\ScaleIO\Gateway\conf\certificates\.keystore`

2. After the upgrade is complete, copy these files back to their original location.

SVM upgrade prerequisites

Verify the existence of OpenSSL packages and install them, if necessary. Increase SVM system disk size to 16 GB, as described in the *Reference* section of this guide.

About this task

NOTE: Do not run install commands if OpenSSL packages are already installed.

Steps

1. To verify if there are already OpenSSL packages on the SVM, run the command:

   ```
   rpm -qa |grep -i openssl
   ```

   Output similar to the following will be returned if OpenSSL exists:

   ```
   openssl1-1.0.1g-0.40.1
   libopenssl1_0_0-1.0.1g-0.40.1
   ```

2. If OpenSSL packages are not installed, run the following commands to install them.

   ```
   rpm -i libopenssl1_0_0-1.0.1g-0.40.1.x86_64.rpm
   ```

   ```
   rpm -i openssl1-1.0.1g-0.40.1.x86_64.rpm
   ```

3. Increase SVM system disk size of all SVMs to 16 GB, as described in the *Reference* section of this guide.

PowerFlex core component prerequisites

Prerequisites for core PowerFlex components

If you intend to use the CHAP component authentication feature after the upgrade, make the necessary preparations to ensure that you upgrade all system components now. Systems containing any components running PowerFlex versions lower than v3.5 do not support CHAP authentication.

NOTE: Upgrade to v3.5 sets the CHAP authentication feature to DISABLED, and passwords will be all-zero. To enable the feature, after the upgrade is complete on all system components, follow the post-installation procedures described in the *Configure and Customize Dell PowerFlex*.

Upgrade the PowerFlex GUI from versions earlier than v3.5

Use these steps to upgrade the PowerFlex GUI from versions earlier than v3.5.

Prerequisites

NOTE: From PowerFlex version 3.5, the PowerFlex GUI is a web-based application installed on a server. Therefore, the new PowerFlex GUI is a fresh installation known as the PowerFlex presentation server, and the GUI client from VxFlex OS versions of the product is neither upgraded nor supported going forward.

- Ensure that the workstation complies with the PowerFlex system requirements.
- Ensure that you have the PowerFlex presentation server installation file. It is included in the PowerFlex complete software package that you downloaded from Dell Technologies support site.

NOTE: It is not recommended to install the PowerFlex presentation server on the same server as the PowerFlex Gateway. If you must install them on the same server, the PowerFlex Gateway must be installed first. The PowerFlex presentation server listens on port 8443, and the PowerFlex Gateway installer checks that port 8443 is available, although it does not use this port.

Steps

1. Install the PowerFlex presentation server on a server in your system:
   - Linux:

     ```
     rpm -i EMC-ScaleIO-mgmt-server-3.6-X.<build>.noarch.rpm
     ```

   - Debian (run with administrator privileges):

     ```
     sudo dpkg -i emc-scaleio-mgmt-server_3.6-X.<build>_amd64.deb
     ```

2. Prior to using the PowerFlex GUI, link the PowerFlex presentation server to the PowerFlex system. For more information, about linking the system, login and use of the PowerFlex GUI, refer to the *Configure and Customize Dell PowerFlex*.

Upgrade the PowerFlex presentation server (PowerFlex GUI) from v3.5 or later

Upgrade the PowerFlex presentation server from v3.5 or later.

Prerequisites

Ensure that you have the package for the PowerFlex presentation server, and access to the node hosting the PowerFlex presentation server.

Steps

1. Save a copy of the PowerFlex presentation server package on the node where PowerFlex presentation server is installed.
2. On the node running the PowerFlex presentation server, run the following command:
   - Linux:

     ```
     rpm -Uvh EMC-ScaleIO-mgmt_server-3.6-X.<build>.noarch.rpm
     ```

   - Debian:

     ```
     dpkg -i emc-scaleio-mgmt_server-3.6-X.<build>.amd64.deb
     ```

3. On a Centos 8-based nodes only, when the installation script has finished, type Q.

Results

Upgrade of the PowerFlex presentation server is complete. Link the PowerFlex presentation server to the MDM cluster, as described in the *Configure and Customize Dell PowerFlex*. Existing credentials can be used, provided they belong to either admin or configure + monitor (LDAP or native) users.

Unregister the old PowerFlex system

Unregister the old PowerFlex system from the vSphere PowerFlex plug-in.

Prerequisites

MDM admin user and password credentials are required for this procedure.

About this task

From the web interface of the vCenter Client, unregister PowerFlex from within the PowerFlex plug-in.

Steps

1. In the **vSphere Web Client** main view, click the **DELL EMC PowerFlex** icon.
2. From the **Navigator** tree on the left, click the **PowerFlex Systems** option.
3. Select the relevant system name and right-click to open the options menu.
4. Select **Unregister PowerFlex system**, enter the MDM admin user and password, and confirm the action.

Unregister the current PowerFlex plug-in

Prerequisites

PowerCLI is required for unregistering the plug-in. Administrator privileges are required.

About this task

In this procedure, the currently installed vSphere PowerFlex plug-in will be removed in preparation for installing the latest version.

Steps

1. From the folder where you extracted the `EMC-ScaleIO-vSphere-plugin-installer-3.6-X.<build>.zip` file, use PowerCLI, as administrator, to run the `VxFlexOSPluginSetup.ps1` script.
2. Enter the vCenter credentials and confirm the script actions. If you are asked to trust an untrusted publisher, type **A** to continue.
3. When prompted to choose a mode, select option **2**, `Unregister ScaleIO plug-in`.
4. Follow the progress on the screen, and when prompted to do so, log out, then log back in to the vSphere web client.
5. Verify that the plug-in is no longer registered (VxFlex OS icon should no longer be visible).

   NOTE: If the icon has not disappeared, even though PowerCLI reports that it was successfully unregistered, reboot the vCenter.

6. When the process is complete, enter **4** to exit the plug-in script.

Register the new PowerFlex plug-in

Register the new vSphere PowerFlex plug-in.

Prerequisites

Before you begin, ensure that there is communication between the vSphere web client server (usually installed on the vCenter) and the web server storing the PowerFlex plug-in.

About this task

Register the new PowerFlex plug-in.

To use your own web server, see Register the PowerFlex plug-in manually.

Steps

1. Copy the following files to a designated host (preferably a vCenter):
   - `ScaleIOVM_3nics_3.6.XXX.XXX.ova`
   - `EMC-ScaleIO-vSphere-plugin-installer-3.6-X.<build>.zip`
2. Extract the contents of the zip file.
3. Using PowerCLI for VMware, set to Run as administrator, run the following script: `VxFlexOSPluginSetup.ps1`.
   a. Enter the vCenter name or IP address, user name, and password.
   b. For `Choose Mode`, select option **1, Register PowerFlex plugin**.
   c. Read the upgrade notice, and enter **`y`** to continue.
   d. For `Select Registration Mode`, select **`Standard`** (simplified, using embedded Tomcat).
      This step may take a few minutes.
   e. If necessary, accept the thumbprint.
4. Log in to the vSphere web client.
   a. If you are already logged in, log out, then log in again.
   b. Verify that the PowerFlex icon appears in the web client.
   c. Open the PowerFlex plug-in in the client, and ensure that it is the correct version.
      If the version did not update, clear your browser cookies, log out, then log in to the PowerFlex plug-in again.
5. In the PowerCLI window, press **Enter** to finish the PowerFlex plug-in download and return to the menu.

PowerFlex component overview

Workflow overview for manual upgrade of PowerFlex components

1. Ensure PowerFlex system is running the same version on all nodes.
2. Upgrade the PowerFlex Gateway to v3.6.
3. Upload the RHEL7 core packages for v3.6 to the PowerFlex Gateway.
4. Initiate the upgrade flow from the PowerFlex Gateway to v3.6.
5. Upgrade the vSphere PowerFlex plug-in (register new PowerFlex plug-in for v3.6).
6. Upgrade SDCs either manually or via the PowerFlex plug-in to v3.6 (one node at a time).

Upgrade the PowerFlex system

Now that SVM replacement is complete (if it was necessary), commence the upgrade of the PowerFlex system.

Prerequisites

Ensure that you have all the PowerFlex RHEL OL7 packages for the PowerFlex version currently installed in your system, and the latest PowerFlex v3.6 RHEL OLx packages.

Ensure that you know the media type (SSD or HDD) of the devices in the system's Storage Pools. You will need to supply this information during the upgrade process. This version of PowerFlex does not allow mixed device types in Storage Pools, so ensure that your Storage Pools are configured accordingly before you upgrade your system.

Ensure that you know the IP addresses and credentials for the PowerFlex Gateway, MDM and LIA.

About this task

Steps

1. In command line, upgrade the RPM of the PowerFlex Gateway to v3.6:

```
rpm -Uvh <RHELx_GATEWAY_PACKAGE>
```

2. From the PowerFlex plug-in, launch the SVM.
3. Log in as an admin user.
4. Using a web browser, go to the PowerFlex Gateway's IP address, and log into the PowerFlex Installer.
5. In the **Packages** view, upload the currently installed RHEL OL7 packages, and the target version v3.6 RHEL OLx packages.
6. In the **Maintain** view, enter the primary MDM's IP address, MDM Username (admin user), and MDM password. Enter the LIA authentication type and password. Click **Retrieve system topology**.
7. When you are prompted, approve all the certificates.
8. At the bottom of the window, click **Upgrade**.
9. When you are prompted to confirm the upgrade, click **Next**.
10. When you are prompted to select the device media type for each Storage Pool, select the appropriate type for each Storage Pool. To continue, click **Proceed**.
11. Retype the MDM password and select your desired settings for the upgrade.

    NOTE: SDC upgrade is performed using the PowerFlex plug-in, and should not be selected now.

12. Click **Upgrade**.
13. Go to the **Monitor** view to follow the progress of the upgrade and to move from one stage of the process to the next.
14. When the Query phase is complete, click **Start upload phase**.
15. When the Upload phase is complete, click **Start install phase**.
16. When the Install phase is complete, click **Mark operation completed**.
17. In command line, check system status:

```
scli --query_all
```

    In the returned output:
    - The MDM cluster state should be optimal.
    - All SDSs should be connected and running version v3.6.
    - All SDCs should be connected.
18. Proceed to the next task.

Upgrade the SDC on an ESXi host

You can manually upgrade the SDC on an ESXi host. If the ESXi also hosts an SDS component, this procedure must be performed one node at a time.

Steps

1. Upgrade the SDC offline bundle:

```
esxcli software vib install -d <full_path_to_offline_bundle>
```

    where *<full_path_to_offline_bundle>* is the full path to the `sdc-3.6-<build>.X-esx<version>.zip` file.

    NOTE: The SDC offline ESXi 6.0 bundle is the same for all ESXi 6.x versions.

2. Gracefully shut down the ESXi host and then return it to normal operation, as described in the "Reference" section of this guide.

    NOTE: The graceful reboot cycle must be performed one host at a time.

3. Ensure that the driver is loaded:

```
vmkload_mod -l | grep scini
```

Output, similar to the following, indicates that the driver is loaded:

```
scini 2 808
```

The SDC component is installed on the ESXi host.

Upgrade the CentOS operating system on SVMs

Use the PowerFlex Installer to upgrade the SVM's CentOS operating system to the latest security level of operating system security patching. This procedure only upgrades the operating system on the SVM. It does not upgrade PowerFlex.

NOTE: SVM operating system patching is applicable even if the SVM replacement procedure was performed. The latest patch bundle has the latest security patches that were released after the latest version of PowerFlex was officially launched (the refresh of the package is done on a monthly basis based on continuous security scans and input from customers).

Use the PowerFlex Installer to upgrade the CentOS operating system on multiple SVMs. This procedure is required when upgrading an ESXi-based PowerFlex system where SVMs are currently running on CentOS 7.5. (PowerFlex v3.0, or v3.0.0.x)

Ensure that you have the latest SVM operating system upgrade ZIP file. It is available from the support site. You can opt to download the latest file from Dell Support for ScaleIO Drivers.

The `patch_script` file that you require is part of the VMware complete software package (which is in the Complete SW download ZIP) and also as a separate artifact that is refreshed from time to time (usually once a month). The ZIP file name is: `SVM_OS_Patching_package_MMDDYYYY.zip`

Running scripts on hosts—overview

PowerFlex can be used to run user-provided scripts on servers hosting MDM or SDS components. This feature is supported on Linux-based (bare-metal or virtual) nodes only.

The PowerFlex Installer can be used to run a user-provided script on a node where PowerFlex is deployed. This feature can be used for any purpose external to the PowerFlex system, such as running a set of CLI commands, patching an operating system, and more. The feature allows the running of scripts in a safe manner, both from a security and a data integrity perspective.

PowerFlex Installer orchestrates the running of the script, ensuring that SDSs are placed in Maintenance Mode, to protect data during the process. In addition, parallel execution of scripts is only permitted on SDSs located in different Protection Domains. After the scripts have been run on an SDS, it exits Maintenance Mode.

Optionally, servers can be set to reboot after execution of the script. The process can also run a verification script either after the reboot, or after execution of the script, when no reboot is required.

Workflow overview (for reference purposes only)

The execution phase of this feature can be summarized as follows:

1. The system validates the following:
   - The patching script exists on the node after it was uploaded from the PowerFlex Gateway
   - No failed capacity exists
   - Sufficient spare capacity exists
   - The MDM cluster is in a valid state
2. Run the script on one host, using the following priorities:
   a. SDS only hosts, each time on a single SDS, unless the option `In parallel on different Protection Domains` is enabled
   b. Tiebreaker
   c. MDMs

   NOTE: The script will not run on a primary MDM. A switch-over MDM command will be run prior to running the script on a primary MDM. The script will not be run in parallel on multiple MDMs or tiebreakers.
3. SDS enters Maintenance Mode.
4. The script runs on the host.
5. The host reboots (if configured to do so).
6. The validation script runs on the node (if configured to do so).
7. SDS exits Maintenance Mode.

Run a script on one or more hosts—procedures

Run a script on one or multiple hosts where PowerFlex is deployed, using the PowerFlex Installer.

Prerequisites

- Prepare the script file to be run on one or more nodes hosting SDSs or MDMs, and optionally, a verification script to be run automatically after execution of the script. Specific file names are required for use of this feature. PowerFlex Installer can only upload scripts up to 500 MB in size. Name the files as follows:
  - Script: `patch_script`
  - Verification script: `verification_script`
- This feature is only supported on Linux-based (bare-metal or virtual) nodes.
- Ensure that you have the IP address and login credentials for the PowerFlex Gateway and the primary MDM.
- Ensure that you have the login credentials for the LIA.
- Ensure that you have enough disk space to run the script.

About this task

- This procedure is a long operation that cannot be rolled back automatically.
- Certificate approval messages may be displayed during the procedure. Approve all the certificates, and click **Retry failed** to complete the operation.

Steps

1. Configure the PowerFlex Installer to upload files from the PowerFlex Gateway node to all selected nodes.

   NOTE: Alternatively you can copy the script file to each node in the system under the file path `/opt/emc/scaleio/lia/bin/<folder>`. Follow the steps to make sure the script file runs on each of the nodes.

   a. Verify that all components in the PowerFlex system are upgraded to the latest version.
   b. On the PowerFlex Gateway node, go to `/opt/emc/scaleio/gateway/webapps/ROOT/WEB-INF/classes/gatewayUser.properties` file.
   c. Uncomment the following lines (verification script is optional):

   ```
   os.patching.is.upload.needed=true (Default is false)
   os.patching.patch.script.source.path=<enter the full path to the file>
   os.patching.verification.script.source.path=<enter the full path to the file>
   ```

   Example of PowerFlex Gateway on Linux:

   ```
   os.patching.is.upload.needed=true
   os.patching.patch.script.source.path=/opt/patch_script
   os.patching.verification.script.source.path=/opt/verification_script
   ```

   Example of PowerFlex Gateway on Windows:

   ```
   os.patching.is.upload.needed=true
   os.patching.patch.script.source.path=C:\\temp\\patch_script
   os.patching.verification.script.source.path=C:\\temp\\verification_patch
   ```

   d. Restart the PowerFlex Gateway service.

   For example, in ESXi:

   `systemctl stop scaleio-gateway.service` and `systemctl start scaleio-gateway.service`

   e. Copy the `patch_script` and `verification_script` files to the relevant folder on the PowerFlex Gateway node.
   f. In the Security settings of the files, allow read, write and execute permissions.
2. In your browser, navigate to the IP address of the PowerFlex Gateway, and log in.
3. Click the **Maintain** tab.
4. Enter the IP address and login credentials for the primary MDM, and for LIA.
5. At the bottom right of the screen, click **Retrieve system topology**.
   The system topology is displayed.
6. Click **System Logs & Analysis**, and select the **Run Script on Hosts** option.

The **Run script on hosts** dialog box is displayed.

7. Enter the MDM password again.
8. For **Running script on** options, select one from the following:

Option	Description
Entire System	Run the script on all MDM and SDS nodes in the system. If you choose this option, you can also choose whether to run the script at the same time on SDSs that belong to different Protection Domains. To do so, select the check box for **In parallel on different Protection Domains**.
Protection Domain	Run the script on MDM and SDS nodes in a single Protection Domain. Select the required Protection Domain from the drop-down list.
Fault Set	Run the script on MDM and SDS nodes in a single Fault Set. Select the required Fault Set from the drop-down list.
SDS	Run the script on a single SDS. Select the required SDS from the drop-down list.

9. For **Running configuration**, select the **Stop process on script failure** option, if desired.
   If problems occur, see the troubleshooting notes at the end of this procedure.
10. In the **Script time-out** box, enter the number of minutes that should elapse before the PowerFlex Installer stops waiting for a response about the running script, and prints a timeout message.
11. In the **Verification script** box, select one of the following:

Option	Action
Run a verification script after the script. If a reboot is performed, the verification script is executed after the reboot.	Select `Run`
Do not run a verification script after the script	Select `Do not run`

12. In the **Post script action** box, select one of the following:

Option	Action
Reboot the node after execution of the script	Select `Reboot`
Do not reboot the node after execution of the script	Select `Do not reboot`

13. Click **Run Script on Hosts**.
14. Click the **Monitor** tab, - validate state starts.
    During this phase, the system verifies certificates and builds an execution phase for the nodes list.
15. Click **Start** execute phase - the execute phase starts.
16. When the execute phase is complete, click **Mark operation completed**.
    Troubleshooting notes:

    If a failure in executing the patch_script occurs, it may be due to one of the following reasons. Rectify the problem and then try again.

    - Script timeout
    - Script file permissions
    - Syntax error in the script
    - Degraded capacity, or MDM cluster in degraded state
    - An SDS is already in Maintenance Mode

Results

Upgrade of the CentOS operating system on all SVMs except for the PowerFlex Gateway is now complete. Upgrade the PowerFlex Gateway using the steps described in "Deploy and replace the PowerFlex Gateway SVM operating system using the PowerFlex plug-in" in the *Upgrade Dell PowerFlex*.

Troubleshoot upload phase when running scripts on hosts

Overcome the PowerFlex Gateway not being able to upload files using the "Run script on host" feature.

About this task

If the "Upload" phase fails with a timeout error, check if the following appears in the scaleio.log file on the PowerFlex Gateway, and the LIA log file of the relevant node. If the error appears, follow the remediation steps listed below. This issue is caused by a lack of resources in the PowerFlex Gateway's operating system, and in the JAVA process sustaining the upload.

Typical scaleio.log file:

```
2020-07-30 11:35:28,333 [executor-23] INFO  c.e.s.s.d.i.c.UploadFileCommand - activating
serial upload force algorithm due to upload of package UploadFileCommand[9.20.25.23,
9.20.125.23] that has failed due to memory problem java.lang.OutOfMemoryError: Java heap
space
2020-07-30 11:35:28,333 [executor-23] ERROR c.e.s.s.d.i.c.UploadFileCommand - UNEXPECTED
Error java.lang.OutOfMemoryError: Java heap space executing command .UploadFileCommand
(abort) :
```

Typical LIA log file:

```
30/07 11:51:19.035533 7f9f98115db0:liaCmd_CallCmdWithTimer:00253: Executing...
                 Command: chmod 744 /opt/emc/scaleio/lia/bin/patch_script
                 Timeout: [0]Sec (if zero - timeout is default)
30/07 11:51:19.040428 7f9f98115db0:liaNet_RecvRequest:00427: UPLOAD PACKAGE END
30/07 12:19:55.027051 7f9f98198db0:netSec_Recv_CK:00847: ERROR: Failed to read data from
handle 0x7f9f981a3ce0 (ERROR)
30/07 12:19:55.027066 7f9f98198db0:netPath_RcvLoop_CK:01611: WARNING:  :: Connected Live
SERVER path 0x7f9f981dd950 of portal 0x7f9f981ff170 net 0xa3a3c0 socket 11 inflights 0
HS:0 socket receive failed, Status ERROR
30/07 12:19:55.027076 7f9f98198db0:netCon_AbortInflightRpcsUptoDummyNode:05869: Aborted
0 inflight RPC messages due to disconnect.
```

Steps

1. Check if the PowerFlex Gateway's operating system has sufficient memory: minimum 8 GiB, recommended 12 GiB.
   - In case of SVM this requires shutdown of the SVM to modify the memory
   - This can be done at any time as long as no gateway orchestration is running
   - If "Run Script on Host" is running abort the operation prior to shutdown of the OS
2. After adding memory we need to increase the Java settings:
   a. Using file editor (such as vi) open the file: /opt/emc/scaleio/gateway/bin/setenv.sh and modify the following line:
      From:

      ```
      CATALINA_OPTS="-Xms128m -Xmx2048m"
      ```

      To:

      ```
      CATALINA_OPTS="-Xms128m -Xmx4096m"
      ```

   b. Restart the gateway service:
      Linux:

      ```
      systemctl scaleio-gateway restart
      ```

      Windows: Restart the service from the **services.msc** window.
3. Log in to the PowerFlex Gateway, and run the "Run Script on Host" feature with upload capability again.

Upgrade the PowerFlex Gateway operating system

Upgrade the PowerFlex Gateway CentOS operating system with the latest security patches.

About this task

Upgrade the CentOS operating system on the SVM hosing the PowerFlex Gateway.

Steps

1. Copy the `patch_script` file to the SVM hosting the PowerFlex Gateway.
2. In the Security settings of the file, allow read, write, and execute permissions.
3. Run the file, using the command:

   ```
   ./patch_script
   ```

   The script will run on the SVM, and will display a message when the operation is complete.
4. Restart the SVM.

After upgrading an ESXi system

Tasks that can be performed after upgrading.

After the upgrade, you can perform many other tasks, including:

- Add virtual IP addresses, which are mandatory for PowerFlex plug-in setups from PowerFlex v2.0 and later. For more information, see *Configure and Customize Dell PowerFlex*.
- Migrate an RDM-based PowerFlex system to DirectPath-based. This option is described in the "Manual migration of an RDM-based PowerFlex system to DirectPath-based" section of this guide.
- Install RFcache, if necessary.

  NOTE: From v3.0 and later, the commands and workflow for adding RFcache are different from earlier versions. For more information, see the *Configure and Customize PowerFlex Guide* and the *PowerFlex CLI Reference Guide*.
- Take advantage of the new features added to this release.

  These options, and others, are described in the *Configure and Customize PowerFlex Guide*.

Miscellaneous topics for manual upgrades in ESXi environments

Miscellaneous topics for manual upgrades in ESXi environments

Upgrade the hardware version of a Storage VM

Use VMware tools to upgrade the hardware version of PowerFlex SVMs.

Prerequisites

- Upgrade VMware tools.

About this task

If the current hardware version is already the most recent version, the upgrade options in the VMware tools will not be available, and this procedure is not necessary.

Steps

1. Start the vSphere Web Client and log in to the vCenter Server
2. Power off the virtual machine.
3. Right-click the virtual machine and select the menu option to upgrade virtual hardware.
   In the vSphere Web Client, the option is **Compatibility** > **Upgrade VM Compatibility**.
4. Click **Yes** to continue.

Results

The virtual hardware is upgraded to the latest supported version.

Upgrade the SDC on an ESXi host

You can manually upgrade the SDC on an ESXi host. If the ESXi also hosts an SDS component, this procedure must be performed one node at a time.

Steps

1. Upgrade the SDC offline bundle:

   ```
   esxcli software vib install -d <full_path_to_offline_bundle>
   ```

   where *<full_path_to_offline_bundle>* is the full path to the `sdc-3.6-<build>.X-esx<version>.zip` file.

   NOTE: The SDC offline ESXi 6.0 bundle is the same for all ESXi 6.x versions.

2. Gracefully shut down the ESXi host and then return it to normal operation, as described in the "Reference" section of this guide.

   NOTE: The graceful reboot cycle must be performed one host at a time.

3. Ensure that the driver is loaded:

   ```
   vmkload_mod -l | grep scini
   ```

   Output, similar to the following, indicates that the driver is loaded:

   ```
   scini 2 808
   ```

   The SDC component is installed on the ESXi host.

Modifying SDC parameters

Optionally, modify SDC parameters such as the virtual IP address of an MDM cluster.

About this task

The modification causes an SDC to use a different MDM virtual IP address or to update the SDC performance settings and configuration.

Steps

1. Copy the drv_cfg tool to the node.
   The drv_cfg tool is not a part of the artifacts available on the support site. Contact Dell Technologies support to access this tool and the guidelines for how to use it.
2. If needed, update the SDC performance settings and the existing configuration.
   This step is needed only if you change the settings after an upgrade.

3. View the complete list of SDC performance settings for the scini driver:

```
esxcli system module parameters list -m scini
```

4. Update the scini settings:

```
esxcli system module parameters set -m scini -p "...."
```

CAUTION: You must specify all parameters in this command when using the -p option. Any parameter that existed, that is not specified now will be replaced with 0 or an empty string.

Example for modifying virtual IP address:

```
esxcli system module parameters set -m scini -p
"IoctlIniGuidStr=7ea77fc9-6125-44c1-9746-9d20053aed93
IoctlMdmIPStr=9.10.21.141,9.10.21.142,9.10.121.142,9.10.121.141"
```

5. You can us the following cat command to get the full string output. This enables you to add or modify only the parameters you need.

```
cat /etc/vmware/esx.conf |grep scini/
 /vmkernel/module/scini/options
= "IoctlIniGuidStr=7ea77fc9-6125-44c1-9746-9d20053aed93
IoctlMdmIPStr=9.201.112.254,9.201.111.254"
```

Copy/paste the info between the quotation marks (" ") and modify; otherwise, it will return additional values.

Results

Selected SDC parameters are modified.

Manual migration of an RDM-based PowerFlex system to DirectPath-based

On PowerEdge servers, use this procedure to manually migrate a PowerFlex system from RDM-based device management to DirectPath-based (passthrough) device management. IO is not interrupted when following the instructions provided.

Prerequisites

- The current PowerFlex system must be v2.6.x or later with RDM-based device management. If you are migrating systems that are earlier than v2.0.1.4, you must install the RAID controller tools before adding devices, as described in "Install RAID controller tools."
- You need to be able to access the vCenter.
- You need to be able to access the PowerFlex management clients.
- It is recommended to schedule this procedure during a maintenance window.

About this task

This manual procedure must be performed on every SDS node in the system. First, you enable the passthrough mode on the ESX node, then add PCI devices to the Storage VM (SVM) on that server. Then, repeat this procedure for the next SDS node.

Steps

1. Detach RDM devices from the SVM:
   a. Use the PowerFlex GUI or CLI to select any SDS in the system.
   b. Enter the SDS into maintenance mode.
   c. From the vCenter, power-off the SVM.
   d. From the vCenter, edit the SVM and remove all the attached RDM disk devices.
      Use the remove - not the remove and delete - option.
2. Enable passthrough on the ESX node:

a. From the vCenter **Configuration** > **Advanced settings** view, select **Enable passthrough**.
b. Select the controller (RAID, SAS, or HBA) that is managing the disks used by PowerFlex.

3. Configure the BIOS of the ESX node for passthrough mode:
   a. In the vCenter Host view, enter the node into maintenance mode.
   b. Once maintenance mode has started, reboot the node.

   The selected ESX node is configured for DirectPath/passthrough mode.
4. Add PCI devices to the SVM:
   a. After the ESX node reboot completes, remove the node from maintenance mode.
   b. Edit the SVM, and add PCI devices.
   c. Power on the SVM.
   Added PCI devices are now displayed in the SVM.
   d. From PowerFlex, remove the SDS from maintenance mode.
   e. Wait for rebuild and rebalance activities to complete.
5. After all rebuild and rebalance activities are finished on one SDS, repeat this entire procedure on every SDS, one-at-a-time.

Next steps

If you are migrating systems that are earlier than v2.0.1.4, you must install the RAID controller tools before adding devices, as described in "Install RAID controller tools."

You can now use the PowerFlex management clients to add more devices to your SDSs.

Install RAID controller tools

After migrating an RDM-based system to DirectPath, install the RAID controller tools to each SVM to enable adding SDS devices.

About this task

This procedure must be performed before using the vSphere plug-in to add devices. If the base PowerFlex system was v2.0.1.4 or later, you do not need to do this procedure.

Perform these steps on every SVM in the system.

Steps

1. Download and install StorCLI 1-15-05_StorCLI: Download and install StorCLI
2. Download and install PercCLI: PERCCLI Utility For All PERC9 Controllers
3. Download and install MegaCLI: Download and install MegaCLI
4. Repeat these steps for every SVM in the PowerFlex system.

Results

You can use the PowerFlex vSphere plug-in to add SDS devices.

SVM manual memory allocation

Manual memory allocation for SVMs is required in a number of scenarios. However, when using the plug-in for a clean deployment, SVM memory allocation is performed automatically.

About this task

The formula for calculating SVM memory requirements can be found in the *Getting to Know PowerFlex Guide*.

In the following cases, SVM memory allocation must be performed manually:

- Manual deployment on VMware.
- Extending an existing SVM with a new PowerFlex role/component, whether this is being done with the plug-in or manually.

  Workaround: Perform all the parts of step 1 and step 2 before extending the additional role/component on the SVM. Perform the steps on one SVM at a time.

- Changing the SDS performance profile, post deployment.

  Workaround: Perform all the parts of step 1, one SVM at a time.

Steps

1. For SVMs that are SDS-only, perform the following:
   a. Move the SDS to maintenance mode (MM).
   b. Shut down the SVM.
   c. Increase SVM memory, according to the formula below.
   d. Power up the SVM.
   e. Exit MM.
2. For SVMs that are MDM (primary, secondary, or TB, may contain SDS, also):
   a. Start with secondary MDM and TBs:
      i. Move the SDS to maintenance mode.
      ii. Shut down the SVM.
      iii. Increase SVM memory, according to the formula below.
      iv. Power up the SVM.
      v. Exit maintenance mode.
   b. Proceed with the primary MDM:
      i. Switch ownership, so the primary MDM is now a secondary MDM.
      ii. Move the SDS to maintenance mode.
      iii. Shut down the SVM.
      iv. Increase SVM memory to the recommended size.
      v. Power up the SVM.
      vi. Exit maintenance mode.

IV

Firmware Upgrade

Upgrade RAID controller or server firmware to the appropriate firmware version.

7

Upgrade of LSI RAID Controller Firmware and Driver

Upgrade the firmware and driver on LSI RAID controllers in VxRack Node 100 Series systems.

LSI RAID controller upgrade on ESXi

Upgrade LSI RAID controller firmware and driver on ESXi-based systems.

Perform the following tasks:

1. Preparation
2. Uploading firmware, driver and storcli installers
3. Upgrading the driver
4. Upgrading the firmware
5. Post-upgrade tasks

NOTE: This procedure applies to PowerFlex installations on ESXi nodes that are NOT using the DirectPath ability, that is, where the RAID controller driver runs on the ESXi node.

Preparation

Prepare your ESX-based system for the upgrade. Ensure that all relevant parties have been notified about these maintenance activities.

Prerequisites

Shut down, VMotion or evacuate all VMs from the host, because it will be restarted during the upgrade procedure. Ensure that this is done gracefully, as described in the Reference part of this guide. Perform the restart only when directed to do so in the procedure. If the servers hosting SDCs also host SDSs, perform the procedure one node at a time.

Steps

1. Put the SDS into maintenance mode, then power off the SVM.
2. Put the ESX host into maintenance mode.

Uploading firmware, driver and storcli installers

Upload the RAID controller firmware, driver, and storcli installers to the host. Save them in the location `/tmp`.

Steps

Download the following packages from the links listed below, unzip them, and save the required files in the location `/tmp` on your host.

a. storcli:

Download storcli

Save the file `vmware-esx-storcli-1.19.04.vib` on your host.

b. LSI_MR3 Driver:

For ESX 6.0: LSI MegaRAID Driver for VMware ESXi 6.0

Save the file `lsi-mr3-6.611.05.00-1OEM.600.0.0.2768847.x86_64` on your host.

For ESX 5.5: LSI MegaRAID Driver for VMware ESXi 5.5

Save the file `lsi-mr3-6.611.05.00-1OEM.550.0.0.1391871.x86_64.vib` on your host.

c. LSI 3108 RAID Controller Firmware:
LSI 3108 RAID Controller Firmware v4.650.00-6121

Save the file `mr3108fw.rom` on your host.

Upgrading the driver

Upgrade the LSI RAID controller driver.

Prerequisites

Ensure that you have uploaded the installer files to `/tmp` on the ESX host.

Steps

1. Install storcli using the `vib` file from the `Vmware-NDS` folder:

```
esxcli software vib install -f -v /tmp/vmware-esx-storcli-1.19.04.vib
```

2. Install the lsi_mr3 driver:

```
esxcli software vib install -v /tmp/lsi-
mr3-6.611.05.00-1OEM.600.0.0.2768847.x86_64.vib
```

3. Enable the lsi_mr3 driver:

```
esxcli system module set --enabled=true --module=lsi_mr3
```

4. If installed, disable the megaraid_sas driver:

```
esxcli system module set --enabled=false --module=megaraid_sas
```

NOTE: While this driver module upgrade requires a restart to take effect, it will be performed in a later step.

Upgrading the firmware

Upgrade the LSI RAID controller firmware.

Prerequisites

Ensure that you have uploaded the installer files to `/tmp` on the ESX host.

About this task

The storcli executable is located in `/opt/lsi/storcli/storcli`.

Steps

1. Pull and record the current controller configuration:

```
storcli /c0 show all
```

Note the current firmware and driver versions shown in the output (similar to the following example):

```
Version :
=======
Firmware Package Build = 24.7.0-0026
```

```
Firmware Version = 4.270.00-3972
Bios Version = 6.22.03.0_4.16.08.00_0x060B0200
NVDATA Version = 3.1411.00-0009
Ctrl-R Version = 5.08-0006
Preboot CLI Version = 01.07-05:#%0000
Boot Block Version = 3.06.00.00-0001
Driver Name = megaraid_sas
Driver Version = 06.807.10.00-rh1
```

2. Disable CacheCade and flush all cached data to disk:

```
storcli /c0 /vall set ssdcaching=off
```

NOTE: This command will fail for SSDs. As it only applies to HDDs, this is expected behavior, and is not a problem.

3. Allow time for the cached data to write completely to disk. When this is done, VDs will show "-" in the Cac column of `storcli /c0 show` output. Once the Cac column value shows as "-" for all VDs, proceed to the next step.
Typical output:

```
-------------------------------------------------------------
DG/VD TYPE  State Access Consist Cache Cac sCC       Size Name
-------------------------------------------------------------
1/1   RAID0 Optl  RW     Yes     RWBD  -   ON    1.090 TB VD01
2/2   RAID0 Optl  RW     Yes     RWBD  -   ON    1.090 TB VD02
3/3   RAID0 Optl  RW     Yes     RWBD  -   ON    1.090 TB VD03
```

4. Delete the CacheCade VD:

NOTE: Do not delete the CacheCade device until all cached data has been written to disk. Otherwise, data loss could occur.

```
storcli /c0 /v0 del cc
```

5. Enable booting with pinned cache:

```
storcli /c0 set bootwithpinnedcache=on
```

6. Upgrade the controller firmware:

```
storcli /c0 download file=/tmp/mr3108fw.rom
```

7. Restart the host to apply the firmware change:
   a. Restart the host.

   NOTE: This restart will apply all of the driver and firmware upgrades.

   b. Confirm that the controller firmware and driver have been updated to the target version:

   ```
   storcli /c0 show all | grep "Firmware Package Build\|Driver
   Name\|Driver Version"
   ```

   Output similar to the following should be displayed:

   ```
   Firmware Package Build = 24.15.0-0016
   Driver Name = lsi-mr3 (Megaraid_sas on Linux)
   Driver Version = 6.611.05.00 (6.811.02 on Linux)
   ```

8. Recreate the CacheCade device, then set its rdcache parameter:
   a. Type the command:

   ```
   storcli /c0 add vd cc type=raid0 drives=E:0-1 wb
   ```

b. Obtain the new VD ID in the Name column, in the row corresponding to the Cac0 type volume, using the command:

```
storcli /c0 show all
```

This step is required because the VD ID may have changed as a result of the upgrade.

c. Type the command:

```
storcli /c0 /v<x> set rdcache=nora
```

where <x> represents the VD ID obtained in the previous step.

For example, if the VD ID in the Name column is VD1, type the following:

```
storcli /c0 add vd cc type=raid0 drives=E:0-1 wb
storcli /c0 /v1 set rdcache=nora
```

9. Activate CacheCade for the HDDs:

```
storcli /c0 /vall set ssdcaching=on
```

Post-upgrade tasks

Gracefully restart the node and return it to normal operation, as described in the Reference part of this guide. Then perform the same procedures on the next node.

Steps

1. Gracefully restart the node and return it to normal operation, as described in the Reference part of this guide.
2. Wait until rebuild and rebalance activities are complete, and then proceed to the next node.

LSI RAID controller upgrade on Linux

Upgrade LSI RAID controller firmware and driver on Linux-based systems.

Perform the following tasks:

1. Preparation
2. Uploading firmware, driver and storcli installers
3. Upgrading the driver
4. Upgrading the firmware
5. Post-upgrade tasks

Preparation

Prepare your Linux-based system for the upgrade. Ensure that all relevant parties have been notified about these maintenance activities.

Steps

1. Put the SDS into maintenance mode.
2. Host OS preparation: shut down services and applications as needed, in preparation for maintenance.

Uploading firmware, driver and storcli installers

Upload the RAID controller firmware, driver, and storcli installers to the host. Save them in the location `/tmp`.

Steps

Download the following packages from the links listed below, unzip them, and save the required files in the location `/tmp` on your host.

a. storcli:

StorCLI Utility v1.19.04

Save the file `storcli-1.19.04-1.noarch.rpm` on your host.

b. Megaraid_sas driver:

MegaRAID Linux Driver v6.11-06.811.02.00-2

Save the file `kmod-megaraid_sas-06.811.02.00_xyz.x86_64.rpm` on your host (distribution/release specific; see installer archive).

c. LSI 3108 RAID Controller Firmware:

LSI 3108 RAID Controller Firmware v4.650.00-6121

Save the file `mr3108fw.rom` on your host.

Upgrading the driver

Upgrade the LSI RAID controller driver.

Prerequisites

Ensure that you have uploaded the installer files to `/tmp` on the Linux host.

If the SDS is running PowerFlex v2.x, ensure that it is in maintenance mode.

Steps

1. Upgrade the megaraid_sas 6.811.02 driver:

```
rpm -Uvh /var/tmp/kmod-megaraid_sas-06.811.02.00_el7.2-2.x86_64.rpm
```

NOTE: This installs the driver to the base kernel version for the relevant distribution/release.

2. Stop the SDS:

```
/opt/emc/scaleio/sds/bin/delete_service.sh
```

3. Unload the driver:

```
modprobe -r megaraid_sas
```

4. Copy the driver that you just installed to the base kernel version folder for your distribution/release, and overwrite the current driver file:

```
cp
/lib/modules/3.10.0-327.el7.x86_64/extra/megaraid_sas/megaraid_s
as.ko /lib/modules/`uname
-r`/kernel/drivers/scsi/megaraid/megaraid_sas.ko
```

```
cp: overwrite
"/lib/modules/3.10.0-327.4.5.el7.x86_64/kernel/drivers/scsi/mega
```

```
raid/megaraid_sas.ko"?
Y
```

5. Load the driver:

```
modprobe megaraid_sas
```

6. Confirm that the just-installed driver is loaded:

```
modinfo megaraid_sas |grep version
```

```
version: 06.811.02.00
```

Upgrading the firmware

Upgrade the LSI RAID controller firmware.

Prerequisites

Ensure that you have uploaded the installer files to `/tmp` on the Linux host.

About this task

The storcli executable is located in `/opt/MegaRAID/storcli/storcli64`.

Steps

1. Pull and record the current controller configuration:

```
storcli64 /c0 show all
```

Note the current firmware and driver versions shown in the output (similar to the following example):

```
Version :
=======
Firmware Package Build = 24.7.0-0026
Firmware Version = 4.270.00-3972
Bios Version = 6.22.03.0_4.16.08.00_0x060B0200
NVDATA Version = 3.1411.00-0009
Ctrl-R Version = 5.08-0006
Preboot CLI Version = 01.07-05:#%0000
Boot Block Version = 3.06.00.00-0001
Driver Name = megaraid_sas
Driver Version = 06.807.10.00-rh1
```

2. Disable CacheCade and flush all cached data to disk:

```
storcli64 /c0 /vall set ssdcaching=off
```

NOTE: This command will fail for SSDs. As it only applies to HDDs, this is expected behavior, and is not a problem.

3. Allow time for the cached data to write completely to disk. When this is done, VDs will show "-" in the Cac column of `storcli64 /c0 show` output. Once the Cac column value shows as "-" for all VDs, proceed to the next step.
Typical output:

```
---------------------------------------------------------------
DG/VD TYPE  State Access Consist Cache Cac sCC       Size Name
---------------------------------------------------------------
1/1   RAID0 Optl  RW     Yes     RWBD  -   ON    1.090 TB VD01
2/2   RAID0 Optl  RW     Yes     RWBD  -   ON    1.090 TB VD02
3/3   RAID0 Optl  RW     Yes     RWBD  -   ON    1.090 TB VD03
```

4. Delete the CacheCade VD:

   NOTE: Do not delete the CacheCade device until all cached data has been written to disk. Otherwise, data loss could occur.

   ```
   storcli64 /c0 /v0 del cc
   ```

5. Enable booting with pinned cache:

   ```
   storcli64 /c0 set bootwithpinnedcache=on
   ```

6. Upgrade the controller firmware:

   ```
   storcli64 /c0 download file=/tmp/mr3108fw.rom
   ```

7. Restart the host to apply the firmware change:
   a. Restart the host.

      NOTE: This restart will apply all of the driver and firmware upgrades.

   b. Confirm that the controller firmware and driver have been updated to the target version:

      ```
      storcli64 /c0 show all | grep "Firmware Package Build\|Driver
      Name\|Driver Version"
      ```

      Output similar to the following should be displayed:

      ```
      Firmware Package Build = 24.15.0-0016
      Driver Name = lsi-mr3 (Megaraid_sas on Linux)
      Driver Version = 6.611.05.00 (6.811.02 on Linux)
      ```

8. Recreate the CacheCade device, then set its rdcache parameter:
   a. Type the command:

      ```
      storcli64 /c0 add vd cc type=raid0 drives=E:0-1 wb
      ```

   b. Obtain the new VD ID in the Name column, in the row corresponding to the Cac0 type volume, using the command:

      ```
      storcli64 /c0 show all
      ```

      This step is required because the VD ID may have changed as a result of the upgrade.
   c. Type the command:

      ```
      storcli64 /c0 /v<x> set rdcache=nora
      ```

      where <x> represents the VD ID obtained in the previous step.

      For example, if the VD ID in the Name column is VD1, type the following:

      ```
      storcli64 /c0 add vd cc type=raid0 drives=E:0-1 wb
      storcli /c0 /v1 set rdcache=nora
      ```

9. Activate CacheCade for the HDDs:

   ```
   storcli64 /c0 /vall set ssdcaching=on
   ```

Post-upgrade tasks

Post-upgrade tasks for the SDS: gracefully restart the node and return it to normal operation, as described in the Reference part of this guide. Then perform the same procedures on the next node.

Steps

1. Gracefully restart the node and return it to normal operation, as described in the Reference part of this guide.
2. Wait until rebuild and rebalance activities are complete, and then proceed to the next node.

8

VxFlex Ready Node Server Firmware Upgrades

Upgrade the firmware and drivers on VxFlex Ready Node servers.

Single server firmware and BIOS upgrade to a specific version

The procedures in this section provide instructions for upgrading the Dell firmware and BIOS to a specific version on a single VxFlex Ready Node server.

Perform the procedures in the order in which they are presented.

Upgrade the Dell firmware and BIOS—PowerEdge R650/R6525/R750

When upgrading the firmware and BIOS, first upgrade the iDRAC firmware. After this is done, upgrade the BIOS and any other firmware that require a node reboot cycle.

Prerequisites

Ensure that you have the following:

- A management server or service laptop from which to connect to the Dell Lifecycle Controller
- The Dell Upgrade Package (DUP)
- The iDRAC password for your server

If you are performing an upgrade on a node that is already in operation, perform a graceful shutdown on the PowerFlex node before commencing the upgrade. Refer to the "Reference Topics" part of this guide for more information.

About this task

NOTE: Use the BIOS upgrade and reboot instructions for any non-iDRAC firmware that requires an upgrade.

Steps

1. In your Internet browser, go to https://*<iDRAC_IP_address>*, and log in.

   Use root as the username and password as the password.

   The iDRAC GUI is displayed.
2. Click the **Maintenance** tab.
3. Navigate to **System Update** tab, and upload the DUP.
4. Select the `.exe` file from the **Update Details** and select **Install**.
   A job queue is created and the system will restart as part of the next step. After the system reset, the DUP update gets staged through the Lifecycle Controller (LC).
5. Select **Apply at Next Reboot**.
6. If you are upgrading the BIOS or additional firmware, gracefully reboot the server, as described in "Gracefully shut down or reboot a node" in the "Post-upgrade and common tasks" section of this guide.

Upgrade the component firmware using

If required, upgrade the firmware version of the new component to the supported version using Dell .

Prerequisites

NOTE: To manually upgrade the firmware using the Dell Lifecycle Controller, skip to the next task.

Ensure that:

- You have network connectivity to the system.
- You know the IP address of the iDRAC port for the node.
- You know the username and password for accessing the iDRAC (default username/password are root/Scaleio123).

About this task

Steps

1. From your Internet browser, go to https://*<iDRAC_IP_address>*,

   where *<iDRAC_IP_address>* is the IP address of the node's iDRAC port.

2. Verify that the iDRAC consoleis for the correct system based on the Service Tag and the iDRAC IP address you entered in the URL, and then in the login window enter the user name and password and click **Login**.
3. If the node is not yet powered on, from the main dashboard, select **Power on System**, and then click **OK**.

   NOTE: System startup may take a few minutes due to system inventory refresh and drive scans on the node.

   If the node is already powered on, skip to Step 5.

4. Wait for the node to go through the bootup cycle. You can monitor the power up in the main dashboard's **Virtual Console** window.
5. From the iDRAC menu bar, select the **System** tab, and then click **Inventory** > **Firmware Inventory**.
6. Check the firmware version of the new component against the driver and support matrix. If a firmware upgrade is required, continue to the next step, or skip to the next task to upgrade the firmware manually.

   NOTE: If no firmware upgrade is required, proceed directly to the task immediately after the instructions for manually upgrading the firmware.

7. If necessary, configure a laptop or server with access to the iDRAC network.
8. Download the *PowerFlex custom node 15G OME Catalog* from Dell Technologies support site and follow the *Managing PowerFlex Custom Node Using Dell OpenManage Enterprise* catalog upgrade guides.

Next steps

For additional information, see *Managing PowerFlex Custom Node Using Dell OpenManage Enterprise* at Dell Technologies support site.

Upgrade the Dell firmware and BIOS—PowerEdge R640/R740xd/R840

When upgrading the firmware and BIOS, first upgrade the iDRAC firmware. After this is done, upgrade the BIOS and any other firmware that require a node reboot cycle.

Prerequisites

Ensure that you have the following:

- A management server or service laptop from which to connect to the Dell Lifecyle Controller
- The Dell Upgrade Package (DUP)
- The iDRAC password for your server

If you are performing an upgrade on a node that is already in operation, perform a graceful shutdown on the PowerFlex node before commencing the upgrade. Refer to the "Reference Topics" part of this guide for more information.

About this task

NOTE: Use the BIOS upgrade and reboot instructions for any non-iDRAC firmware that requires an upgrade.

Steps

1. In your Internet browser, go to https://<iDRAC_IP_address>, and log in.
   Use root as the username and password as the password.
   The iDRAC GUI is displayed.
2. Click the **Maintenance** tab.
3. Navigate to **System Update** tab, and upload the DUP.
4. Select the `.exe` file from the **Update Details** and select **Install**.
   A job queue is created and the system will restart as part of the next step. After the system reset, the DUP update gets staged through the Lifecycle Controller (LC).
5. Select **Apply at Next Reboot**.
6. If you are upgrading the BIOS or additional firmware, gracefully reboot the server, as described in "Gracefully shut down or reboot a node" in the "Post-upgrade and common tasks" section of this guide.

Upgrade the Dell firmware and BIOS—PowerEdge R630/R730xd

When upgrading the firmware and BIOS, first upgrade the iDRAC firmware. After this is done, upgrade the BIOS and any other firmware that require a node reboot cycle.

Prerequisites

- If you are performing an upgrade on a node that is already in operation, perform a graceful shutdown on the PowerFlex node before commencing the upgrade. Refer to the "Reference Topics" part of this guide for more information.
- You need a management server or service laptop from which to connect to the Dell Lifecyle Controller.

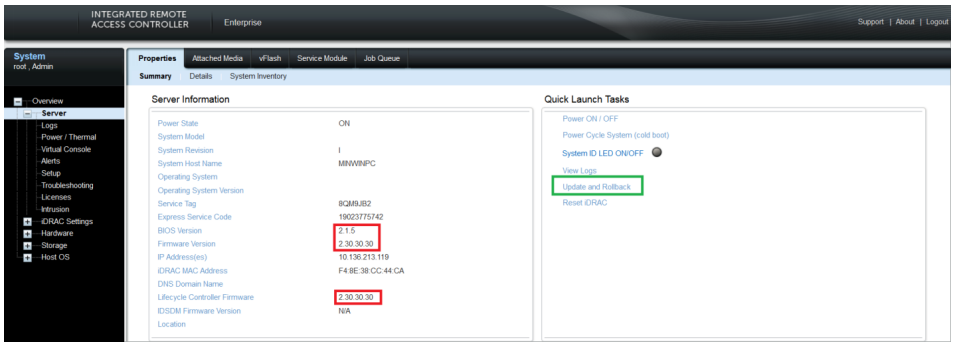
About this task

In the following example, the iDRAC firmware is upgraded from version 2.30.30.30 to 2.40.40.40, and the BIOS is upgraded from version 2.1.5 to 2.2.5.

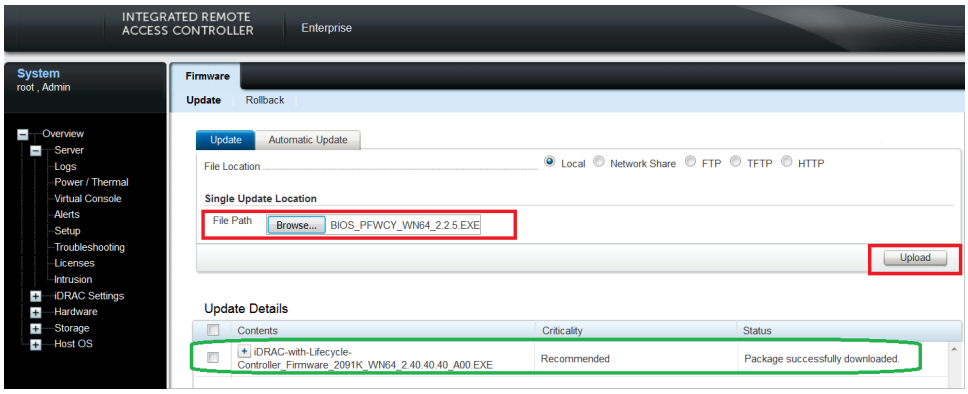
NOTE: Use the BIOS upgrade and reboot instructions for any non-iDRAC firmware that requires an upgrade.

Steps

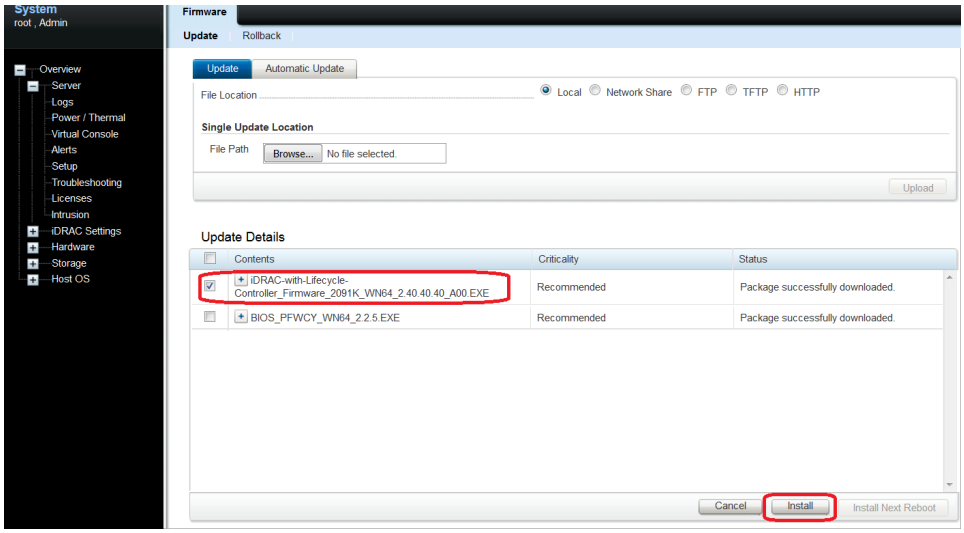
1. From your Internet browser, go to https://<iDRAC_IP_address> and then log in.
   Use root as the username and password as the password.
   The iDRAC **Server Overview** page appears. The **Properties** tab displays the currently installed BIOS and firmware versions.



2. From the **Quick Launch** tasks, click **Update and Rollback**.
   The **Firmware** page appears:



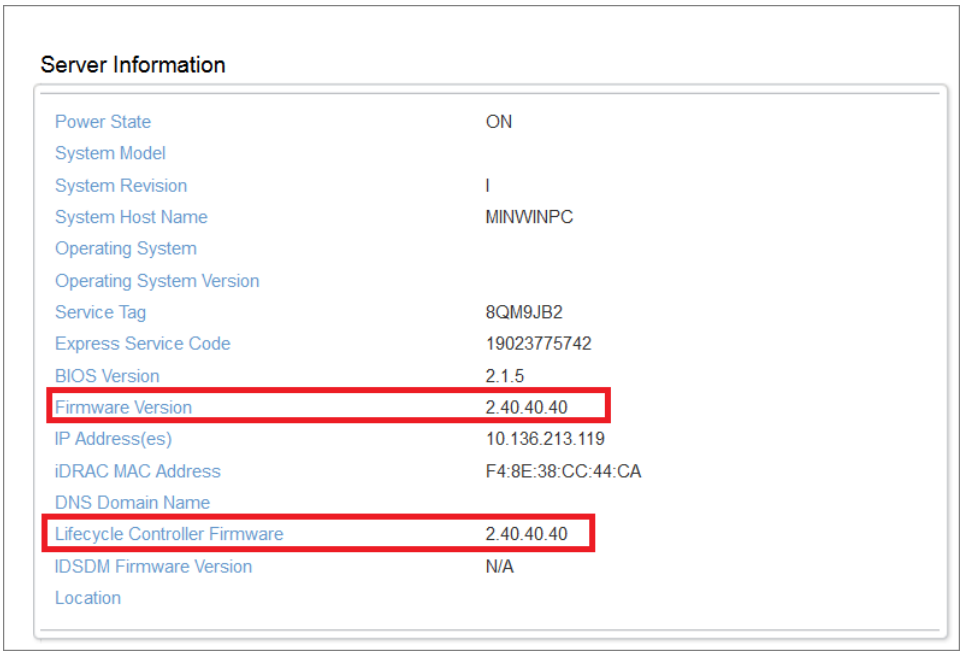
3. On the **Update** tab, click **Browse** and select the iDRAC firmware package—in this example, `iDRAC-with-Lifecycle-Controller_Firmware_ 2.40.40.40.exe`.
4. Click **Upload**.
5. Select the iDRAC firmware file and click **Install** to start the iDRAC firmware update:



The iDRAC is updated and the web session is reset.

6. Reconnect to the iDRAC and log in.
7. In the **Server Overview** page, verify that the iDRAC controller was updated.

   The following example shows that the firmware was updated to version 2.40.40.40:



NOTE: If you are only updating the iDRAC firmware, there is no need to reboot the node. Instead, you can move to the next node to update the iDAC firmware, repeating steps 1 - 7 for each additional node.

8. To upgrade the BIOS or other firmware versions, launch a KVM session to the node.

   NOTE: A reboot is required when upgrading the BIOS or other firmware, and the KVM session provides a window into the BIOS update process.

   The Java applet downloads.
9. Enable the pop-up so that the SVM session can open.
10. On the iDRAC **Update** tab, click **Browse** and select the correct BIOS upgrade file—in this example, `BIOS_PFWCY_WN64_2.2.5.exe`.
11. Click **Upload**.
12. After the file has uploaded, click **Install**.

    The BIOS upgrade is queued for installation.
13. Select **Apply at Next Reboot**.
14. Repeat steps 10 to 13 for each additional firmware you want to upgrade.
15. If you are upgrading the BIOS or additional firmware, gracefully reboot the server, as described in "Gracefully shut down or reboot a node" in the "Post-upgrade and common tasks" section of this guide.

Multiple firmware upgrades on one or more servers

The procedures in this section provide instructions for upgrading multiple Dell firmwares on one or more working VxFlex Ready Node servers.

Perform the procedures in the order in which they are presented, for every server requiring a firmware update.

Open the KVM console - R630/R730xd servers

Open the KVM console on a VxFlex Ready Node server.

Prerequisites

Ensure that:

- The system environment meets the prerequisites for using the KVM console.
- You know the IP address of the iDRAC port.

- You know the username and password for accessing the iDRAC (default username/password are root/calvin).

Steps

1. From your Internet browser, go to https://<*iDRAC_IP_address*>.
2. In the **DELL Console Login** window, type the user name and password, and click **Login**.
3. In the main KVM console window, select the **Server** node in the navigation pane.
   The **System Summary** dashboard is displayed.
4. In the **Virtual Console Preview** pane, click **Launch** to start a console session.
   A popup security warning screen is displayed.
5. Select **Accept** and click **Run**.
   The **Java Console** window opens and provides you with console access to the server.

   If this is the first time that you are opening a console, additional warning and confirmation prompts may appear. Click **OK** to grant approvals in these prompts.

   The subsequent steps depend on your browser selection and how it is configured. If downloads run automatically, the console window appears. If not, follow the instructions given in the next step to open the console window.
6. Depending on your browser, perform the following steps:

Browser	Steps
Firefox	a. Click **Launch Virtual Console**. The **"What should Firefox do with this file?"** window appears, with the **Open with Java(™) Web Start Launcher** option selected. b. Click **OK**. c. Scroll through the successive pop-up windows, then click **Run** to launch the console.
Google Chrome	a. Click **Show all downloads**. b. In the **Downloads** window, at the warning message, click **Keep**. c. In the **Downloads** list, click the downloaded file URL. d. At the warning message, click **Keep** or **Keep anyway**. e. Click the `jviewer.jnlp` file. If a security message appears, click **Run**. The management console window is displayed.
Internet Explorer	Click **Open** to run the `jviewer.jnlp` file.

Results

The KVM console is open and ready for use.

Open the KVM console - VxFlex Ready Node R640/R740xd servers

Open the KVM console on a VxFlex Ready Node server.

Prerequisites

Ensure that:

- The system environment meets the prerequisites for using the KVM console.
- You know the IP address of the iDRAC port.
- You know the username and password for accessing the iDRAC (default username/password are root/Scaleio123).

Steps

1. From your Internet browser, go to https://<*iDRAC_IP_address*>.
2. In the **DELL Console Login** window, type the user name and password, and click **Login**.
3. From the dashboard, click **Launch Virtual Console** to start a console session.
   A popup security warning screen is displayed.
4. Select **Accept** and click **Run**.

The **Java Console** window opens and provides you with console access to the server.

If this is the first time that you are opening a console, additional warning and confirmation prompts may appear. Click **OK** to grant approvals in these prompts.

The subsequent steps depend on your browser selection and how it is configured. If downloads run automatically, the console window appears. If not, follow the instructions given in the next step to open the console window.

5. Depending on your browser, perform the following steps:

Browser	Steps
Firefox	a. Click **Launch Virtual Console**. The **"What should Firefox do with this file?"** window appears, with the **Open with Java(™) Web Start Launcher** option selected. b. Click **OK**. c. Scroll through the successive pop-up windows, then click **Run** to launch the console.
Google Chrome	a. Click **Show all downloads**. b. In the **Downloads** window, at the warning message, click **Keep**. c. In the downloads list, click the downloaded file URL. d. At the warning message, click **Keep** or **Keep anyway**. e. Click the `jviewer.jnlp` file. If a security message appears, click **Run**. The management console window is displayed.
Internet Explorer	Click **Open** to run the `jviewer.jnlp` file.

Results

The KVM console is open and ready for use.

Updating the BIOS, firmware and settings

VxFlex Ready Node deployments require specific versions of drivers, BIOS, and firmware that have been qualified by Dell EMC. If the servers do not have the correct versions, you must update them.

A variety of factors can influence a mismatch between the required versions and the versions installed on the servers, such as firmware updates post server shipment, or a FRU replacement with a different firmware version than in the warehouse. For example, if you have replaced the server system board, the FRU's BIOs and iDRAC firmware versions will be different.

You are therefore required to verify that all server drivers, BIOS, and firmware meet the required versions, as published in the VxFlex Ready Node Driver and Firmware Matrix, located at Dell Support for ScaleIO Ready Node PowerEdge 13G, before deploying a server in the VxFlex Ready Node environment.

To perform any updates needed to meet VxFlex Ready Node requirements, use the VxFlex Ready Node Hardware Update Bootable ISO ("Hardware ISO"). The Hardware ISO is based on the Dell OpenManage Deployment Toolkit (DTK). The DTK provides a framework of tools necessary for the configuration of PowerEdge servers. For PowerFlex, a custom script has been injected, along with specific qualified BIOS/firmware update packages.

For additional information regarding the Hardware ISO, see the reference section.

Preparing the node for a graceful reboot

Prepare each node to reboot in a graceful fashion by first entering the node into maintenance mode.

Prerequisites

Ensure that you have admin rights for accessing the PowerFlex GUI. If necessary, the customer can give you the credentials.

Steps

1. Move all applications to a different node:
   - On an ESX node that is not a cluster member, and that is not configured for HA and DRS, migrate the VMs to another ESX.

- On a Linux or a Windows node, migrate the applications (or the VMs, if the node is running a hypervisor).

  NOTE: In non-hypervisor environments, ask the customer for assistance in moving applications from the node.

2. Log in to the PowerFlex GUI as an admin user.
3. In PowerFlex **Backend** view, select **By SDSs** table view.

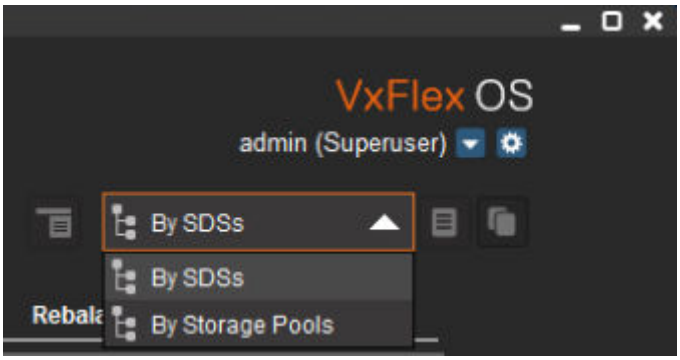


Figure 1. Table view toggle options

4. Right-click the SDS node you are rebooting, and select **Enter Maintenance Mode**.
5. In the **Enter maintenance mode** window, ensure that there are no errors, and then click **OK**.
6. When the operation finishes successfully, click **Close**.

   The node's IP address appears with a wrench next to it.

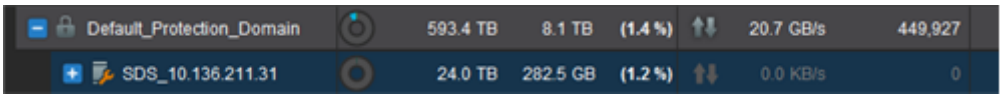


Figure 2. SDS displayed in maintenance mode

7. On an ESX node, enter maintenance mode:
   a. Log in to the vCenter via the vSphere Client or Web Client, and locate the relevant ESX IP address.
   b. Select the SVM, and from the **Basic Tasks** pane select **Shut down the virtual machine**.
   c. When the SVM is off, right-click the node and select **Enter Maintenance Mode**.
8. Obtain customer permission to reboot the node, and then gracefully reboot the node using the relevant API for the operating system.

   NOTE: On a Linux node, no checks are required for a graceful reboot after entering the SDS into maintenance mode.

Upgrading the firmware using the iDRAC virtual console

The iDRAC Virtual KVM console and Virtual CDROM features, provided by the iDRAC Enterprise license, eliminate the need for physical access to the VxFlex Ready Node servers. The Hardware ISO can be attached as a remote/Virtual CDROM image and configured to perform hardware updates to the VxFlex Ready Node firmware.

About this task

NOTE: For additional instructions regarding version updates following SATADOM replacement, see Update the SATADOM firmware (VxFlex Ready Node R630/R730xd only).

NOTE: This procedure is not relevant for PowerEdge R650 and R750. Only OpenManage Enterprise (OME) or manual firmware updates are supported for these models.

Steps

1. Configure a laptop with access to the iDRAC network.

2. Download the VxFlex Ready Node Hardware Update Bootable ISO from the VxFlex Ready Node page and make it accessible on the network share folder.
3. Connect to the iDRAC Virtual KVM console.

   For instructions, see the topic relevant for your server: Open the KVM console - R630/R730xd servers or Open the KVM console - VxFlex Ready Node R640/R740xd servers.
4. Attach the Hardware ISO to Virtual Media.
   a. From the **Virtual Media** menu, select **Connect Virtual Media**.
   b. From the **Virtual Media** menu, select **Map CD/DVD**.
   c. Click **Browse**, and select the Hardware ISO.
5. From the **Next Boot** menu, select **Virtual CD/DVD/ISO** and configure the Next Boot option so that the server will boot to the iDRAC Virtual CDROM.
6. Power-cycle the server using the appropriate **Power** menu option, and allow the server to boot to the virtual media.
7. Repeat steps 3 to 6 for each VxFlex Ready Node server that requires BIOS or firmware updates and configuration. All VxFlex Ready Node servers can be updated in parallel.
8. When the updates are completed, refresh the iDRAC browser screen, log in to the iDRAC, and re-launch the virtual console as needed.
9. Wait for the configuration and firmware updates to complete. The server console screen will indicate when the script is complete.

   WARNING: Do not reboot the VxFlex Ready Node server while the update process is being performed!

   The iDRAC will be reset several times during the update process. This causes the iDRAC virtual console viewer to close, virtual media to disconnect, and the iDRAC browser window to be unavailable for several minutes during each reset. The hardware update scripts will continue to run from RAM on the server.

   The update script will generate a log indicating whether each configuration and firmware flash is successful.
10. (Optional) Check each VxFlex Ready Node server's log for successful competition:
    a. After the update script completes, press Alt+F2 to access a user console, and then Enter to log in.
    b. Check the log contents for errors:

       ```
       less /bundleapplicationlogs/apply_components.log
       ```

       For more information, see Troubleshooting the Hardware ISO .
    c. Press q to exit the log viewer, and then Alt+F1 to access the original console screen.
11. Reboot the VxFlex Ready Node server and allow the update and configuration jobs to complete. Power-cycle the server using the appropriate **Power** menu option.

Return the node to operation

To return the node to operation, perform the following steps:

About this task

Steps

1. Wait for the node to power on.

   The OS will boot up for Windows and Linux operating systems. For Windows and Linux nodes, all PowerFlex processes will start up automatically.
2. For Linux and Windows nodes, skip to the next step. On an ESX node:
   a. From the vSphere Client, ensure that the node is displayed as on and connected in both **Hosts** and **Clusters** view.
   b. Right-click the node and select **Exit Maintenance Mode**.
   c. Expand the server and select the PowerFlex VM. If the SVM does not power on automatically, power it on manually.
3. After the node is up, perform the following checks in the PowerFlex GUI:
   a. In the **Alerts** view, make sure that no SDS disconnect message appears.
   b. If the node was an MDM cluster member, in the Dashboard **Management** tile, verify that the cluster is no longer degraded.
   c. In the **Frontend** tab > **SDCs** view, check the SDC to which the node IP is assigned, and make sure that it is connected.

4. In the PowerFlex GUI **Backend > Storage** view, in **By SDSs** table view, right-click the SDS and select **Exit Maintenance Mode**.
5. In the **Action** window, click **OK**.
6. Wait for the rebalance operations to finish.
   The node is now operational and application I/O can be started on the node. For ESX nodes, you can migrate VMs to the node.
7. For each VxFlex Ready Node server, after the updates are finalized, clear the iDRAC job queue using the iDRAC GUI:
   a. From your Internet browser, go to https://*<iDRAC_IP_address>*.
   b. In the **DELL Console Login** window, type these credentials:
      - username: root
      - password: calvin (for Dell-supplied nodes) or password (for EMC-supplied nodes)
   c. Click **Login**.
   d. In the iDRAC GUI navigation pane, select the **Server** node, and then select the **Job Queue** tab.
   e. Ensure that all jobs have completed successfully. Any job failures may require re-running the bootable ISO, or further troubleshooting.
   f. Select the checkbox for all of items in the **Job Queue** list, and then click **Delete**.

Next steps

Repeat for each node requiring a firmware upgrade.

Upgrade NVIDIA GPU firmware and drivers

NVIDIA GPU upgrade procedures require NVIDIA support.

Prerequisites

Both the GRID Enterprise software and SUMS licenses are available for purchase via Dell EMC and should have been purchased with the GPU-enabled VxFlex Ready Node node. For additional information, contact Customer Support.

Before contacting NVIDIA support, ensure that you have both a valid NVIDIA GRID Enterprise software license and the SUMS (Support, Update, and Maintenance Subscription) license. The SUMS license is required for you to open a support case with NVIDIA.

Steps

1. Upgrade the firmware and drivers according to the instructions from NVIDIA support.
2. When the upgrade is complete, gracefully reboot the server.

V

Reference Topics

Some procedures may be required during, before, or after upgrade scenarios.

9

Post-Upgrade and Other Common Tasks

There are a number of post-upgrade tasks and other optional tasks that can be performed after the upgrade is complete.

Shutdown or restart a node gracefully

When performing tasks on a node that require it to be shutdown or restarted, do so gracefully.

Operating system upgrades and patches, as well as other maintenance activities, like part replacement, require shutting down or rebooting a node.

Gracefully shut down or reboot a node

Prepare the server for a patching or maintenance operation (such as a part replacement) by entering the node into maintenance mode and shutting down/rebooting the node in a graceful fashion.

Prerequisites

Ensure that you have admin rights for accessing the PowerFlex GUI. If necessary, the customer can give you the credentials.

Steps

1. When shutting down/rebooting a node that is a primary MDM (manager), it is recommended that you manually switch MDM ownership to a different node:
   a. From the PowerFlex CLI (SCLI), run:

   ```
   scli --query_cluster
   ```

   NOTE: SCLI is installed as part of the MDM component and can be found in the following path:
   - ESXi (SVM) — `scli`
   - Linux — `scli`

   b. If the node's IP addresses are included in the `--query_cluster` output, the faulty node has a role of either MDM or tiebreaker (TB), in addition to its SDS role.

   If the node's IP address is located in the primary MDM role, a switch-over action is required.

   c. Switch MDM ownership to a different node:

   ```
   scli -switch_mdm_ownership (-new_master_mdm_id <ID> | --new_master_mdm_ip <IP> | --
   new_master_mdm_name <NAME>)
   ```

   The node remains in the cluster. The cluster will be in degraded mode after it is powered off, until the faulty component or patch operation in the node is fixed and the node is powered back on.

   d. Verify that the cluster status shows that the node is not the primary MDM anymore:

   ```
   scli --query_cluster
   ```

   Output similar to the following should appear, with the relevant node configuration and IP addresses for your deployment:

   ```
   Cluster:
       Mode: 5_node, State: Normal, Active: 5/5, Replicas: 3/3
       Virtual IP Addresses: 9.20.10.100, 9.20.110.100
   Master MDM:
       ID: 0x775afb2a65ef1f02
   ```

```
        IP Addresses: 9.20.10.104, 9.20.110.104, Management IP Addresses:
10.136.215.239, Port: 9011, Virtual IP interfaces: sio_d_1, sio_d_2
        Version: 2.0.13000
Slave MDMs:
    ID: 0x5b2e9f273b7af9b0
        IP Addresses: 9.20.10.105, 9.20.110.105, Management IP Addresses:
10.136.215.223, Port: 9011, Virtual IP interfaces: sio_d_1, sio_d_2
        Status: Normal, Version: 2.0.13000
    ID: 0x5828f65b15e778f1
        IP Addresses: 9.20.10.102, 9.20.110.102, Management IP Addresses:
10.136.215.232, Port: 9011, Virtual IP interfaces: sio_d_1, sio_d_2
        Status: Normal, Version: 2.0.13000
Tiebreakers:
    ID: 0x6618e0b804644ca4
        IP Addresses: 9.20.10.101, 9.20.110.101, Port: 9011
        Status: Normal, Version: 2.0.13000
    ID: 0x12534ccb3d28fee3
        IP Addresses: 9.20.10.103, 9.20.110.103, Port: 9011
        Status: Normal, Version: 2.0.13000
```

In the example output, the primary MDM IP addresses are:

```
IP Addresses: 9.20.10.104, 9.20.110.104, Management IP Addresses: 10.136.215.239
```

The secondary IP addresses are:

```
IP Addresses: 9.20.10.105, 9.20.110.105, Management IP Addresses: 10.136.215.223
IP Addresses: 9.20.10.102, 9.20.110.102, Management IP Addresses: 10.136.215.232
```

2. Move all applications to a different node:
   - On an ESXi node that is not a cluster member, and that is not configured for HA and DRS, migrate the VMs to another ESXi.
   - On a Linux node, migrate the applications (or the VMs, if the node is running a hypervisor).

     NOTE: In non-hypervisor environments, ask the customer for assistance in moving applications from the node.
3. Log in to the PowerFlex GUI as an admin user.
4. In the left pane, click **Configuration** > **SDSs**.
5. In the right pane, select the relevant SDS and click **More** > **Enter Maintenance Mode**.
6. In the **Enter SDS <name> into Maintenance Mode** dialog box, select one of the following options:
   - **Instant**
   - **Protected**
7. Click **Enter Maintenance Mode**.
8. In the confirmation message dialog box, click **Dismiss**.
9. On an ESXi node:
   a. Log in to the vCenter via the vSphere Web Client, and locate the relevant ESXi IP address.
   b. Select the SVM, and from the **Actions** > **Power** menu, select **Shut Down Guest OS**.
   c. When the SVM is off, right-click the ESXi node and select **Enter Maintenance Mode**.
10. If you are applying a patch:
    a. Run the patch.
    b. Reboot the node, if necessary.
11. For part replacement or to shut down/reboot a node:
    a. Obtain customer permission to shut down the node.
12. Gracefully shut down/reboot the node using the relevant API for the operating system.

    NOTE: On a Linux node, no checks are required for a graceful shutdown after entering the SDS into maintenance mode.

Return the node to operation

To return the node to operation, perform the following steps:

Steps

1. Power on the node, or if rebooting, wait for the node to start booting.
   The OS will boot up for Linux operating systems. For Linux nodes, all PowerFlex processes will start up automatically.
2. For an ESXi node, perform the following:
   a. From the vSphere Web Client, ensure that the node is displayed as on and connected in both **Hosts** and **Clusters** view.
   b. Right-click the node and select **Exit Maintenance Mode**.
   c. Expand the server and select the Storage VM (SVM). If the SVM does not power on automatically, power it on manually.
3. After the node is up, perform the following checks in the PowerFlex GUI:
   a. In the left pane, click **Alerts**. In the right pane, make sure that no SDS disconnect message appears.
   b. In the left pane, click **Dashboard**, and in the right pane, click **Health**. Verify that the SDCs are healthy, and if the node was an MDM cluster member, verify that the MDM cluster is no longer degraded.
4. In the left pane, click **Configuration** > **SDSs**.
5. In the right pane, select the relevant SDS and click **More** > **Exit Maintenance Mode**.
6. In the **Exit SDS <name> from Maintenance Mode** dialog box, click **Exit Maintenance Mode**.
7. In the confirmation message dialog box, click **Dismiss**.
8. Wait for the rebuild/rebalance operations to finish.
   The node is now operational and application I/O can be started on the node. For ESXi nodes, you can migrate VMs to the node.

Disabling IPMI

If you are not using IPMI over LAN for monitoring or management using third-party tools, disable IPMI on all nodes in the system in order to close a security vulnerability.

Disable IPMI on a R630/R730xd server

Use the following procedure to disable IPMI on PowerFlex R630/R730xd nodes.

About this task

You can run this procedure at any time post-deployment.

Prerequisites

Ensure that:

- You have network connectivity to the server.
- You know the IP address of the iDRAC port.
- You know the password for accessing iDRAC as root). If necessary, the customer can give you the credentials.

Steps

1. Open a new browser session and log in to iDRAC.
2. On the left menu, click **iDRAC settings** > **Network**.
3. On the **Network** page upper menu, click **IPMI Settings**.
   The page jumps to the **IPMI settings** area.
4. In the **Value** column, clear the **Enable IPMI Over LAN** check box, and then click **Apply**.
   The **Network** page refreshes.
5. On the upper menu, click **IPMI Settings** again, and then confirm that the **Enable IPMI Over LAN** check box is cleared .
6. Log out of iDRAC.
7. Repeat the above steps on every R630/R730xd node in the system.

Disable IPMI on PowerFlex R640/R740xd/R840 nodes

Use the following procedure to disable IPMI on PowerFlex R640/R740xd/R840 nodes.

About this task

You can run this procedure at any time post-deployment.

Prerequisites

Ensure that:

- You have network connectivity to the server.
- You know the IP address of the iDRAC port.
- You know the password for accessing iDRAC as root). If necessary, the customer can give you the credentials.

Steps

1. Open a new browser session and log in to iDRAC.
2. On the main menu, click **iDRAC settings** > **Connectivity**.
3. In the **Connectivity** tab, select **Network** > **IPMI Settings**.
   The IPMI configuration settings appear.
4. Change the **Enable IPMI Over LAN** option to **Disabled**, and then click **Apply**.
5. In the **Success** message, click **OK**.
6. Log out of iDRAC.
7. Repeat the above steps on every R640/R740xd/R840 node in the system.

Check NVDIMM capacity compatibility for future upgrades

Check your system for the recommended NVDIMM capacity, to prepare it in advance for future upgrades of Fine Granularity Storage Pools. You will not be able to upgrade your system to versions higher than v3.0.1 unless sufficient NVDIMMs are installed in the system.

Prerequisites

- Ensure that you have the tool for checking upgrade readiness: Click the following KB link which hosts the readiness checker script **PowerFlex Pre-Upgrade Readiness Checker Script**.
- Ensure that the server from which you will run the tool has the following:
  - Linux operating system (the tool cannot be used on Windows-based servers)
  - Access to the MDM
  - Python version 2.7 or higher must be installed
  - PowerFlex CLI is installed
- Ensure that you know the IP addresses of the MDMs, and know the user name and password required for accessing them.
- Ensure that your user role includes monitor privileges. Monitor privileges are used to query the system for SDSs that require additional NVDIMMs.

About this task

If your PowerFlex system does not contain enough NVDIMM capacity to support Fine Granularity Storage Pools in PowerFlex versions later than v3.0.1, your system will generate alerts to warn you about this issue. You will not be able to upgrade your system to versions higher than v3.0.1 until you have addressed this issue. You can check your system now to find out if it has enough NVDIMM capacity. If there is insufficient capacity, Dell EMC recommends that you contact your account manager now in order to prepare your hardware in advance for future software versions.

Steps

1. Save the `FlexOS_PreUpgrade_Readiness_Checker.py` tool on the MDM, or on a node with connectivity with the MDM.

2. In command line, run the command:

```
FlexOS_PreUpgrade_Readiness_Checker.py [-h] --username <USERNAME> [--password
<PASSWORD>] [--cli_executable <CLI_BIN>] [--use_nonsecure_communication] [--mdm_ip
<MDM_IP_ADDRESSES>]
```

where:

<USERNAME> is the user name used to query the MDM

<PASSWORD> is the user's password

<CLI_BIN> is the location of the PowerFlex CLI on the server

<MDM_IP_ADDRESSES> is a comma-separated list of MDM IP addresses

The tool will output one of the following messages:

- `Your system has insufficient NVDIMM capacity on SDS {XXX} to support future version upgrades. The required total NVDIMM capacity for the upgrade is {YYY}. Contact your account manager for more information.`

  (This message may appear several times, depending on the number of SDSs with insufficient NVDIMM capacity.)

- `System is ready for upgrade`

If the output is `System is ready for upgrade`, no further actions are required now. Your system contains enough NVDIMM capacity to support Fine Granularity storage acceleration in future software versions.

If the output is `Your system has insufficient NVDIMM capacity on SDS {XXX} to support future version upgrades. The required total NVDIMM capacity for the upgrade is {YYY}. Contact your account manager for more information.`, continue to the next step.

3. Use one of the following methods to determine which SDSs need more NVDIMM capacity.

Option	Procedure

CLI	a. Prepare a list of the SDSs in your system. You can use the `--query_all_sds` to collect this information. For example:
	`scli --query_all_sds`
	b. Using the CLI, run the following command for every SDS that uses NVDIMM acceleration for Fine Granularity storage:
	`scli --query_sds (--sds_id <ID> | --sds_name <NAME> | --sds_ip <IP>)`
	For example:
	`scli --query_sds --sds_name sds1948`
	Look for output similar to the following:
	`Acceleration device information (total 2 devices)`
	`1: Name: N/A Path: /dev/dax0.0 Original-path: /dev/dax0.0 ID dfef27800010000`
	`Acceleration Pool: accp1, Capacity: 15.7 GB (16052 MB), Used: 21.7 GB (22170 MB), State: Normal`
	c. If the `Used` value is greater than the `Capacity` value, as shown in the output example above, more NVDIMM capacity is required in order to upgrade the system.
	d. Make a note of all the SDSs where more NVDIMM capacity is required.

GUI	a. From the PowerFlex GUI, open the **Alerts** pane, and look for alerts for insufficient NVDIMM capacity for future version upgrades. b. Prepare a list of all the SDSs where these alerts occur.

4. Contact your account manager for sales and technical assistance.

Configure the SDS for NVDIMM health awareness

Optionally, configure the SDS to enable periodic monitoring of bad blocks in NVDIMM devices.

Prerequisites

Root access permissions are required for this procedure.

About this task

To enable this feature in the PowerFlex system, perform this procedure on every SDS that uses NVDIMM acceleration devices. This feature is disabled by default.

Steps

1. In command line, run the command:

```
echo "acc_dev_ndctl_error_detection_enable=1" >> /opt/emc/scaleio/sds/cfg/conf.txt
```

2. Restart the SDS:

```
/opt/emc/scaleio/sds/bin/delete_service.sh
```

```
/opt/emc/scaleio/sds/bin/create_service.sh
```

3. Wait until system rebuild/rebalance is complete, and then repeat steps 1 and 2 on the next SDS, until all SDSs are configured.

Extracting PowerFlex packages

Extract PowerFlex Ubuntu (DEB) or CoreOS (BSX) packages.

About this task

PowerFlex packages for Ubuntu and CoreOS must be extracted for some installation and upgrade scenarios.

Steps

1. Extract a package:

OS	Description

CoreOS	a. Untar the package:
	`tar -xvf EMC-ScaleIO-<component>-3.6-X.<build>.CoreOS.x86_64.tar`
	b. Extract the package to yield the BSX file:
	`./siob_extract EMC-ScaleIO-<component>-3.6-X.<build>.CoreOS.x86_64.siob`

Ubuntu	a. Untar the package:
	`tar -xvf EMC-ScaleIO-<component>-3.6-X.<build>.Ubuntu.x86_64.tar`
	b. Extract the package to yield the DEB file:
	`./siob_extract EMC-ScaleIO-<component>-3.6-X.<build>.Ubuntu.x86_64.siob`

2. Repeat the previous step for all packages that need to be extracted.

Upgrading the PowerFlex Gateway when a custom certificate is used

If a custom security certificate is used on the PowerFlex Gateway (Windows and Linux environments), you must save a copy of the file where certificates are stored (`*.keystore` file) and the `catalina.properties` file before you upgrade the PowerFlex Gateway. After the upgrade is complete, you must copy these files back to their original location.

The default file locations, per operating system, are:

Linux:

`/opt/emc/scaleio/gateway/conf/catalina.properties`

`/opt/emc/scaleio/gateway/conf/certificates/.keystore`

Windows (64 bit):

`C:\Program Files\EMC\ScaleIO\Gateway\conf\catalina.properties`

`C:\Program Files\EMC\ScaleIO\Gateway\conf\certificates\.keystore`

Troubleshoot PowerFlex plug-in registration issues

You can use logs to assist in troubleshooting problems that may occur during registration of the vSphere PowerFlex plug-in.

Search for issues in the log files. To find relevant log entries, search for `scaleio` in the log file. You can also use the troubleshooting procedure to completely remove and clean-up the PowerFlex plug-in files, and then reregister the PowerFlex plug-in. In PowerFlex plug-in downgrade scenarios, complete removal and clean-up procedures, followed by vCenter restart, are mandatory.

The vSphere web client (Virgo) logs are located in the following directories:

Table 9. vCenter operating system paths

vCenter	Operating system	Path to file
6.x	Windows	C:\ProgramData\VMware\vCenterServer\logs\vsphere-client\logs
	Linux	/var/log/vmware/vsphere-client/logs

Other relevant logs:

Table 10. vCenter operating system paths

vCenter	Operating system	Path to file
6.x	Windows	Plug-in logs and other files: C:\Users\vspherewebclientsvc\AppData\Roaming\VMware\scaleio Plug-in location: C:\ProgramData\VMware\vCenterServer\cfg\vsphere-client\vc-packages\vsphere-client-serenity
	Linux	Plug-in logs and other files: /etc/vmware/vsphere-client/vc-packages/scaleio Plug-in location: /etc/vmware/vsphere-client/vc-packages/vsphere-client-serenity

For more information about log collection, refer to *PowerFlex Log Collection Technical Notes*

Troubleshooting procedure for PowerFlex plug-in registration issues

To solve PowerFlex plug-in registration issues, perform a complete removal and cleanup, and then reregister the PowerFlex plug-in.

About this task

Completely clean your system of the PowerFlex plug-in in order to solve registration issues.

Steps

1. Close the existing PowerCLI sessions and open a new one (as administrator).
2. Using the PowerCLI script, unregister the PowerFlex plug-in.
3. Stop the vSphere web client service.
   On Linux, for example:

   ```
   service-control --stop vsphere-client
   ```

4. Delete the contents of the folder where the PowerFlex plug-in is stored.
5. Delete the contents of the PowerFlex plug-in's logs folder, or the folder itself.
   On Linux, for example: `/etc/vmware/vsphere-client/vc-packages/scaleio`
6. Clean the Virgo logs folder:
   On Linux, for example: `/var/log/vmware/vsphere-client/logs`
7. Start the vSphere web client service.
   For example, on Linux:

   ```
   service-control --start vsphere-client
   ```

8. Clear your web browser's cache/cookies (or use a different web browser).
9. Using the PS1 script in PowerCLI, register the PowerFlex plug-in via PowerCLI remember

   NOTE: Do not to press ENTER until you perform the log-in into your vSphere web client (that completes the registration).

10. Once you see the PowerFlex plug-in's icon in the vSphere web client, you can press ENTER in the PowerCLI session.

Unregister the PowerFlex plug-in

Remove a registered vSphere PowerFlex plug-in.

About this task

To remove the currently registered PowerFlex plug-in, perform the following:

Steps

1. Run the script to remove the PowerFlex plug-in:
   a. From the folder where you extracted the current PowerFlex plug-in ZIP file (for example: `EMC-ScaleIO-vSphere-plugin-installer-3.6-X.<build>.X.zip`), use PowerCLI to run the PowerFlex plug-in script (for example: `VxFlexOSPluginSetup.ps1`).
   b. Select option `2, Unregister PowerFlex plug-in`.
2. Enter the vCenter credentials and confirm the script actions.
3. Log out, then log back in to the vSphere web client.
4. The PowerFlex plug-in is no longer registered.

Register the PowerFlex plug-in manually

About this task

This topic describes an advanced way to use PowerCLI to register the vSphere PowerFlex plug-in on a vCenter.

The PowerFlex plug-in is provided as a ZIP file that can be downloaded by the vSphere web client servers in your environment. The ZIP file can be downloaded directly from the Online Support site, or, if the web servers do not have internet access, from a file server.

Before you begin, ensure that there is communication between the vSphere web client server and the web server storing the PowerFlex plug-in.

Steps

1. Upload the ZIP file to either an HTTP or an HTTPS server.

   If you are uploading the ZIP file to an HTTP server, perform the following:

   a. On the computer where the vSphere Web client is installed, locate the `webclient.properties` file.

   Table 11. vCenter operating system paths

vCenter	Operating system	Path to file
6.x	Windows 2008/2012	C:\ProgramData\VMware\vCenterServer\cfg\vsphere-client
	Linux	/etc/vmware/vsphere-client/

   b. Add the following line to the file:

   `allowHttp=true`

   c. Restart the VMware vSphere Web Client service.
2. Using PowerCLI for VMware, and set to **Run as administrator**, run `Set-ExecutionPolicy AllSigned`.
3. Close PowerCLI, and reopen it, set as **Run as administrator**.
4. Extract `EMC-ScaleIO-vSphere-plugin-installer-3.6-X.<build>.X.zip`
5. Upload `EMC-ScaleIO-vSphere-plugin-installer-3.6-X.<build>.X.zip` to the HTTP/HTTPS server.
6. From PowerCLI, run the following script: `VxFlexOSPluginSetup.ps1`
   a. Enter the vCenter name or IP address, user name, and password.
   b. For **Select Mode**, choose option `1, Register VxFlex OS plug-in`.
   c. For **Select Registration Mode**, choose `Advanced`.
   d. Enter the full URL of the plug-in ZIP file.

      If the ZIP file is located on a PowerFlex Gateway, enter the following URL: https://*<VxFlex_OS_GATEWAY_IP_ADDRESS>*.

      If you are manually placing the zip file on the server, place it in the `/opt/emc/scaleio/ams/webapps/ROOT/resources/scaleio_repository` folder.

   e. If necessary, accept the thumbprint.
7. Log out, and log back in to the vSphere web client.

   Example for HTTP server:

```
.\registerVxFlexOSPlugin.ps1 -vcenter 10.103.109.16
    -userUrl "http://10.76.60.14/sample/EMC-ScaleIO-vSphere-plugin-
installer-3.0-100.134.zip"
```

   Example for HTTPS server:

   .\registerVxFlexOSPlugin.ps1 -vcenter 10.103.109.16

```
-userUrl
"https://10.76.61.139/sample/EMC-ScaleIO-vSphere-plugin-installer-3.0-100.134.zip"
-thumbprint CA:66:49:D0:CE:D9:8C:A0:D0:93:E3:83:DE:59:25:5F:79:E1:53:B6
-adminEmail test.email@emc.com
```

The script registers the plug-in and the following message appears:

```
Registering VxFlexOS extension...
```

```
The extension was registered successfully
```

Store Tomcat's keystore password in the lockbox

After PowerFlex Gateway installation or upgrade, Tomcat's keystore password is written to file in plain text. Optionally, if your organization requires it, follow this procedure to store it in the lockbox.

About this task

The PowerFlex Gateway's keystore password is stored as plain text in a configuration file, and the keystore contains the PowerFlex Gateway's TLS private key. To prevent the private key from being extracted unencrypted, perform the following procedure.

NOTE: This procedure requires a PowerFlex Gateway restart.

Steps

1. On the PowerFlex Gateway, open the `catalina.properties` file in the following location:
   - Linux: `/opt/emc/scaleio/gateway/conf`
   - Windows: `C:\Program Files\EMC\Scaleio\Gateway\conf`
2. Copy the *keystore.password* value.
   For example, for `keystore.password = ubSF2TqZgQpmLZU2GMvyyAdGgIkt1w`, copy `ubSF2TqZgQpmLZU2GMvyyAdGgIkt1w`.
3. Log in as a root user to the system on which the PowerFlex Gateway is running.
4. Run the following FOSGWTool command in command line:

   NOTE: If a lockbox does not exist yet, add the *--create_default_lockbox* flag to the command in order to create one.

   NOTE: You can overwrite an existing value using the *--overwrite* flag.

   - Linux:

     ```
     /opt/emc/scaleio/gateway/bin/FOSGWTool.sh --store_value_in_lockbox --key
     lb.keystore.password --value <keystore.password value>
     ```

   - Windows:

     ```
     C:\Program Files\EMC\ScaleIO\Gateway\bin\FOSGWTool.bat
                --store_value_in_lockbox --key lb.keystore.password --value
     <keystore.password value>
     ```

5. Edit the `server.xml` file:
   a. Using a text editor, open the file:
      - Linux: `/opt/emc/scaleio/gateway/conf/server.xml`
      - Windows: `C:\Program Files\EMC\Scaleio\Gateway\conf\server.xml`
   b. Change the entry `keystorePass="${keystore.password}"` to `keystorePass="${lb.keystore.password}"`, and save the file.

   The *${keystore.password}* reference has been replaced with a reference to the key name that you used to store the password in the lockbox.
6. Remove the plain text password from the `catalina.properties` file.
   a. Using a text editor, open the `catalina.properties` file mentioned in Step 1.
   b. Delete the password value for *keystore.password*, and save the file.
7. Restart the PowerFlex Gateway service:
   - Windows: From the Windows **Services** window, restart the **EMC ScaleIO Gateway**.

- Linux: Type the following command:

```
etc/init.d/scaleio-gateway restart
```

Results

The plain text password has been moved from the Tomcat keystore to the lockbox.

Prepare lockbox files for Java upgrade

Before changing the Java version of a node that is running the PowerFlex Gateway or AMS of PowerFlex v2.5 or later, you must prepare lockbox-related files.

About this task

The lockbox in PowerFlex v2.5 saves files in the Java folder of the PowerFlex Gateway and the AMS. These files need to be saved before any Java version update, then pasted back into the folder.

Steps

1. From the `java\jre<VERSION>\lib\ext` (for Windows) or `java/jre<VERSION>/lib/ext` (for Linux) Java folder, copy these files to a different folder:
   - `commons-lang3-3.6.jar`
   - `cryptoj-6.2.3.jar`
2. Update the Java version.
3. Paste these files back to the folder from where you copied them.

Updating the SDC parameters

About this task

This section describes how to use the VMware plug-in to update the SDCs with system parameters that are needed to maintain SDC-MDM communication.

Steps

1. From the plug-in **Advanced tasks** menu, click **Update SDC parameters**, and follow instructions to complete that process.
2. Ensure that SDC parameters were updated by running this command on each ESX:

```
cat /etc/vmware/esx.conf |grep scini|grep -i mdm
```

Switching to secured authentication mode

Switch the authentication mode of PowerFlex from non-secured to secured mode.

This can be relevant when upgrading from a previous version that was running in non-secure mode, or after performing a clean install with the PowerFlex Installer and choosing non-secure authentication, via the advanced installation options.

Secure authentication can only be enabled on operating systems that support Open SSL v1.1 or later.

You need to update the following areas:

- Internal secure mode (between the MDM and SDS)
- External secure mode (between the management clients and MDM)
- LIA authentication (between the PowerFlex Installer and the LIA)

Proceed to the section that matches your operating system environment.

Physical Linux servers

About this task

This topic describes how to change the authentication mode from non-secure to secure on physical Linux servers.

Before beginning, verify the version of the OpenSSL on the server, by running this command:

```
rpm -qa | grep -i openssl
```

If the version is 1.1 or later, you can continue. If it is not, secure authentication cannot be enabled.

Steps

1. Log in to the SCLI, in a non-secure fashion:
   `scli --login --username <USER> --password <PASSWORD> --use_nonsecure_communication`
2. Change to internal secure mode, by running the following command:
   `scli --set_component_authentication_properties --use_authentication --use_nonsecure_communication`
3. Change to external secure mode, by running the following command:
   `scli --set_management_client_communication --enable_client_secure_communication --use_nonsecure_communication`
4. Enable secure LIA authentication, by performing the following:

   NOTE: This procedure can take quite some time, and there is no roll-back.

   a. Log in to the PowerFlex Installer (https://*<GW_Server_IP>*)
   where *<GW_Server_IP>* is the IP address of the server where you installed the PowerFlex Gateway package.
   b. Click the **Maintain** tab.
   c. From the **Maintenance operation** screen, perform the following:
      i. Type the primary MDM credentials.
      ii. Type the LIA password.
      iii. In **Advanced Options**, select **Allow non-secure communication with LIAs**.
      iv. Click **Retrieve system topology**.
      v. If a certificate approval is required, approve it.
      vi. Type the authentication credentials again.
      vii. Click **Retrieve system topology**.
      viii. Click **Security settings** and select **Enable LIAs security**.
      ix. Type the authentication credentials again.
      x. Confirm the request, and click **Enable LIAs security**.
      xi. If a certificate approval is required, approve it.
      xii. When complete, click **Mark operation completed**.

VMware servers

About this task

This topic describes how to change the authentication mode from non-secure to secure on VMware servers.

To enable secure authentication mode, perform the following:

Steps

1. Log in to the SCLI, in a non-secure fashion:
   `scli --login --username <USER> --password <PASSWORD> --use_nonsecure_communication`
2. Change the internal secure mode, by running the following command:
   `scli --set_component_authentication_properties --use_authentication --use_nonsecure_communication`

3. Change the external secure mode, by running the following command:
   `scli --set_management_client_communication --enable_client_secure_communication --use_nonsecure_communication`
4. Enable secure LIA authentication, by performing the following:
   a. Open the PowerFlex PowerFlex Installer:
      i. From the plug-in **VxFlex OS Systems** screen, right-click the system for which you want to perform this operation.
      ii. Select **Open PowerFlex Gateway**.

      NOTE: This operation can take some time.

      iii. From the **Welcome** screen, enter the default credentials.
   b. From the main menu, click **Maintain**.
   c. In **Advanced Options**, select **Allow non-secure communication with LIAs**.
      The **Maintenance operation** screen appears.
      i. Click **Security settings** and select **Enable LIAs security**.
      ii. Confirm the request, and click **Enable LIAs security**.
      iii. If a certificate approval is required, approve the certificates.
5. From the main plug-in window, click **PowerFlex systems**.
6. Right-click the system to register, and select **Reregister PowerFlex system**.
7. Enter the user name and password of the system, and click **OK**.
8. Approve the primary MDM certificate by clicking **Accept**.
9. From the plug-in **VxFlex OS Systems** screen, right-click the system and accept the secondary MDM certificate.

Configure Secure Remote Services or email notifications after upgrading

After the system upgrade, Secure Remote Services for remote support, or email notifications for alert monitoring, can be configured.

- If Secure Remote Services or email notifications were not enabled for your system prior to the upgrade, or if you want to change the settings, edit the `gatewayUser.properties` file, and restart the gateway service, as described in "Customer support" in the "Post-deployment tasks" section of the *Configure and Customize Dell PowerFlex*.
- If Secure Remote Services were enabled for your system prior to the upgrade, the feature will still be enabled after the upgrade. No further configurations are required.

  NOTE: The property which controls Secure Remote Services in the `gatewayUser.properties` file was changed, and is now called `features.notification_method`. The flag value required for enabling Secure Remote Services was changed from `true` to `srs`. These are changed automatically during the upgrade process.

- If running Secure Remote Services versions 3.40 or earlier upgrade to version 3.52 to enable continued connectivity and support for Secure Connect Gateway, as described in "Customer support" in the "Post-deployment tasks" section of the *Configure and Customize Dell PowerFlex*.

10

Upgrade a VMware-based PowerFlex system that contains CloudLink-encrypted devices

Upgrade a PowerFlex v2.1.0.4/v2.5/2.6.x VMware-based system that contains CloudLink-encrypted SDS devices to PowerFlex later versions.

Manually upgrade an ESXi-based PowerFlex system with CloudLink

For use cases where automated upgrades are not possible, follow the manual upgrade procedures described in this section. The use case covered in this section is for CloudLink-based systems.

Use the following workflow to upgrade the PowerFlex system:

1. Collect information about system components.
2. Ensure that all prerequisites are met.
3. Upgrade Java.
4. Upgrade the PowerFlex GUI.
5. Upgrade the PowerFlex Gateway.
6. Upgrade the PowerFlex system components:
   - MDM
   - LIA
   - SDS
   - RFcache (xcache)—optional
   - SDC

Collect and record PowerFlex information

Collect and record information about your system's components and CloudLink Center before commencing upgrade procedures.

About this task

Use the procedures that follow to collect and record the information in this table:

Table 12. Resource and credential information

Resource	IP address	Credentials
VMware vSphere production cluster	-	-
Jump server	-	-
PowerFlex root password for SVMs	-	-
PowerFlex admin password	-	-
PowerFlex Gateway	-	-
Production datacenter name	-	-
Production vCenter name	-	-
MDM1 (primary)	-	-
MDM2 (secondary 1)	-	-

Table 12. Resource and credential information (continued)

Resource	IP address	Credentials
MDM3 (secondary 2)	-	-
Tiebreaker1	-	-
Tiebreaker2	-	-
PowerFlex virtual IP subnets	-	-
SDS IP addresses (SVM and RHEL)	-	-
CloudLink Center	-	-

Steps

1. Gather the datacenter and vCenter names:
   a. Log on to the customer vCenter server.
   b. Click the **Home** icon at the top of the window and select **Hosts and Clusters**.
   c. Record the **datacenter** name.
   d. Record the **vCenter** name.
   The vCenter name is displayed above the datacenter name.
2. Gather and record additional PowerFlex cluster and component information:
   a. Use a console application to log in to the primary MDM SVM.
   b. Log in to the PowerFlex CLI:

```
scli --login --username admin --password <PASSWORD>
```

   c. Perform a query to obtain cluster information:

```
scli --query_cluster
```

   d. Gather SDS IP addresses:

```
scli --query_all_sds
```

   e. Gather SDC IP addresses:

```
scli --query_all_sdc
```

3. Verify the state of the PowerFlex system:
   a. Log in to the PowerFlex GUI using the primary MDM IP address and admin user credentials.
   b. On the **Dashboard**, verify the following:
      - There is no rebuild or rebalance running in the background (display should show 0.0 KB/s)
      - Adequate capacity exists for the upgrade
      - No SDSs or SDCs are disconnected
      - SDS devices are error-free
      - The MDM cluster, including tiebreakers, is not in Degraded state
      - There are no active alerts shown in the **Alerts** view
4. Verify available disk space:
   a. Use a console application to log in to each MDM and each tiebreaker.
   b. On each one, run the following command to display available disk space:

```
df -h
```

   Each MDM and tiebreaker must have at least 1.2 GB of available disk space.

Prerequisites for manual upgrades in ESXi environments

In addition to general prerequisites for all PowerFlex systems, ensure that the specific requirements for upgrading PowerFlex manually in ESXI environments are met.

ESXi-based nodes and VMware cluster prerequisites

Ensure that the ESXi-based nodes and the VMware cluster meet the prerequisites before you perform the upgrade.

- When upgrading a node based on ESXi 6.5U2 and higher
  1. Ensure that the following line does not appear in the `/etc/ssh/sshd_config` file of the ESXi host:`kexalgorithms diffie-hellman-group1-sha1, diffie-hellmangroup-exchange-sha1`

     If this line does appear, delete it. The line deletion does not relate to any configuration, and you do not need to restart either the ESXi-based node, nor the sshd service.

  2. Repeat the procedure for the rest of the nodes based on ESXi 6.5U2 and higher.
- The upgrade process requires entering the ESXi-based nodes into maintenance mode. To do so, ensure that Admission Control is not enabled on the VMware cluster:

Table 13. Resource and credential information

Resource	IP address	Credentials
VMware vSphere production cluster	-	-
Jump server	-	-
PowerFlex root password for SVMs	-	-
PowerFlex admin password	-	-
PowerFlex Gateway	-	-
Production datacenter name	-	-
Production vCenter name	-	-
MDM1 (primary)	-	-
MDM2 (secondary 1)	-	-
MDM3 (secondary 2)	-	-
Tiebreaker1	-	-
Tiebreaker2	-	-
PowerFlex virtual IP subnets	-	-
SDS IP addresses (SVM and RHEL)	-	-

Java upgrade prerequisites

Before changing the Java version of a node that is running PowerFlex Gateway, you must prepare lockbox-related files.

Prerequisites

Ensure that you have Oracle JRE 1.8 build 301 or OpenJDK JDK 8 build 302.b08 installed before upgrading your system.

About this task

NOTE: If you want to upgrade OpenJDK to version 11.0.12.0.7, do so after upgrade of the PowerFlex Gateway.

NOTE: DO NOT perform the following procedure when upgrading to OpenJDK to version 11.0.12.0.7.

The lockbox in PowerFlex uses files in the Java folder of the PowerFlex Gateway. These files need to be saved before any Java version update, then pasted back into the Java folder.

Steps

1. On the PowerFlex Gateway, in the `jre\lib\ext` (or `jre/lib/ext` for Windows) Java folder, copy these files to a different folder:
   - `commons-lang3-3.6.jar`
   - `cryptoj-6.2.3.jar`
2. Update the Java version.
3. Paste the files back to the folder from where you copied them.

PowerFlex GUI upgrade prerequisites

Before upgrading PowerFlex components from all previous versions, ensure that Oracle JRE 1.8 build 301, or OpenJDK JDK 8 build 302.b08, is installed on the server that will host the PowerFlex presentation server. The PowerFlex presentation server is an application that is used from PowerFlex v3.5 and higher. The VxFlex OS GUI client that was used in versions earlier than v3.5 will not be upgraded and is not supported going forward. If the PowerFlex presentation server and the PowerFlex Gateway will be installed on the same server, the PowerFlex Gateway must be installed before the PowerFlex presentation server. The PowerFlex presentation server listens on port 8443, and the PowerFlex Gateway checks that port 8443 is available, although it does not use this port..

NOTE: If you want to upgrade OpenJDK to version 11, do so after upgrade of the PowerFlex presentation server.

NOTE: If Java is already installed, check the existing version of Java by running the following command: `java -version`

For example, install the Oracle JRE file relevant for the server's operating system:

```
rpm -i /var/tmp/jre-8uXX-linux-x64.rpm
```

For information about downloading and installing OpenJDK, refer to: How to download and install prebuilt OpenJDK packages

Non-default certificate use before and after a PowerFlex Gateway upgrade

When using a non-default security certificate, you must perform certain actions before and after upgrading the PowerFlex Gateway.

About this task

If a non-default security certificate is used on the PowerFlex Gateway in Windows and Linux environments (for example, if the certificate is signed by the user organization CA), you must save a copy of the certificate and the catalina file before the upgrade, and restore them to their original location after the upgrade.

Steps

1. Before commencing the PowerFlex Gateway upgrade, locate the `*.keystore` file and the `catalina.properties` file, and save a copy of them in another location.

   The default file locations, per operating system, are:
   - Linux:
     - `/opt/emc/scaleio/gateway/conf/catalina.properties`
     - `/opt/emc/scaleio/gateway/conf/certificates/.keystore`
   - Windows (64 bit):
     - `C:\Program Files\EMC\ScaleIO\Gateway\conf\catalina.properties`
     - `C:\Program Files\EMC\ScaleIO\Gateway\conf\certificates\.keystore`
2. After the upgrade is complete, copy these files back to their original location.

SVM upgrade prerequisites

Verify the existence of OpenSSL packages and install them, if necessary. Increase SVM system disk size to 16 GB, as described in the *Reference* section of this guide.

About this task

NOTE: Do not run install commands if OpenSSL packages are already installed.

Steps

1. To verify if there are already OpenSSL packages on the SVM, run the command:

   ```
   rpm -qa |grep -i openssl
   ```

   Output similar to the following will be returned if OpenSSL exists:

   ```
   openssl1-1.0.1g-0.40.1
   libopenssl1_0_0-1.0.1g-0.40.1
   ```

2. If OpenSSL packages are not installed, run the following commands to install them.

   ```
   rpm -i libopenssl1_0_0-1.0.1g-0.40.1.x86_64.rpm
   ```

   ```
   rpm -i openssl1-1.0.1g-0.40.1.x86_64.rpm
   ```

3. Increase SVM system disk size of all SVMs to 16 GB, as described in the *Reference* section of this guide.

PowerFlex core component prerequisites

Prerequisites for core PowerFlex components

If you intend to use the CHAP component authentication feature after the upgrade, make the necessary preparations to ensure that you upgrade all system components now. Systems containing any components running PowerFlex versions lower than v3.5 do not support CHAP authentication.

NOTE: Upgrade to v3.5 sets the CHAP authentication feature to DISABLED, and passwords will be all-zero. To enable the feature, after the upgrade is complete on all system components, follow the post-installation procedures described in the *Configure and Customize Dell PowerFlex*.

Upgrade the PowerFlex GUI from versions earlier than v3.5

Use these steps to upgrade the PowerFlex GUI from versions earlier than v3.5.

Prerequisites

NOTE: From PowerFlex version 3.5, the PowerFlex GUI is a web-based application installed on a server. Therefore, the new PowerFlex GUI is a fresh installation known as the PowerFlex presentation server, and the GUI client from VxFlex OS versions of the product is neither upgraded nor supported going forward.

- Ensure that the workstation complies with the PowerFlex system requirements.
- Ensure that you have the PowerFlex presentation server installation file. It is included in the PowerFlex complete software package that you downloaded from Dell Technologies support site.

NOTE: It is not recommended to install the PowerFlex presentation server on the same server as the PowerFlex Gateway. If you must install them on the same server, the PowerFlex Gateway must be installed first. The PowerFlex presentation server listens on port 8443, and the PowerFlex Gateway installer checks that port 8443 is available, although it does not use this port.

Steps

1. Install the PowerFlex presentation server on a server in your system:
   - Linux:

     ```
     rpm -i EMC-ScaleIO-mgmt-server-3.6-X.<build>.noarch.rpm
     ```

   - Debian (run with administrator privileges):

     ```
     sudo dpkg -i emc-scaleio-mgmt-server_3.6-X.<build>_amd64.deb
     ```

2. Prior to using the PowerFlex GUI, link the PowerFlex presentation server to the PowerFlex system. For more information, about linking the system, login and use of the PowerFlex GUI, refer to the *Configure and Customize Dell PowerFlex*.

Upgrade the PowerFlex GUI from v3.5 or later

Upgrade the PowerFlex presentation server from v3.5 or later.

Prerequisites

Ensure that you have the package for the PowerFlex presentation server, and access to the node hosting the PowerFlex presentation server.

Steps

1. Save a copy of the PowerFlex presentation server package on the node where PowerFlex presentation server is installed.
2. On the node running the PowerFlex presentation server, run the following command:
   - Linux:

     ```
     rpm -Uvh EMC-ScaleIO-mgmt_server-3.6-X.<build>.noarch.rpm
     ```

   - Debian:

     ```
     dpkg -i emc-scaleio-mgmt_server-3.6-X.<build>.amd64.deb
     ```

3. On a Centos 8-based nodes only, when the installation script has finished, type Q.

Results

Upgrade of the PowerFlex presentation server is complete. Link the PowerFlex presentation server to the MDM cluster, as described in the *Configure and Customize Dell PowerFlex*. Existing credentials can be used, provided they belong to either admin or configure + monitor (LDAP or native) users.

Unregister the old PowerFlex system

Unregister the old PowerFlex system from the vSphere PowerFlex plug-in.

Prerequisites

MDM admin user and password credentials are required for this procedure.

About this task

From the web interface of the vCenter Client, unregister PowerFlex from within the PowerFlex plug-in.

Steps

1. In the **vSphere Web Client** main view, click the **DELL EMC PowerFlex** icon.
2. From the **Navigator** tree on the left, click the **PowerFlex Systems** option.
3. Select the relevant system name and right-click to open the options menu.
4. Select **Unregister PowerFlex system**, enter the MDM admin user and password, and confirm the action.

Unregister the current PowerFlex plug-in

Prerequisites

PowerCLI is required for unregistering the plug-in. Administrator privileges are required.

About this task

In this procedure, the currently installed vSphere PowerFlex plug-in will be removed in preparation for installing the latest version.

Steps

1. From the folder where you extracted the `EMC-ScaleIO-vSphere-plugin-installer-3.6-X.<build>.zip` file, use PowerCLI, as administrator, to run the `VxFlexOSPluginSetup.ps1` script.
2. Enter the vCenter credentials and confirm the script actions. If you are asked to trust an untrusted publisher, type **A** to continue.
3. When prompted to choose a mode, select option **2**, `Unregister ScaleIO plug-in`.
4. Follow the progress on the screen, and when prompted to do so, log out, then log back in to the vSphere web client.
5. Verify that the plug-in is no longer registered (VxFlex OS icon should no longer be visible).

   NOTE: If the icon has not disappeared, even though PowerCLI reports that it was successfully unregistered, reboot the vCenter.

6. When the process is complete, enter **4** to exit the plug-in script.

Register the new PowerFlex plug-in

Register the new vSphere PowerFlex plug-in.

Prerequisites

Before you begin, ensure that there is communication between the vSphere web client server (usually installed on the vCenter) and the web server storing the PowerFlex plug-in.

About this task

Register the new PowerFlex plug-in.

To use your own web server, see Register the PowerFlex plug-in manually.

Steps

1. Copy the following files to a designated host (preferably a vCenter):
   - `ScaleIOVM_3nics_3.6.XXX.XXX.ova`
   - `EMC-ScaleIO-vSphere-plugin-installer-3.6-X.<build>.zip`
2. Extract the contents of the zip file.
3. Using PowerCLI for VMware, set to Run as administrator, run the following script: `VxFlexOSPluginSetup.ps1`.
   a. Enter the vCenter name or IP address, user name, and password.
   b. For `Choose Mode`, select option **1, Register PowerFlex plugin**.
   c. Read the upgrade notice, and enter **y** to continue.
   d. For `Select Registration Mode`, select **Standard** (simplified, using embedded Tomcat).
      This step may take a few minutes.
   e. If necessary, accept the thumbprint.
4. Log in to the vSphere web client.
   a. If you are already logged in, log out, then log in again.
   b. Verify that the PowerFlex icon appears in the web client.
   c. Open the PowerFlex plug-in in the client, and ensure that it is the correct version.
      If the version did not update, clear your browser cookies, log out, then log in to the PowerFlex plug-in again.

5. In the PowerCLI window, press **Enter** to finish the PowerFlex plug-in download and return to the menu.

PowerFlex component upgrade overview

Workflow overview for manual upgrade of PowerFlex components

1. Ensure PowerFlex system is running the same version on all nodes.
2. Upgrade the PowerFlex Gateway to v3.6.
3. Upload the RHEL7 core packages for v3.6 to the PowerFlex Gateway.
4. Initiate the upgrade flow from the PowerFlex Gateway to v3.6.
5. Upgrade the vSphere PowerFlex plug-in (register new PowerFlex plug-in for v3.6).
6. Upgrade SDCs either manually or via the PowerFlex plug-in to v3.6 (one node at a time).

Upgrade the PowerFlex system

Start the upgrade of the PowerFlex system.

Prerequisites

Ensure that you have the current version and the target version of the PowerFlex RHEL OL7 packages.

Ensure that you know the media type (SSD or HDD) of the devices in the system's Storage Pools. You will need to supply this information during the upgrade process. This version of PowerFlex does not allow mixed device types in Storage Pools, so ensure that your Storage Pools are configured accordingly before you upgrade your system.

Ensure that you know the IP addresses and credentials for the PowerFlex Gateway, MDM and LIA.

Steps

1. In command line, upgrade the RPM of the PowerFlex Gateway to v3.6:

   ```
   rpm -Uvh <RHEL7_GATEWAY_PACKAGE>
   ```

2. From the PowerFlex plug-in, launch the PowerFlex Gateway SVM. Right-click the PowerFlex system, and select **Open PowerFlex Gateway**.
3. Log in as an admin user of the PowerFlex Gateway.
4. Using a web browser, go to the PowerFlex Gateway's IP address, and log into the PowerFlex Installer.
5. In the **Packages** view, upload the v2.6.1.1 RHEL OL7 packages and the v3.6 RHEL OL7 packages.
6. In the **Maintain** view, enter the primary MDM IP address, MDM Username (admin user), and MDM password. Enter the LIA authentication type and password. Click **Retrieve system topology**.
7. When you are prompted, approve all the certificates.
8. At the bottom of the window, click **Upgrade**, enter the MDM password, and select the following check boxes:
   - **Do not upgrade any SDC**
   - **Allow upgrade, even when already in upgrade state**
   - **Set High Performance profile for MDM, SDS and SDC**
9. Click **Upgrade**.
10. When a link to approve all certificates is displayed, approve all the certificates.
11. When you are prompted to select the device media type for each Storage Pool, select the appropriate type for each Storage Pool. To continue, click **Proceed**.
12. Retype the MDM password and select your desired settings for the upgrade:

    NOTE: SDC upgrade is performed using the PowerFlex plug-in, and should not be selected now.

    a. Select the check box **Do not upgrade any SDCs**
    b. Select the check box **Allow upgrade, even when already in upgrade state**
    c. Select the check box **Set High Performance profile for MDM, SDS and SDC**
13. Click **Upgrade**.
14. Go to the **Monitor** view to follow the progress of the upgrade and to move from one stage of the process to the next.
15. When the Query phase is complete, click **Start upload phase**.

16. When the Upload phase is complete, click **Start install phase**.
17. When the Install phase is complete, click **Mark operation completed**.
18. In command line, check system status:

```
scli --query_all
```

In the returned output:
- The MDM cluster state should be optimal.
- All SDSs should be connected and running version v3.6.
- All SDCs should be connected.

Upgrade the SDC on an ESXi host

You can manually upgrade the SDC on an ESXi host. If the ESXi also hosts an SDS component, this procedure must be performed one node at a time.

Steps

1. Upgrade the SDC offline bundle:

```
esxcli software vib install -d <full_path_to_offline_bundle>
```

where *<full_path_to_offline_bundle>* is the full path to the `sdc-3.6-<build>.X-esx<version>.zip` file.

NOTE: The SDC offline ESXi 6.0 bundle is the same for all ESXi 6.x versions.

2. Gracefully shut down the ESXi host and then return it to normal operation, as described in the "Reference" section of this guide.

NOTE: The graceful reboot cycle must be performed one host at a time.

3. Ensure that the driver is loaded:

```
vmkload_mod -l | grep scini
```

Output, similar to the following, indicates that the driver is loaded:

```
scini 2 808
```

The SDC component is installed on the ESXi host.

Upgrade the CentOS operating system on SVMs

Use the PowerFlex Installer to upgrade the SVM's CentOS operating system to the latest security level of operating system security patching. This procedure only upgrades the operating system on the SVM. It does not upgrade PowerFlex.

NOTE: SVM operating system patching is applicable even if the SVM replacement procedure was performed. The latest patch bundle has the latest security patches that were released after the latest version of PowerFlex was officially launched (the refresh of the package is done on a monthly basis based on continuous security scans and input from customers).

Use the PowerFlex Installer to upgrade the CentOS operating system on multiple SVMs. This procedure is required when upgrading an ESXi-based PowerFlex system where SVMs are currently running on CentOS 7.5. (PowerFlex v3.0, or v3.0.0.x)

Ensure that you have the latest SVM operating system upgrade ZIP file. It is available from the support site. You can opt to download the latest file from Dell Support for ScaleIO Drivers.

The `patch_script` file that you require is part of the VMware complete software package (which is in the Complete SW download ZIP) and also as a separate artifact that is refreshed from time to time (usually once a month). The ZIP file name is: `SVM_OS_Patching_package_MMDDYYYY.zip`

Running scripts on hosts—overview

PowerFlex can be used to run user-provided scripts on servers hosting MDM or SDS components. This feature is supported on Linux-based (bare-metal or virtual) nodes only.

The PowerFlex Installer can be used to run a user-provided script on a node where PowerFlex is deployed. This feature can be used for any purpose external to the PowerFlex system, such as running a set of CLI commands, patching an operating system, and more. The feature allows the running of scripts in a safe manner, both from a security and a data integrity perspective.

PowerFlex Installer orchestrates the running of the script, ensuring that SDSs are placed in Maintenance Mode, to protect data during the process. In addition, parallel execution of scripts is only permitted on SDSs located in different Protection Domains. After the scripts have been run on an SDS, it exits Maintenance Mode.

Optionally, servers can be set to reboot after execution of the script. The process can also run a verification script either after the reboot, or after execution of the script, when no reboot is required.

Workflow overview (for reference purposes only)

The execution phase of this feature can be summarized as follows:

1. The system validates the following:
   - The patching script exists on the node after it was uploaded from the PowerFlex Gateway
   - No failed capacity exists
   - Sufficient spare capacity exists
   - The MDM cluster is in a valid state
2. Run the script on one host, using the following priorities:
   a. SDS only hosts, each time on a single SDS, unless the option `In parallel on different Protection Domains` is enabled
   b. Tiebreaker
   c. MDMs

   NOTE: The script will not run on a primary MDM. A switch-over MDM command will be run prior to running the script on a primary MDM. The script will not be run in parallel on multiple MDMs or tiebreakers.
3. SDS enters Maintenance Mode.
4. The script runs on the host.
5. The host reboots (if configured to do so).
6. The validation script runs on the node (if configured to do so).
7. SDS exits Maintenance Mode.

Run a script on one or more hosts—procedures

Run a script on one or multiple hosts where PowerFlex is deployed, using the PowerFlex Installer.

Prerequisites

- Prepare the script file to be run on one or more nodes hosting SDSs or MDMs, and optionally, a verification script to be run automatically after execution of the script. Specific file names are required for use of this feature. PowerFlex Installer can only upload scripts up to 500 MB in size. Name the files as follows:
  - Script: `patch_script`
  - Verification script: `verification_script`
- This feature is only supported on Linux-based (bare-metal or virtual) nodes.
- Ensure that you have the IP address and login credentials for the PowerFlex Gateway and the primary MDM.
- Ensure that you have the login credentials for the LIA.
- Ensure that you have enough disk space to run the script.

About this task

- This procedure is a long operation that cannot be rolled back automatically.
- Certificate approval messages may be displayed during the procedure. Approve all the certificates, and click **Retry failed** to complete the operation.

Steps

1. Configure the PowerFlex Installer to upload files from the PowerFlex Gateway node to all selected nodes.

   NOTE: Alternatively you can copy the script file to each node in the system under the file path `/opt/emc/scaleio/lia/bin/<folder>`. Follow the steps to make sure the script file runs on each of the nodes.

   a. Verify that all components in the PowerFlex system are upgraded to the latest version.
   b. On the PowerFlex Gateway node, go to `/opt/emc/scaleio/gateway/webapps/ROOT/WEB-INF/classes/gatewayUser.properties` file.
   c. Uncomment the following lines (verification script is optional):

   ```
   os.patching.is.upload.needed=true (Default is false)
   os.patching.patch.script.source.path=<enter the full path to the file>
   os.patching.verification.script.source.path=<enter the full path to the file>
   ```

   Example of PowerFlex Gateway on Linux:

   ```
   os.patching.is.upload.needed=true
   os.patching.patch.script.source.path=/opt/patch_script
   os.patching.verification.script.source.path=/opt/verification_script
   ```

   Example of PowerFlex Gateway on Windows:

   ```
   os.patching.is.upload.needed=true
   os.patching.patch.script.source.path=C:\\temp\\patch_script
   os.patching.verification.script.source.path=C:\\temp\\verification_patch
   ```

   d. Restart the PowerFlex Gateway service.

   For example, in ESXi:

   `systemctl stop scaleio-gateway.service` and `systemctl start scaleio-gateway.service`

   e. Copy the `patch_script` and `verification_script` files to the relevant folder on the PowerFlex Gateway node.
   f. In the Security settings of the files, allow read, write and execute permissions.

2. In your browser, navigate to the IP address of the PowerFlex Gateway, and log in.
3. Click the **Maintain** tab.
4. Enter the IP address and login credentials for the primary MDM, and for LIA.
5. At the bottom right of the screen, click **Retrieve system topology**.
   The system topology is displayed.
6. Click **System Logs & Analysis**, and select the **Run Script on Hosts** option.
   The **Run script on hosts** dialog box is displayed.
7. Enter the MDM password again.
8. For **Running script on** options, select one from the following:

Option	Description
Entire System	Run the script on all MDM and SDS nodes in the system. If you choose this option, you can also choose whether to run the script at the same time on SDSs that belong to different Protection Domains. To do so, select the check box for **In parallel on different Protection Domains**.
Protection Domain	Run the script on MDM and SDS nodes in a single Protection Domain. Select the required Protection Domain from the drop-down list.
Fault Set	Run the script on MDM and SDS nodes in a single Fault Set. Select the required Fault Set from the drop-down list.
SDS	Run the script on a single SDS. Select the required SDS from the drop-down list.

9. For **Running configuration**, select the **Stop process on script failure** option, if desired.
   If problems occur, see the troubleshooting notes at the end of this procedure.
10. In the **Script time-out** box, enter the number of minutes that should elapse before the PowerFlex Installer stops waiting for a response about the running script, and prints a timeout message.

11. In the **Verification script** box, select one of the following:

Option	Action
Run a verification script after the script. If a reboot is performed, the verification script is executed after the reboot.	Select `Run`
Do not run a verification script after the script	Select `Do not run`

12. In the **Post script action** box, select one of the following:

Option	Action
Reboot the node after execution of the script	Select `Reboot`
Do not reboot the node after execution of the script	Select `Do not reboot`

13. Click **Run Script on Hosts**.
14. Click the **Monitor** tab, - validate state starts.
    During this phase, the system verifies certificates and builds an execution phase for the nodes list.
15. Click **Start** execute phase - the execute phase starts.
16. When the execute phase is complete, click **Mark operation completed**.

    Troubleshooting notes:

    If a failure in executing the patch_script occurs, it may be due to one of the following reasons. Rectify the problem and then try again.

    - Script timeout
    - Script file permissions
    - Syntax error in the script
    - Degraded capacity, or MDM cluster in degraded state
    - An SDS is already in Maintenance Mode

Results

Upgrade of the CentOS operating system on all SVMs except for the PowerFlex Gateway is now complete. Upgrade the PowerFlex Gateway using the steps described in "Deploy and replace the PowerFlex Gateway SVM operating system using the PowerFlex plug-in" in the *Upgrade Dell PowerFlex*.

Troubleshoot upload phase when running scripts on hosts

Overcome the PowerFlex Gateway not being able to upload files using the "Run script on host" feature.

About this task

If the "Upload" phase fails with a timeout error, check if the following appears in the scaleio.log file on the PowerFlex Gateway, and the LIA log file of the relevant node. If the error appears, follow the remediation steps listed below. This issue is caused by a lack of resources in the PowerFlex Gateway's operating system, and in the JAVA process sustaining the upload.

Typical scaleio.log file:

```
2020-07-30 11:35:28,333 [executor-23] INFO  c.e.s.s.d.i.c.UploadFileCommand - activating
serial upload force algorithm due to upload of package UploadFileCommand[9.20.25.23,
9.20.125.23] that has failed due to memory problem java.lang.OutOfMemoryError: Java heap
space
2020-07-30 11:35:28,333 [executor-23] ERROR c.e.s.s.d.i.c.UploadFileCommand - UNEXPECTED
Error java.lang.OutOfMemoryError: Java heap space executing command .UploadFileCommand
(abort) :
```

Typical LIA log file:

```
30/07 11:51:19.035533 7f9f98115db0:liaCmd_CallCmdWithTimer:00253: Executing...
                 Command: chmod 744 /opt/emc/scaleio/lia/bin/patch_script
                 Timeout: [0]Sec (if zero - timeout is default)
30/07 11:51:19.040428 7f9f98115db0:liaNet_RecvRequest:00427: UPLOAD PACKAGE END
30/07 12:19:55.027051 7f9f98198db0:netSec_Recv_CK:00847: ERROR: Failed to read data from
```

```
handle 0x7f9f981a3ce0 (ERROR)
30/07 12:19:55.027066 7f9f98198db0:netPath_RcvLoop_CK:01611: WARNING:  :: Connected Live
SERVER path 0x7f9f981dd950 of portal 0x7f9f981ff170 net 0xa3a3c0 socket 11 inflights 0
HS:0 socket receive failed, Status ERROR
30/07 12:19:55.027076 7f9f98198db0:netCon_AbortInflightRpcsUptoDummyNode:05869: Aborted
0 inflight RPC messages due to disconnect.
```

Steps

1. Check if the PowerFlex Gateway's operating system has sufficient memory: minimum 8 GiB, recommended 12 GiB.
   - In case of SVM this requires shutdown of the SVM to modify the memory
   - This can be done at any time as long as no gateway orchestration is running
   - If "Run Script on Host" is running abort the operation prior to shutdown of the OS
2. After adding memory we need to increase the Java settings:
   a. Using file editor (such as vi) open the file: /opt/emc/scaleio/gateway/bin/setenv.sh and modify the following line:
   From:
   ```
   CATALINA_OPTS="-Xms128m -Xmx2048m"
   ```
   To:
   ```
   CATALINA_OPTS="-Xms128m -Xmx4096m"
   ```
   b. Restart the gateway service:
   Linux:
   ```
   systemctl scaleio-gateway restart
   ```
   Windows: Restart the service from the **services.msc** window.
3. Log in to the PowerFlex Gateway, and run the "Run Script on Host" feature with upload capability again.

Upgrade the PowerFlex Gateway operating system

Upgrade the PowerFlex Gateway CentOS operating system with the latest security patches.

About this task

Upgrade the CentOS operating system on the SVM hosing the PowerFlex Gateway.

Steps

1. Copy the `patch_script` file to the SVM hosting the PowerFlex Gateway.
2. In the Security settings of the file, allow read, write, and execute permissions.
3. Run the file, using the command:
   ```
   ./patch_script
   ```
   The script will run on the SVM, and will display a message when the operation is complete.
4. Restart the SVM.

Upgrade CloudLink

Refer to *Dell PowerFlex CloudLink for SDS Services* for the appropriate CloudLink version, after completing the PowerFlex upgrade procedures.

This upgrade includes a sleep parameter of 60 as described in the "Migrate the PowerFlex nodes in the new CloudLink path" section of the appropriate Cloudlink version. For more information, see PowerFlex: Devices in error state when upgrading CloudLink with PFxM.

After upgrading an ESXi system

Tasks that can be performed after upgrading.

After the upgrade, you can perform many other tasks, including:

- Add virtual IP addresses, which are mandatory for PowerFlex plug-in setups from PowerFlex v2.0 and later. For more information, see the *Configure and Customize Dell PowerFlex*.
- Migrate an RDM-based PowerFlex system to DirectPath-based. This option is described in "Manual migration of an RDM-based PowerFlex system to DirectPath-based" in this guide.
- Install RFcache, if necessary.

  NOTE: From v3.0 and later, the commands and workflow for adding RFcache are different from earlier versions. For more information, see the *Configure and Customize Dell PowerFlex* and the *Dell PowerFlex CLI Reference Guide*.

- Take advantage of the new features added to this release. These options, and others, are described in the *Configure and Customize Dell PowerFlex*.

11

Upgrade the operating system to a major ESXi version

This section explains how to upgrade PowerFlex from ESXi 6.x to ESXi 7.x. You can also use it to upgrade from ESXi 6.x to a later version of ESXi 6.x. This process is only supported on PowerFlex v3.5.x and later.

Upgrading the PowerFlex operating system to a major ESXi version

Beginning with PowerFlex v3.5, the PowerFlex system supports ESXi 7.0 as an operating system for SDCs in a compute-only node configuration. When upgrading the hypervisor to this version, you must first upgrade the PowerFlex system to v3.5.x or higher and then perform the tasks in the following upgrade procedure.

NOTE: This procedure does not support upgrading hyperconverged nodes running on ESXi 6.x to ESXi 7.x . Hyperconverged nodes running on ESXi 7.x are only supported in PowerFlex Manager. For information, refer to the PowerFlex product offerings.

- The PowerFlex ESXi 6.0 SDC ZIP package (for example, `sdc-2.5.0.180-esx6.0.zip`) is compatible with all ESXi 6.x versions supported for the relevant version of PowerFlex.
- The PowerFlex ESXi 7.x SDC ZIP package is compatible with all ESXi 7.x versions supported for the relevant version of PowerFlex.

NOTE: You can also use this procedure to upgrade from ESXi 6.x to a later ESXi 6.x version or patch, without the need to upgrade the SDC. In this case, skip the steps for upgrading the SDC from ESXi 6.x to ESXi 7.0.

ESXi upgrade prerequisites

Before upgrading the ESXi system to ESXi 7.x, ensure that the system meets the following prequisites:

- PowerFlex v3.5 or higher is installed on the system.
- The system is fully operational and includes an MDM cluster
- All SDSs and SDCs are connected
- The vCenter version matches the target ESXi version for this upgrade. For example, if you are upgrading the compute-only nodes to ESXi 7.0, vCenter needs to be at version 7.0.
- If an ESXi cluster is defined, DRS is configured correctly and the vMotion network is operational on all nodes.

Perform all operations in the PowerFlex GUI as an admin user.

Upload the upgrade packages to the PowerFlex nodes

Before performing the upgrade, copy the ESXi upgrade files to the PowerFlex nodes you are upgrading or to a shared volume.

Prerequisites

Ensure that you have access to the following:

- Standby virtual machine for downloading the VMware ZIP file and SDC package
- Shared volume or datastore location with approximately 400 MB free disk space to accommodate the ESXi upgrade package.

About this task

The following flow uses the ESXi 7.0 package as an example.

Steps

1. Download the relevant VMware upgrade package from the VMware Download VMware vSphere page.
2. Using SCP, copy the Dell-customized VMware ZIP archive from the file path created in the previous step to a shared datastore in a PowerFlex volume or to the `/vmfs/volumes/scaleio_datastore/` folder:

   For example, for ESXi 7.0, the ZIP archive name is `VMware-VMvisor-Installer-7.0.0-15843807.x86_64-DellEMC_Customized-A00.zip`.

   NOTE: The name of the folder might be `scaleio_datastoreX`, where *X* is a number.

3. Copy the SDC zip archive, `sdc-x.x.x.xxx-esx7.0.zip`, from the file path created in the first step to the local folder.

   The file name contains the version and build number of the version to which you are upgrading.

   NOTE: Changes to the SDC package will be performed only as needed to avoid unnecessary reboots.

Next steps

Proceed to the next task to prepare for upgrading ESXi on the first node.

NOTE: The following three tasks are performed consecutively on a single node. After returning the upgraded node to operation, you will perform the three tasks on the next node.

Prepare for the ESXi upgrade

Perform the following steps on a single PowerFlex node to prepare it for the ESXi upgrade. After you prepare the node, you will perform the upgrade and return the node to operational state. Only after you return the node to operation will you perform this task on the next node you are upgrading.

Steps

1. Check that the system is in an optimal state.
   a. In the PowerFlex GUI left pane, click **Alerts**, and then in the right pane confirm that no SDS disconnect messages appear.
   b. If the node is an MDM cluster member, in the left pane select **MDM** > **Cluster Settings**, and in the right pane verify that the cluster's state is optimal.
   c. In the left pane, click **Configuration** > **SDCs**, and then in the right pane verify that the SDCs are connected.
2. For each SDC, record the SDC's IP address.
3. When upgrading the node to a later version of ESXi 6.x:

   When upgrading compute-only nodes, skip to the next step.

   a. In the left pane, click **Configuration** > **SDSs**.
   b. In the right pane, locate the node you are upgrading, and then record the SDS's IP address.
   c. In the right pane, select the check box for the relevant SDS, and then in the upper-right menu click **More** > **Enter Maintenance Mode**.
   d. In the **Enter SDS <SDS name> into Maintenance Mode** dialog box, select **Protected** .
   e. Click **Enter Maintenance Mode**.
   f. Wait until the SDS's state displays as **Maintenance**.
   g. In the pop-up dialog box, click **Dismiss**.
4. Log in to vCenter using the vSphere client, and perform the following actions:
   a. Correlate the SDC data IP you recorded in step 2 to the ESXi host's relevant VMkernel IP address.
   b. When upgrading a hyperconverged node, correlate the relevant SDS data IP address to the SVM on the same host.

   You will need this information later to help identify the host management's IP address.
5. Log in to vCenter using the vSphere client, and locate the relevant IP address or SVM name (which should include the host serial number).

   NOTE: Verify that the SVM IP address displayed in vCenter is the same as the SVM management IP you recorded.

6. Select the relevant SVM, and in the **Basic Tasks** pane click **Shut down the virtual machine**.

7. Log back in to the PowerFlex GUI, and in the **Alerts** pane verify that an alert was received that the SDS is disconnected and that, if the SVM is a cluster member, vCenter also received an alert from the MDM.
8. In vCenter, right-click the node and select **List Maintenance Mode** > **Enter Maintenance Mode**.
9. In the **Confirm Maintenance Mode** dialog box, clear the **Move powered-off and suspended virtual machines to other hosts in the cluster** check box, and then click **OK**.
   - If DRS and vMotion are configured, the VMs should start to migrate to other nodes.
   - If DRS is not configured, but vMotion is, migrate the VMs to other nodes manually and repeat the previous step to enter maintenance mode.
10. If a warning message appears requesting you to migrate the VMs, click **OK** and skip to the next step. In this case, click **OK** and proceed with the procedure.
11. Using SSH, connect to the ESXi node using the management IP address, and log in as root user with the relevant password.
12. Back up the SDC GUID of the node to be upgraded:

    NOTE: You can also use this procedure to upgrade from ESXi 6.x to a later ESXi 6.x version or patch, without the need to upgrade the SDC. In this case, skip the steps for the ESXi 7.x SDC.

    a. View the complete list of SDC settings for the scini driver:

    ```
    cat /etc/vmware/esx.conf |grep scini/
    ```

    Output similar to the following appears:

    ```
    /vmkernel/module/scini/options =
    "IoctlIniGuidStr=7ea77fc9-6125-44c1-9746-9d20053aed93
    IoctlMdmIPStr=9.201.112.254,9.201.111.254"
    ```

    NOTE: The GUID is displayed in quotation marks in the command output. You will need it after SDC installation following the ESXi upgrade.

    b. Copy the entire command output and paste it to Notepad, along with the SDC IP so you can identify to which SDC the GUID belongs.

    For example:

    ```
    10.234.214.229 SDC settings:
    ```

    ```
    /vmkernel/module/scini/options =
    "IoctlIniGuidStr=7ea77fc9-6125-44c1-9746-9d20053aed93
    IoctlMdmIPStr=9.201.112.254,9.201.111.254"
    ```

13. Remove the SDC from the ESXi 6.x node:

    NOTE: You can also use this procedure to upgrade from ESXi 6.x to a later ESXi 6.x version or patch, without the need to upgrade the SDC. In this case, skip the steps for the ESXi 7.x SDC.

    a. Using SSH, check the ESXi version installed on the host:

    ```
    esxcli system version get
    ```

    Output similar to the following appears:

    ```
    Product: VMware ESXi
    Version: 6.5.0
    Build: Releasebuild-xxxxxxx
    ```

    b. Check the version of the SDC:

    ```
    esxcli software vib list |grep sdc
    ```

    Output similar to the following appears:

    ```
    scaleio-sdc-esx6.x             3.0-1000.180                    EMC
    VMwareAccepted    2020-02-05
    ```

c. Remove the SDC from the ESXi 6.x host using one of the following commands:

```
esxcli software vib remove --vibname scaleio-sdc-esx6.x
```

or

```
esxcli software vib remove -n scaleio-sdc-esx6.x
```

The output should display the results of the command. For example:

```
Removal Result
Message: The update completed successfully, but the system needs to be rebooted for
the changes to be effective.
Reboot Required: true
VIBs Installed:
VIBs Removed: VMware_bootbank_scaleio-sdc-esx6.5_2.6-XXX.0
VIBs Skipped:
```

Next steps

Proceed to the next task to upgrade the ESXi and SDC versions on this node.

Upgrade the ESXi and SDC versions on the node

Perform the following tasks to upgrade the ESXi and SDC versions on the node you just prepared in the previous task.

Steps

1. To optionally list the profile of the ESXi zip archive, run:

   Perform the following tasks to upgrade the ESXi and SDC versions on the node you just prepared in the previous task.

   ```
   esxcli software sources profile list -d /vmfs/volumes/scaleio-datastore/<VMWare.zip>
   ```

   For example:

   ```
   esxcli software sources profile list -d /vmfs/volumes/scaleio-datastore/VMware-
   VMvisor-Installer-7.0.0-15843807.x86_64-DellEMC_Customized-A00.zip
   ```

   Output similar to the following appears:

   ```
   Name                              Vendor  Acceptance Level
   --------------------------------  ------  ----------------
   DellEMC-ESXi-7.0-15843807-A00  Dell    PartnerSupported
   ```

2. Run the following command as a single line to upgrade the ESXi version:

   ```
   esxcli software profile update -p DellEMC-ESXi-7.0-15843807-A00 -d /vmfs/volumes/
   sio_datastore/VMware-VMvisor-
   VMware-VMvisor-Installer-7.0.0-15843807.x86_64-DellEMC_Customized-A00.zip
   ```

   The command assumes the path is under `/tmp`.

   NOTE: If you are upgrading from an ESXi version based on a Dell customized image to a VMware-based patch that is not Dell customized, use the `--force` switch command syntax to force the upgrade to ignore certain dependencies. For example:

   ```
   esxcli software profile update -p ESXi-7.0U2a-17867351-standard -d /vmfs/volumes/
   DS140/VMware-ESXi-7.0U2a-17867351-depot.zip --force
   ```

   After the upgrade, you can upgrade the specific VIBs manually if they exist in the VxFlex Ready Node driver pack.

NOTE: The `--force` switch command syntax is also required in cases where the SDC is being upgraded from an ESXi version based on a Dell customized image to a VMware-based patch that is not Dell customized,.

After the upgrade has completed successfully, the following output appears, followed by a list of the upgraded packages:

```
Update Result
Message: The update completed successfully, but the system needs to be rebooted for
the changes to be effective.
Reboot Required: true
```

3. On the ESXi system where you are installing the SDC, set the acceptance level:

```
esxcli software acceptance set --level=PartnerSupported
```

4. Install the SDC:

NOTE: You can also use this procedure to upgrade from ESXi 6.x to a later ESXi 6.x version or patch, without the need to upgrade the SDC. In this case, skip the steps for the ESXi 7.x SDC and proceed directly to reboot.

```
esxcli software vib install -d <Full_Path>
```

where *<Full_Path>* is the full path to the location of the SDC upgrade zip archive.

For example:

```
esxcli software vib install -d /vmfs/volumes/sio_datastore/sdc-3.5.0.XXX-esx7.X.zip
```

NOTE: If in step 2 you upgraded from a Dell customized image to a VMware-based patch that is not Dell customized, use the `--force` switch command syntax to force the upgrade to ignore certain dependencies. For example:

```
esxcli software vib install -d /vmfs/volumes/sio_datastore/sdc-3.5.0.XXX-
esx7.X.zip --force
```

5. Reboot the node.

Next steps

Proceed to the next task to return the node to operational state.

Return the node to operational state

Perform the following steps to return the node you upgraded in the previous task to operational state:

Steps

1. Ping the ESXi management IP address to verify that the node is connected in the network.
2. From the vSphere client, right-click the node and select **Connection** > **Connect** to manually connect the node to vCenter.
   Because the ESXi version of the node changed, the node could not reconnect automatically to vCenter. If "could not connect to HOST" alerts appear, wait a few minutes for the alerts to disappear, and if necessary try to connect again.
3. Ensure that the node is displayed as ON and connected in both **Hosts** and **Clusters** view.
4. Right-click the host and select **Maintenance Mode** > **Exit Maintenance Mode**.
5. For a hyperconverged upgrade flow, in vCenter, expand the server and select the Storage VM (SVM). Power on the SVM manually.
6. After the node has powered up, log back in to the PowerFlex GUI.
7. For a hyperconverged flow, in the left pane click **Alerts**, and in the right pane confirm that no SDS disconnect message appears.

NOTE: After the SVM has powered on, it might take approximately 10 to 20 seconds for the SDS to power up and remove the disconnection alerts.

8. For a hyperconverged upgrade flow, remove the SDS from maintenance mode:

   NOTE: When upgrading a compute-only node (SDC), skip this step.

   a. In the left pane, click **Configuration** > **SDSs**.
   b. In the right pane, select the check box for the relevant SDS, and then in the upper menu click **More** > **Exit Maintenance Mode**.
   c. In the **Exit SDS <SDS name> from Maintenance Mode** dialog box, click **Exit Maintenance Mode**.
   d. Wait until the SDS's state displays as **Healthy**.
   e. In the pop-up window, click **Dismiss**.
9. Verify the following in the PowerFlex GUI:

   a. For a hyperconverged upgrade flow, if the node was an MDM cluster member, in the left pane click **MDM** > **Cluster settings**, and in the right pane verify that the cluster's **State** appears as **Healthy**.
   b. In the left pane click **Configuration** > **SDCs**, and in the right pane check the SDC to which the node IP is assigned and verify that it is connected.

      NOTE: If the SDC was upgraded, ensure that the same GUID is used in the new SDC. If the SDC GUI is incorrect, refer the "Modify parameters on ESXi servers" in the *Configure and Customize Dell EMC PowerFlex v3.x* guide.
10. Wait for the rebuild/rebalance operations to complete.
    The node is now operational.
11. Repeat the upgrade process for each node, starting with Prepare for the ESXi upgrade, until all nodes are upgraded and the system has been upgraded to the desired ESXi version with the relevant SDC version.

Next steps

NOTE: Verify that all nodes have been upgraded before proceeding to the post-upgrade steps.

Post-upgrade steps

After you have upgraded the system to the desired ESXi version, perform the following tasks according to the instructions in this guide:

1. Verify that you have upgraded all nodes to the desired ESXi version.
2. If the system is not using the latest PowerFlex version, upgrade the system to the latest version.

12

DTK - Hardware Update Bootable ISO

The Dell OpenManage Deployment Toolkit (DTK) includes a set of utilities, sample scripts, and sample configuration files that you can use to deploy and configure the Dell systems. This section provides additional information on using the VxFlex Ready Node Hardware Update Bootable ISO ("Hardware ISO") to update drivers, BIOS, and firmware on VxFlex Ready Node servers.

Dell OpenManage DRAC Tools (RACADM)

An integrated Dell Remote Access Controller (iDRAC) with Lifecycle Controller is embedded in every VxFlex Ready Node server. The RACADM command-line utility provides a scriptable interface that enables you to configure the iDRAC either locally or remotely. The utility runs on the management station and the managed system.

The RACADM utility supports the following interfaces:

- Local - Supports running RACADM commands from the managed server's operating system. To run local RACADM commands, install the OpenManage software on the managed server. Only one instance of Local RACADM can be executed on a system at a time.
- SSH or Telnet (also known as Firmware RACADM) - Firmware RACADM is accessible by logging in to iDRAC using SSH or Telnet.
- Remote - Supports running RACADM commands from a remote management station such as a laptop or desktop running Linux. To run Remote RACADM commands, install the DRAC Tools utility from the OpenManage software on the remote computer.

NOTE: For information regarding RACADM commands, see the *RACADM CLI Guide* at: http://topics-cdn.dell.com/pdf/idrac7-8-lifecycle-controller-v2.40.40.40_Reference Guide_en-us.pdf.

Update the hardware using remote RACADM

You can install and execute the Dell RACADM tool from any management system with access to the iDRAC network. The remote RACADM command set is useful in this situation to mount and execute the Hardware ISO to a large number of VxFlex Ready Node servers. This prevents the need for multiple web browser windows and manual keyboard and mouse clicks for each server.

About this task

NOTE: For additional instructions regarding version updates following SATADOM replacement on R630/R730xd servers, see "Update the SATADOM firmware".

Steps

1. Configure a laptop or server with access to the iDRAC network.
2. Install the Dell DRAC tools, including RACADM:
   a. In any Internet browser, go to the RACADM Command Line Interface for DRAC page in the Dell Systems Management wiki.
   b. Follow the instructions for downloading and installing Remote RACADM for your operating system.
3. Download the VxFlex Ready Node Hardware Update Bootable ISO from the ScaleIO Ready Node-PowerEdge 13G VxFlex Ready Node and make it accessible on the network share folder.
4. Create a CIFS or NFS network share that includes the Hardware ISO. The share will need to be accessible to the iDRAC network.

   Refer to operating system vendor guidelines for instructions on how to create CIFS (Windows) or NFS (Linux) shares.
5. Open a terminal/command prompt with superuser/administrator accessibility. This prompt should allow execution of the RACADM utility.

6. Mount the Hardware ISO to the iDRAC from the remote share, where the following command is all on one line:

```
racadm -r <dracIP> -u root -p <password> remoteimage -c -u <myuser> -p <mypass> -l
//<myip>/SIORN/<file_name>.iso
```

Where:

- *<dracIP>* is the iDRAC IP address
- *<password>* is the password for the server
- *<myuser>* is the NFS/CIFS share username
- *<mypass>* is the NFS/CIFS share password
- *<myip>* is the NFS/CIFS server IP

For example:

```
racadm -r <dracIP> -u root -p <password> remoteimage -c -u <myuser> -p <mypass> -l
//<myip>/SIORN/ScaleIO-Ready-Node-Hardware-Update-for-Dell_13G_2019_June_A00.iso
```

```
racadm -r <dracIP> -u root -p <password> remoteimage -c -u <myuser> -p <mypass> -l
//<myip>/SIORN/VxFlex-Ready-Node-Hardware-Update-for-Dell_14G_2018_May_A00.iso
```

NOTE: The default password is calvin for R630/R730xd nodes and Scaleio123 for R640/R740xd/R840 nodes.

7. Enable the iDRAC VirtualCD to boot once using these Remote RACADM commands:

```
racadm -r <dracIP> -u root -p <password> set iDRAC.ServerBoot.BootOnce Enabled
racadm -r <dracIP> -u root -p <password> set iDRAC.ServerBoot.FirstBootDevice VCD-DVD
```

8. Verify that the server is in a state in which it can be power-cycled. (The operating system should be in maintenance mode, as needed).
9. Power-cycle the VxFlex Ready Node server:

```
racadm -r <dracIP> -u root -p <password> serveraction graceshutdown
```

10. Repeat steps 6 - 9 for each VxFlex Ready Node server needing the BIOS and firmware updates and configuration. All servers can be updated in parallel.
11. Wait for the configuration and firmware updates to complete. The server console screen will indicate when the script is complete.

WARNING: Do not reboot the VxFlex Ready Node server while the update process is being performed!

The iDRAC will be reset several times during the update process. This causes the iDRAC virtual console viewer to close, virtual media to disconnect, and the iDRAC browser window to be unavailable for several minutes during each reset. The hardware update scripts will continue to run from RAM on the server.

The update script will generate a log indicating whether each configuration and firmware flash is successful.

12. (Optional) Check each VxFlex Ready Node server's log for successful competition:
    a. Connect to the iDRAC KVM console screen.
    b. After the update script completes, press Alt+F2 to access a user console, and then Enter to log in.
    c. Check the log contents for errors:

```
less /bundleapplicationlogs/apply_components.log
```

    For more information, see Troubleshooting the Hardware ISO .

    d. Press q to exit the log viewer, and then Alt+F1 to access the original console screen.
13. Reboot the servers and allow the update and configuration jobs to complete. You can do this remotely using RACADM (see Step 9) or locally from the server console screen.

14. For each server, check the iDRAC job queue to verify that the iDRAC jobs created by the bootable ISO have completed successfully:

```
racadm -r <dracIP> -u root -p <password> jobqueue view
```

Any job failures may require running the bootable ISO again, or further troubleshooting.

15. For each server, after the hardware updates are finalized, clean up the iDRAC job queue:

```
racadm -r <dracIP> -u root -p <password> jobqueue delete --all
```

16. Continue with operating system installation and configuration, PowerFlex deployment, and other required tasks.

Update the SATADOM firmware (VxFlex Ready Node R630/R730xd only)

The Dell factory should provide the SATADOM boot device with the correct firmware already loaded. However, in some cases, it may be necessary to update the firmware loaded on the SATADOM boot device.

About this task

For convenience, SATADOM firmware version S130710K is included on the VxFlex Ready Node Hardware ISO.

WARNING: The SATADOM firmware update process permanently erases all data from the SATADOM device. Proceed with caution!

Steps

1. Connect to the iDRAC Virtual KVM console.
   For instructions, see Open the KVM console - R630/R730xd servers.
2. Boot the VxFlex Ready Node server to the Hardware ISO.
   For instructions, see Upgrading the firmware using the iDRAC virtual console.
3. After the update script completes, press Alt-F2 to access a user console.
4. Press Enter to login.
5. Execute the SATADOM firmware:

```
/tmp/S130710K_read_log_issue/satadom.sh
```

6. Press y to begin the SATADOM update.
7. Press Alt+F1 to access the original console screen, and then press Enter to reboot the server.

Recommended BIOS and firmware settings

This section describes the BIOS, firmware, and configuration settings included in the Hardware ISO.

PowerFlex ID module

The Hardware ISO runs a script that automatically flashes the ID module, as needed.

This action rebrands a PowerEdge server naming, and is intended to make it compatible with the VxFlex Ready Node offering.

The VxFlex Ready Node ID module should be installed only on VxFlex Ready Node servers. In the event of a system board failure, the Hardware ISO can assist with the reinstallation of the ID module on the replacement system board.

BIOS and firmware

The Hardware ISO runs a script that automatically forces the server to install the necessary firmware updates.

This firmware is consistent with the qualified Dell VxFlex Ready Node Driver and Firmware Matrix, located at:

- PowerEdge R630/R730xd: Dell Technologies support site
- VxFlex Ready Node R640/R740xd servers: Dell Technologies support site

Some of the firmware listed in the table is dependent on the VxFlex Ready Node hardware configuration. The Hardware ISO attempts to apply all firmware updates, but only those updates that are compatible will be installed.

Applying settings using RACADM

The individual firmware files are also available on the Dell Online Support site, and can easily be installed using the following remote RACADM command:

```
racadm -r <dracIP> -u root -p <password> update -f <filename.exe>
```

Where:

- *<dracIP>* is the iDRAC IP address
- *<password>* is the password for the server
- *<filename.exe>* is the name of the Dell Windows update packages

NOTE: The default password is calvin for R630/R730xd nodes and Scaleio123 for R640/R740xd/R840 nodes.

Configuration settings

The Hardware ISO runs a script that automatically configures the BIOS and iDRAC settings listed in the table below. Some settings are dependent on the server model.

Table 14. Hardware ISO configuration settings

Description	Setting	Value
Server BIOS Boot Sequence	BIOS.BiosBootSettings.BootSeq	HardDisk.List.1-1
Hard Disk Boot Order	BIOS.BiosBootSettings.HddSeq	Disk.SATAEmbedded.J-1,RAID.Integrated.1-1 or Disk.SATAEmbedded.D-1,RAID.Integrated.1-1
Server Boot Mode	BIOS.BiosBootSettings.BootMode	Bios
SRIOV Global Enablement	BIOS.IntegratedDevices.SriovGlobalEnable	Enabled
Memory Performance Tuning	BIOS.MemSettings.SnoopMode	EarlySnoop
System Change Tracking	BIOS.MiscSettings.InSystemCharacterization	Disabled
CPU Virtualization Features	BIOS.ProcSettings.ProcVirtualization	Enabled
CPU Cores	BIOS.ProcSettings.ProcCores	All
CPU X2APIC Mode	BIOS.ProcSettings.ProcX2Apic	Enabled
CPU Turbo Engagement	BIOS.ProcSettings.ControlledTurbo	Disabled
System Power Profile	BIOS.SysProfileSettings.SysProfile	PerfOptimized
OS to iDRAC Pass-through Mode	iDRAC.OS-BMC.PTMode	usb-p2p

Table 14. Hardware ISO configuration settings (continued)

Description	Setting	Value
OS to iDRAC Enablement	iDRAC.OS-BMC.AdminState	Enabled
iDRAC DHCP Enablement	iDRAC.IPv4.DHCPEnable	Disabled
iDRAC Default Credentials	iDRAC.Tuning.DefaultCredentialWarning	Disabled
iDRAC IPMI Enablement	iDRAC.IPMILan.Enable	Enabled
iDRAC Alert Enablement	iDRAC.IPMILan.AlertEnable	Enabled
iDRAC IPv6 Enablement	iDRAC.IPv6.Enable	Enabled

Applying settings using RACADM

The individual settings can also be applied using the remote RACADM command:

```
racadm -r <dracIP> -u root -p <password> set <setting> <value>
```

Where:

- *<dracIP>* is the iDRAC IP address
- *<password>* is the password for the server
- *<setting>* is the BIOS/iDRAC setting name
- *<value>* is the BIOS/iDRAC setting value

NOTE:

When setting the BIOS configuration, include this command:

```
racadm -r <dracIP> -u root -p <password> jobqueue create BIOS.Setup.1-1 <value>
```

Troubleshooting the Hardware ISO

This section describes troubleshooting procedures for problems you may encounter while using the Hardware ISO.

Troubleshoot general iDRAC failures

When problems occur with iDRAC or Lifecycle Controller jobs, you can delete all jobs with a single iDRAC command. All of the completed jobs, plus any orphaned pending jobs, are deleted, and the data manager service on the iDRAC is restarted.

Steps

1. Clear the iDRAC job queue:

   ```
   racadm -r <dracIP> -u root -p <password> jobqueue delete -i JID_CLEARALL_FORCE
   ```

   Where:

   - *<dracIP>* is the iDRAC IP address
   - *<password>* is the password for the server

   NOTE: The default password is calvin for R630/R730xd nodes and Scaleio123 for R640/R740xd/R840 nodes.

2. Wait 120 seconds.
   The iDRAC is unable to process any other jobs during this time.
3. Reset the iDRAC

   ```
   racadm -r <dracIP> -u root -p <password> racreset
   ```

The iDRAC becomes accessible on the network 3 to 5 minutes after the reset.

iDRAC virtual console issues

Within the iDRAC virtual console window, if the keyboard or some of the keys are not responding, perform the following checks:

- Ensure on your keyboard that the Scroll Lock button is off.
- In the iDRAC virtual console, ensure that the **Keyboard/Mouse Attach State** is set to **Auto-attached**.

For persistent iDRAC virtual console issues, see Troubleshoot general iDRAC failures.

iDRAC virtual media issues

The following are solutions to problems that may occur when using iDRAC virtual media:

1. If the iDRAC is stuck, one of these actions may assist with recovery:
   - In the server's **Attached Media** screen, disconnect the iDRAC Remote File Share from the iDRAC browser.
   - Disconnect the iDRAC remote image:

     ```
     racadm -r <dracIP> -u root -p <password> remoteimage -d
     ```

     Where:

     - *<dracIP>* is the iDRAC IP address
     - *<password>* is the password for the server

   NOTE:

   The default password is calvin for R630/R730xd nodes and Scaleio123 for R640/R740xd/R840 nodes.

2. Issues when booting to iDRAC virtual media may cause CPU machine check errors at POST. When this occurs, clear the iDRAC job queue and reset the iDRAC, as described in Troubleshoot general iDRAC failures.

DTK A08: BIOS set commands fail

When running DTK A08 on R630/R730xd servers with iDRAC FW version 2.50.50.50 or higher, the BIOS set commands may fail.

The DTK is unable to configure the BIOS settings, and the following error message may appear:

```
ERROR: BOOT016: Input source argument value for the boot device is incorrect or not
found among the boot devices on the system.
```

Rebranding operation fails

When updating the Personality Module on R640 or R740xd servers using the DTK, the operation may fail and the following error message may appear:

```
UEFI0052 Unable to complete the rebranding operation.
```

To resolve the problem, upload the PM file manually:

1. Contact Customer Support for the PowerFlex PM file.
2. Follow the instructions in the *PowerFlex Upgrade Guide* for "Single server firmware and BIOS upgrade to a specific version" to upload the file.

Check the logs for error messages

You can view VxFlex Ready Node Hardware Update Bootable ISO logs after the update script completes.

Steps

1. Press Alt-F2 to access a user console, and then press Enter.
2. Open the log to check the contents for errors:

```
less /bundleapplicationlogs/apply_components.log
```

3. You can also view the script for the Hardware ISO, which is useful in helping to identify and troubleshoot log entries:

```
less /opt/dell/toolkit/systems/drm_files/apply_bundles.sh
```

Results

The script attempts to configure several boot order commands, regardless of the hardware configuration of the VxFlex Ready Node server. This allows the script to support multiple hardware platform configurations.

Therefore, it is normal to see these error messages within the logs:

```
Apply R730xd bootorder
[Key=BIOS.Setup.1-1#BiosBootSettings]
RAC1017: Successfully modified the object value and the change is in
       pending state.
       To apply modified value, create a configuration job and reboot
       the system. To create the commit and reboot jobs, use "jobqueue"
       command. For more information about the "jobqueue" command, see RACADM
       help.

ERROR: BOOT016: Input source argument value for the boot device is incorrect or
        not found among the boot devices on the system.

ERROR: BOOT016: Input source argument value for the boot device is incorrect or
        not found among the boot devices on the system.

ERROR: BOOT016: Input source argument value for the boot device is incorrect or
        not found among the boot devices on the system.

[Key=BIOS.Setup.1-1#BiosBootSettings]
RAC1017: Successfully modified the object value and the change is in
       pending state.
:
```

Firmware updates may also display and log the following message:

```
This update is not compatible with your system configuration.
```

```
iDRAC Settings Complete
Proceeding with server BIOS and firmware flash.
Collecting inventory...
Running validation...

This Update Package is not compatible with your system configuration.

Collecting inventory...
Running validation...
```

These firmware are included on the ISO in order to support various VxFlex Ready Node platforms. The log messages do not necessarily indicate a failure.

Additional resources

This section contains information regarding additional resources that may be helpful useful when using the Hardware ISO.

PowerFlex resources

VxFlex Ready Node deployments have specific guidelines regarding server installation, rack and stack procedures, and power and networking requirements.

The *VxFlex Ready Node Hardware Installation Guide* describes how to install the physical components of a VxFlex Ready Node system. For additional information regarding the VxFlex Ready Node product, documentation, advisories, downloads, and white papers, visit ScaleIO Ready Node-PowerEdge 13GVxFlex Ready Node.

Dell Lifecycle Controller (LC)

With the launch of the Dell PowerEdge 13th-generation servers in September 2014, Dell enhanced embedded management without the need to install a software-based agent within the host operating system. 14th-generation servers continue this functionality.

At the heart of the servers' embedded management is the iDRAC with Lifecycle Controller (LC) technology. This technology allows users to perform useful tasks such as configuring BIOS and hardware settings, deploying operating systems, updating drivers, changing RAID settings, and saving hardware profiles. Together, they provide a robust set of management functions that can be leveraged throughout the entire server lifecycle.

Supported versions are iDRAC8 (13th generation servers) or iDRAC9 (14th generation servers).

For more information, visit the Lifecycle Controller wiki homepage.

Dell OpenManage Deployment Toolkit (DTK)

The Dell OpenManage Deployment Toolkit (DTK) includes a set of utilities, sample scripts, and sample configuration files that you can use to deploy and configure Dell systems.

You can use the DTK to build script-based and RPM-based installation for deploying large number of systems on a pre-operating system environment in a reliable way, without changing their current deployment processes. Using DTK you can install operating systems on Dell systems in BIOS mode R630/R730xd or BIOS or Unified Extensible Firmware Interface (UEFI) mode (R640/R730xd/R840).

For more information, visit the Dell OpenManagement Deployment Toolkit wiki homepage.

Set up the iDRAC IP address and BIOS

Set up the iDRAC IP address and set up or validate the BIOS on the VxFlex Ready Node servers.

Prerequisites

Ensure that you have access to, or have the details for:

- The KVM console
- The server iDRAC IP address
- The server iDRAC subnet mask
- The gateway IP address
- The VLAN ID of the iDRAC, if the VLAN is used

NOTE: The iDRAC IP address may be an IPv4 or an IPv6 address.

About this task

Defaults:

- iDRAC username - root

- iDRAC password - The default password is calvin for R630/R730xd nodes and Scaleio123 for R640/R740xd/R840 nodes.
- BIOS password - emcbios

During the iDRAC IP and BIOS setup, use the following operations:

- Use the arrow keys to navigate in the BIOS screens.
- Use the + and - keys on the keyboard to change the option selection in the BIOS screens.
- Use SPACE or ENTER keys to change the settings.
- The ENTER key opens a list to the desired values.

Steps

1. Open the KVM console.

   For console operations (KVM access), ensure that you have either a VGA tool kit/Crash Cart to allow physical console connection from a laptop computer to a server, or a computer screen and keyboard connection to the rack.
2. Press F2 to access the main menu.
3. From the **System Setup Main Menu** screen, select the **iDRAC Settings** menu option.
4. Configure network settings:
   a. In the **iDRAC Settings** screen, select **Network**.
   b. In the **iDRAC Settings--Network** pane, verify the following parameter values:
      - **Enable NIC** = **Enabled**
      - **NIC Selection** = **Dedicated**
   c. From the **IPv4 Settings** pane, configure the IPv4 parameter values for the iDRAC port:
      - **Enable IP IPv4** = **Enabled**
      - **Enable DHCP** = **Disabled**
      - **Static IP Address** = Static IP address
      - **Static Gateway** = Gateway IP address
      - **Static Subnet Mask** = Subnet mask IP address
   d. From the **IPv6 Settings** pane, configure the IPv6 parameter values for the iDRAC port.
   e. From the **IPMI Settings** pane, verify the following parameter values:
      - **Enable IPMI Over LAN** = **Enabled**
      - **Channel Privilege Level Limit** = **Administrator**
   f. If you are working in a VLAN setup, access the **VLAN Configuration** pane and configure the VLAN ID parameters.
   g. When the parameter set up is complete, click **Back** to display the **iDRAC Settings** screen.
5. From the **iDRAC Settings** screen, click **Finish**, **Yes**, and then **OK** to return to the **System Setup Main Menu** screen.
6. In the **System Setup Main Menu** screen, select the **System BIOS** menu.
7. In the **System BIOS Settings** screen, verify that the processor settings are correct.

   If the settings are incorrect, configure them as follows:
   a. Select **Processor Settings**.
   b. In the **System BIOS Settings--Processor Settings** pane, verify the following parameter values:
      - **Virtualization Technology** = **Enabled**
      - **Number of Cores Per Processor** = **All**
   c. Click **Back** to return to the **System BIOS Settings** screen.
8. Configure boot settings,:
   a. Select **Boot Settings**.
   b. In the **System Boot Settings--Boot Settings** pane, verify that **Boot Mode** is set to **BIOS (R630/R730xd) or UEFI (R640/R740xd/R840)**.
   c. Select the **BIOS Boot Settings** link.
   d. In the **System BIOS Settings--Boot Settings--BIOS Boot Settings** pane, verify that:
      - In the **Boot Sequence** list, **Hard drive C:** appears as the first item.
      - In the **Hard-Disk Drive Sequence** list, the **SATADOM-ML 3SE (R630/R730xd) or BOSS (R640/R740xd/R840)** device appears as the first item.
   e. Click **Back** twice to return to the **System BIOS Settings** screen.
9. Configure integrated devices:
   a. Select **Integrated Devices**.

      The **System BIOS Settings--Integrated Devices** screen appears.

b. Set **SR-IOV Global Enable** to **Enabled**.
c. Verify that the **Internal USB Port** parameter is set to **Off**.
d. Click **Back** to return to the **System BIOS Settings** screen.

10. From the **System BIOS Settings** screen, click **Finish**, **Yes**, and then **OK** to return to the **System Setup Main Menu** screen.
11. Select **Finish** to exit the BIOS and apply all settings post boot.

Results

The iDRAC IP and server BIOS address configuration is complete.

Verify the status of the system hardware, storage controller, and disks - R630/R730xd servers

Use the following procedure to verify the status of the system hardware, storage controller, and disks in a VxFlex Ready Node R630/R730xd server.

Prerequisites

Ensure that you know:

- The IP address of the iDRAC port
- The username and password for the iDRAC portal (default username and password are root and password)

Steps

1. From a browser, go to http://*<iDRAC_IP_address>*.
2. In the **DELL Console Login** window, type the user name and password, and then click **Login**.
   The **System Summary** dashboard displays the high-level status of all hardware devices in the **Server Health** pane.

   In an ideal scenario, all hardware sensors are green.
3. In the **Quick Launch Tasks** pane, select **View Logs**.
   The **System Event Log** is displayed with color-coded severity levels.
4. Ensure that any power supply- and fan-related events in the event log are non-repetitive. Repetitive events may be due to the intermittent nature of faults, such as poor physical connections.

   For repetitive events, it is recommended that you remove the relevant hardware module and replace it in its socket.
5. In the navigation pane, select **Storage** > **Controllers**.

   The controller-related information is displayed in the **Health and Properties** table, with the controller type shown in the **Name** column.

   For example:

   ```
   PERC H730 Mini (Embedded)
   ```

6. In the navigation pane, select **Storage** > **Physical Disks**.

   Information about the physical disks is displayed in a table in the **Health and Properties** pane.
7. Verify that no disk is in failed state.

   If any of the disks is failed, refer to the relevant Disk Replace FRU.
8. In the navigation pane, select **Server**.
9. In the **System Inventory** pane, verify that the server drivers and firmware in the **Firmware Inventory** list match the required versions, as published in the VxFlex Ready Node Driver and Firmware Matrix. To access the Driver and Firmware Matrix, go to ScaleIO Ready Node-PowerEdge 13G.
   If the driver and firmware versions do not match the matrix, you must update them using the DTK - Hardware Update Bootable ISO.

Verify the status of the system hardware, storage controller, and disks - R640/R740xd servers

Verify the status of the system hardware, storage controller, and disks in a VxFlex Ready Node R640/R740xd server.

Prerequisites

Ensure that you know:

- The IP address of the iDRAC port
- The username and password for the iDRAC portal (default username and password are root and password)

Steps

1. From a browser, go to http://*<iDRAC_IP_address>*.
2. In the **DELL Console Login** window, enter the user name and password, and then click **Login**.
   The **Dashboard** displays the high-level status of all hardware devices in the **System Health** pane.

   In an ideal scenario, all hardware sensors are green.
3. Navigate to **Maintenance** > **System event log.**
   The **System Event Log** pane is displayed with color-coded severity levels.
4. Ensure that any power supply- and fan-related events in the event log are non-repetitive. Repetitive events may be due to the intermittent nature of faults, such as poor physical connections.

   For repetitive events, Dell recommends that you remove the relevant hardware module and replace it in its socket.
5. In the navigation pane, select **Storage** > **Overview** > **Controller**.

   A table displays the controller-related information, with the controller type in the **Name** column. For example:

   ```
   PERC H740P Mini (Embedded)
   ```

   ```
   PERC H740P Adapter (PCI Slot 4)
   ```

6. In the navigation pane, select **Storage** > **Physical Disks**.

   A table displays information regarding the physical disks, and an option to **Blink**/**Unblink** the selected disk.
7. Verify that no disk is in failed state.

   If any of the disks is failed, refer to the relevant Disk Replace FRU.
8. In the navigation pane, select **System**.
9. In the **Inventory** pane, verify that the server drivers and firmware in the **Firmware Inventory** list match the required versions, as published in the VxFlex Ready Node Driver and Firmware Matrix. To access the Driver and Firmware Matrix, go to Dell Technologies support site.
   If the driver and firmware versions do not match the matrix, you must update them using the DTK - Hardware Update Bootable ISO.

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Increasing SVM system disk size

Increase SVM system disk size on all SVMs prior to upgrading your PowerFlex system.

Increase SVM system disk size

Increase SVM system disk size to at least 16 GB before upgrading your PowerFlex system.

Prerequisites

- Login credentials to the vCenter are required.
- Credentials for SSH root access to each SVM are required.

About this task

The system disk size of the SVM must be increased from 8 GB to 16 GB. This procedure must be performed on all SVMs in your system. All the steps can be performed online, and no reboot is required.

Steps

1. Login to vCenter, and on each SVM, click **Edit settings** and increase the size of the system disk from 8 GB to 16 GB.
2. SSH to the SVM using root credentials.
3. Check if the rpm file `cloud-utils-growpart-0.29-5.el7.noarch.rpm` is installed, using the command:

   ```
   rpm -qa |grep growpart
   ```

   Example of expected output:

   ```
   cloud-utils-growpart-0.29-5.el7.noarch
   ```

   NOTE: In the future this package or a newer one will be part of the SVM patch bundle.
4. If the package is not installed on the SVM, download it from the following link: cloud-utils-growpart-0.29-5.el7.noarch.rpm. Then install it:
   a. Install the package using the command:

      ```
      rpm -ivh <package name>
      ```
5. Execute the command:

   ```
   echo 1>/sys/class/block/sda/device/rescan
   ```

   The increased drive size will be reflected in the operating system, and the `growpart` command will now work as expected.
6. Grow the system partition (it should be `/dev/sda1`)
   a. Get the system partition information by running the command:

      ```
      df -h /
      ```

      Typical output:

      ```
      Filesystem      Size  Used Avail Use% Mounted on
      /dev/sda1        8G  7.6G  0.4G  95% /
      ```

b. Run the following command to expand the partition:

```
growpart /dev/sda 1
```

Output example:

```
CHANGED: partition=1 start=2048 old: size=17817600 end=17819648 new: size=335523
                                                                              51 end=33554399
```

7. Run the following command to resize the file system:

```
xfs_growfs -d /dev/sda1
```

Output example:

```
meta-data=/dev/sda1              isize=512    agcount=4, agsize=556800 blks
                                 =                       sectsz=512   attr=2,
projid32bit=1
                                 =                       crc=1        finobt=0
spinodes=0
data                        =                       bsize=4096   blocks=2227200,
imaxpct=25
=                        sunit=0      swidth=0 blks
naming                   =version 2              bsize=4096   ascii-ci=0 ftype=1
log                           =internal               bsize=4096   blocks=2560,
version=2
                                 =                       sectsz=512   sunit=0 blks,
lazy-count=1
realtime                 =none                   extsz=4096   blocks=0, rtextents=0
data blocks changed from 2227200 to 4194043
```

8. Check the new size of the system disk by running the command:

```
df -h /
```

Output example:

```
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1        16G  8.3G  7.7G  52% /
```

9. Repeat this procedure on all SVMs in the PowerFlex system.

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Upgrading systems with more than 512 SDCs, and configuring LIA LDAP in large systems

Upgrade systems that contain more than 512 SDCs or configure LIA LDAP in large-scale systems, by performing the process in batches. Up to 512 SDCs can be included in each batch. The `gatewayUser.properties` file on the PowerFlex Gateway controls ignore lists which are used to exclude components from an upgrade or configuration procedure. Thus, several batches can be processed; each time a different set of components is processed or ignored.

Upgrade systems with more than 512 SDCs, or configure LIA LDAP in large systems

During nondisruptive upgrades, LIA LDAP configuration, and other actions in large-scale systems, the PowerFlex Gateway constantly requests status updates from all the nodes, even if the operations on those nodes are still pending or have already been completed. This scenario can affect PowerFlex Gateway performance. These recommendations can prevent such performance issues.

Prerequisites

Obtain the IP addresses of all the components in the PowerFlex system.

About this task

Perform the upgrade or configuration process in batches. In each pass, one set of components is upgraded, and the rest are ignored using an ignore list in the `gatewayUser.properties` file. Prepare several versions of this IP address ignore list, and on each pass of the upgrade or configuration process, comment out all ignore lists except for one list.

The first batch to be upgraded or configured must contain MDMs and SDSs. Subsequent batches must contain MDMs and up to 512 SDCs per batch (MDMs can be ignored for LIA LDAP configuration).

Steps

1. Create IP address ignore lists in the `gatewayUser.properties` file.
   a. Using a text editor, open the `gatewayUser.properties` file, located in the following directory on the PowerFlex Gateway server:
      - Windows—`C:\Program Files\EMC\ScaleIO\Gateway\webapps\ROOT\WEB-INF\classes\`
      - Linux—`/opt/emc/scaleio/gateway/webapps/ROOT/WEB-INF/classes`
   b. Add a new line containing the property *im.ip.ignore.list=* for each batch of components that will be *ignored* during processing. Add a comma-separated list of IP addresses to the property. The first batch to be processed must include MDMs and SDSs, so they should not be included in the first ignore list. Subsequent batches should include all MDMs and up to 512 SDCs (for LIA LDAP configuration, MDMs can also be ignored after the first batch). Comment out the ignore lists for the second, and subsequent, batches, using the # symbol at the beginning of a line. Optionally, for clarity, add comments describing each ignore list.

```
# NDU ignore list for batch 1
# sdc_group_01,sdc_group_02,sdc_group_03,sdc_group_04
im.ip.ignore.list=<IP_ADDRESSES_SDC_GROUP_1><IP_ADDRESSES_SDC_GROUP_2><IP_ADDRESSES_
SDC_GROUP_3><IP_ADDRESSES_SDC_GROUP_4>
```

```
# NDU ignore list for batch 2
# all_sds,sdc_group_02,sdc_group_03,sdc_group_04
```

```
#
im.ip.ignore.list=<IP_ADDRESSES_ALL_SDS><IP_ADDRESSES_SDC_GROUP_2><IP_ADDRESSES_SDC_
GROUP_3><IP_ADDRESSES_SDC_GROUP_4>
```

```
# NDU ignore list for batch 3
# all_sds,sdc_group_01,sdc_group_03,sdc_group_04
#
im.ip.ignore.list=<IP_ADDRESSES_ALL_SDS><IP_ADDRESSES_SDC_GROUP_1><IP_ADDRESSES_SDC_
GROUP_3><IP_ADDRESSES_SDC_GROUP_4>
```

```
# NDU ignore list for batch 4
# all_sds,sdc_group_01,sdc_group_02,sdc_group_04
#
im.ip.ignore.list=<IP_ADDRESSES_ALL_SDS><IP_ADDRESSES_SDC_GROUP_1><IP_ADDRESSES_SDC_
GROUP_2><IP_ADDRESSES_SDC_GROUP_4>
```

```
# NDU ignore list for batch 5
# all_sds,sdc_group_01,sdc_group_02,sdc_group_03
#
im.ip.ignore.list=<IP_ADDRESSES_ALL_SDS><IP_ADDRESSES_SDC_GROUP_1><IP_ADDRESSES_SDC_
GROUP_2><IP_ADDRESSES_SDC_GROUP_3>
```

Example:

```
# NDU ignore list for batch 1
# sdc_group_01,sdc_group_02,sdc_group_03,sdc_group_04
im.ip.ignore.list=1.1.25.10,2.2.25.10,1.1.25.103,2.2.25.103,1.1.25.104,2.2.25.104,1.
1.25.106,...
```

```
# NDU ignore list for batch 2
# all_sds,sdc_group_02,sdc_group_03,sdc_group_04
#
im.ip.ignore.list=1.1.90.20,4.4.90.20,3.3.90.20,2.2.90.20,1.1.90.21,4.4.90.21,2.2.90
.21,...
```

```
# NDU ignore list for batch 3
# all_sds,sdc_group_01,sdc_group_03,sdc_group_04
#
im.ip.ignore.list=1.1.90.20,4.4.90.20,3.3.90.20,2.2.90.20,1.1.90.21,4.4.90.21,2.2.90
.21,...
```

```
# NDU ignore list for batch 4
# all_sds,sdc_group_01,sdc_group_02,sdc_group_04
#
im.ip.ignore.list=1.1.90.20,4.4.90.20,3.3.90.20,2.2.90.20,1.1.90.21,4.4.90.21,2.2.90
.21,...
```

```
# NDU ignore list for batch 5
# all_sds,sdc_group_01,sdc_group_02,sdc_group_03
#
im.ip.ignore.list=1.1.90.20,4.4.90.20,3.3.90.20,2.2.90.20,1.1.90.21,4.4.90.21,2.2.90
.21,...
```

c. Save and close the file.
d. Restart the PowerFlex Gateway service:
   - Windows—From the Windows **Services** window, restart the **EMC ScaleIO Gateway**.
   - Linux—Type the following command:

     ```
     service scaleio-gateway restart
     ```

2. Run the standard upgrade procedure on the first batch of components.
3. Edit the `gatewayUser.properties` file described in step 1.
   a. Comment out the first ignore list.
   b. Uncomment the second ignore list.

Example:

```
# NDU ignore list for batch 1
# sdc_group_01,sdc_group_02,sdc_group_03,sdc_group_04
#
im.ip.ignore.list=1.1.25.10,2.2.25.10,1.1.25.103,2.2.25.103,1.1.25.104,2.2.25.104,1.
1.25.106,...
```

```
# NDU ignore list for batch 2
# all_sds,sdc_group_02,sdc_group_03,sdc_group_04
im.ip.ignore.list=1.1.90.20,4.4.90.20,3.3.90.20,2.2.90.20,1.1.90.21,4.4.90.21,2.2.90
.21,...
```

c. Save and close the file.
d. Restart the PowerFlex Gateway service:
   - Windows—From the Windows **Services** window, restart the **EMC ScaleIO Gateway**.
   - Linux—Type the following command:

     ```
     service scaleio-gateway restart
     ```

4. Run the upgrade or configuration process again for the second batch.
5. If there are more than two batches of components to be processed, edit the `gatewayUser.properties` file again.
   a. Comment out the second batch of components.
   b. Uncomment the third batch of components.
   c. Save and close the file.
   d. Restart the PowerFlex Gateway service:
      - Windows—From the Windows **Services** window, restart the **EMC ScaleIO Gateway**.
      - Linux—Type the following command:

        ```
        service scaleio-gateway restart
        ```

6. Run the upgrade or configuration process again for the third batch.
7. Repeat the procedures in steps 5 and 6 for the remaining batches.

Results

Upgrade or configuration of the system is complete.